Resolvent Convergence under Heterogeneous Second-Moment Profiles and Quadratic-Form Control

Cosme Louart*

Abstract

Let $X = (x_1, \dots, x_n) \in \mathbb{R}^{p\times n}$ have independent real columns with finite second moments, not necessarily centered or identically distributed. For $z \in \mathbb{C} \setminus [0,\infty)$, we compare

$$G^z = \left(\frac{1}{n}XX^\top - zI_p\right)^{-1}$$

with a deterministic matrix $\tilde{G}^z$ defined by the second-moment profile $(\Sigma_i)_{i\in[n]}$, where $\Sigma_i = \mathbb{E}[x_i x_i^\top]$.

For deterministic matrices A, we prove finite-dimensional L^q bounds for $\mathrm{Tr}(A(G^z - \tilde{G}^z))$ in terms of moments of centered quadratic forms, without assuming independence among the coordinates of a column. Operator-test bounds impose no trace-growth or aspect-ratio restriction and display explicitly the effect of the sizes and repetitions of the second-moment matrices $(\Sigma_i)_{i\in[n]}$. Under $\sqrt{p}\sup_{i\in[n]}|\Sigma_i|_{\mathrm{HS}} = O(n)$, the Hilbert–Schmidt bounds involve approximation of the profile by commuting positive-semidefinite matrices, with explicit dependence on the approximation error and the dimension of their span. We also obtain corresponding bounds for $\mathrm{Tr}(AG^zBG^z)$, with deterministic A, B. All bounds are locally uniform away from $[0,\infty)$.

Keywords: sample covariance matrices; deterministic equivalents; resolvent estimates; heterogeneous covariance profiles; dependent coordinates; nonasymptotic bounds.
MSC2020 subject classifications: 60B20, 15B52, 62H10.

Introduction

We study deterministic approximations of the resolvent of a sample second-moment matrix. Let $X = (x_1, \dots, x_n) \in \mathbb{R}^{p\times n}$ have independent columns with finite second moments, not necessarily centered or identically distributed, and set

$$\Sigma_i := \mathbb{E}[x_i x_i^\top], \qquad G^z := \left(\frac{1}{n}XX^\top - zI_p\right)^{-1}, \quad z \in \mathbb{C} \setminus [0,\infty).$$

*School of Data Science, The Chinese University of Hong Kong (Shenzhen), Shenzhen, China

The collection $(\Sigma_i)_{i\in[n]}$ is called the second-moment profile. We approximate G^z by a deterministic matrix $\tilde{G}^z$ depending only on this collection, and bound $\mathrm{Tr}(A(G^z-\tilde{G}^z))$ in L^q for deterministic A. This includes normalized traces, which describe the empirical spectral distribution, and directional quantities obtained with $A = uu^\top$.

The Marchenko–Pastur theorem [MP67] and its extensions provide the classical description of sample covariance spectra; see [BS10, PS11]. Deterministic equivalents provide approximations defined at each matrix dimension, notably for variance-profile and information-plus-noise models [DS07, HLN07]. Concentration of quadratic forms provides a way to go beyond independence of coordinates, as in Bai–Zhou [BZ08] and Yaskov [Yas16]. Related results treat heterogeneous second-moment matrices [BGC16, MWY23] and quantitative resolvent estimates under concentration assumptions [Cho22, FMPW26].

Here we obtain finite-dimensional bounds that separate the dependence on the matrices Σ_i, the test A, and the fluctuations of the centered quadratic forms

$$x_i^\top B x_i - \mathrm{Tr}(\Sigma_i B).$$

The matrices Σ_i may all differ and need not commute, and no limiting assumption is imposed on their collection. Independence is required only between columns. The estimates give convergence criteria according to the size of these quadratic-form fluctuations, uniformly for z in a fixed compact subset of $\mathbb{C}\setminus[0,\infty)$.

The results are organized along two axes. The first is the norm used to measure A. The Hilbert–Schmidt comparison combines $|A|_{\mathrm{HS}}$, bounds on $|\Sigma_i|_{\mathrm{HS}}$, and moments of quadratic forms with $|B|_{\mathrm{HS}} \le 1$. The operator comparison instead uses $|A|_{\mathrm{op}}$, the nuclear norms $|\Sigma_i|_* = \mathrm{Tr}(\Sigma_i)$, and the corresponding moments for $|B|_{\mathrm{op}} \le 1$. The former can exploit smaller quadratic-form fluctuations, while the latter can benefit from a smaller test operator norm and allows an unrestricted aspect ratio p/n. The rates also reflect relations within the second-moment profile $(\Sigma_i)_{i\in[n]}$: groups of equal matrices enter the operator bound, whereas approximation by commuting matrices enters the Hilbert–Schmidt bound.

The second axis is the moment order q of the centered quadratic forms. For each norm, we give bounds for $q \ge 1$ and refined bounds for $q \ge 2$. At $q = 1$, only first absolute moments are required, even when the centered quadratic forms have no finite second moment. For $q \ge 2$, the refined estimates contain a fluctuation term with a factor $n^{-1/2}$, together with bias terms depending on the profile and the quadratic-form second moments. For i.i.d. entries with a fixed centered unit-variance law, the first-moment argument recovers weak convergence in probability to the Marchenko–Pastur law as $p \to \infty$ and $p/n \to c \in [0,\infty)$, without higher entry moments (Corollary 1.13).

The proofs separate fluctuations from bias and combine leave-one-out identities with stability of the deterministic fixed-point equations. Successive class replacements handle heterogeneous collections $(\Sigma_i)_{i\in[n]}$ in the operator comparison. This replacement procedure produces resolvents with general matrices Z in place of zI_p. We therefore establish our results for matrix parameters Z whose numerical range is separated from $\mathbb{R}_+$. Additive deformation and Cauchy's formula then also give deterministic approximations for resolvent sandwiches $G^z B G^z$, with arbitrary deterministic complex B. At negative spectral parameters, resolvents are regularized inverses; these sandwiches occur, in particular, in ridge-regression prediction errors.

Section 1 states the results, compares them with related work, and gives examples, with selected rates summarized in Table 1. Sections 2 and 3 contain the deterministic and probabilistic proofs, respectively. Subsection 3.4 treats resolvent sandwiches; auxiliary results are collected in the appendices.

Notation

Set

$$\mathbb{R}_+ := [0,\infty), \qquad \mathbb{H} := \{z\in\mathbb{C} : \Im z > 0\}.$$

Let δ_0 denote the Dirac mass at zero. For $S \subset \mathbb{C}$, its convex conic hull is denoted by

$$\operatorname{cone}(S) := \left\{ \sum_{\ell=1}^{m} a_\ell s_\ell : m \in \mathbb{N},\ a_\ell \in \mathbb{R}_+,\ s_\ell \in S \right\}.$$

For $q, r > 0$, $q \wedge r$ denotes their minimum. We write $[n] := \{1, \dots, n\}$. For $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, let $\mathcal{M}_p(\mathbb{K}) := \mathbb{K}^{p \times p}$ and abbreviate $\mathcal{M}_p := \mathcal{M}_p(\mathbb{C})$. For $S \subset \mathbb{C}$, set

$$\mathcal{D}_n(S) := \{\operatorname{Diag}(d_1, \dots, d_n) : d_i \in S\}.$$

Here $\mathbf{i} = \sqrt{-1}$, $M^\top$ is the transpose, and $M^* = \overline{M}^\top$ is the conjugate transpose. For Hermitian matrices $A, B \in \mathcal{M}_n$, the Loewner order $A \preceq B$ means $x^*(B - A)x \geq 0$ for every $x \in \mathbb{C}^n$; we write $B \succeq A$ for the reverse order. For any square matrix A, set

$$\Re A := \frac{A + A^*}{2}, \qquad \Im A := \frac{A - A^*}{2\mathbf{i}}.$$

For $x \in \mathbb{C}^n$, $D = \operatorname{Diag}(x)$ has entries $D_i = x_i$. The Euclidean, operator, Hilbert–Schmidt, and nuclear norms are denoted by $|\cdot|$, $|\cdot|_{\mathrm{op}}$, $|\cdot|_{\mathrm{HS}}$, and $|\cdot|_*$, respectively. Thus

$$|M|_{\mathrm{HS}}^2 = \operatorname{Tr}(MM^*), \qquad |M|_{\mathrm{HS}} = \sup_{|A|_{\mathrm{HS}} \leq 1} |\operatorname{Tr}(A^*M)|, \qquad |M|_* = \sup_{|A|_{\mathrm{op}} \leq 1} |\operatorname{Tr}(A^*M)|.$$

For $Z \in \mathcal{M}_p$, its numerical range is

$$\mathcal{W}(Z) := \{u^* Z u : u \in \mathbb{C}^p,\ |u| = 1\}.$$

It is compact and convex by the Toeplitz–Hausdorff theorem. For $1 \leq q < \infty$, L^q denotes the space of real or complex random variables ξ satisfying $\mathbb{E}\,[|\xi|^q] < \infty$, equipped with the norm

$$\|\xi\|_{L^q} := \left(\mathbb{E}\,[|\xi|^q]\right)^{1/q}.$$

For a finite matrix profile $\Sigma = (\Sigma_i)_{i \in [n]}$, set

$$|\Sigma|_\nu^\infty := \max_{i \in [n]} |\Sigma_i|_\nu, \qquad \nu \in \{\mathrm{op}, \mathrm{HS}, *\}. \tag{0.1}$$

For vectors and diagonal matrices, $|u|_\infty := \max_i |u_i|$ and $|D|_\infty := \max_i |D_i| = |D|_{\mathrm{op}}$. An index of the specialized notation appears in Appendix A.

Contents

Introduction **1**

Notation **2**

1 Main results and examples **4**
1.1 Deterministic results 4
1.2 Probabilistic results 5
1.2.1 Quadratic-form moduli and directional estimates 5
1.2.2 Operator-norm route 6
1.2.3 Hilbert–Schmidt-norm route 8
1.2.4 Resolvent sandwiches 9
1.3 Rates, comparisons, and examples 10
1.3.1 Comparison with earlier results 10
1.3.2 Selected illustrations 11
1.3.3 Independent entries 14

2 Deterministic equivalents for sectorial matrix parameters **15**
2.1 Matrix sectors and two-sided resolvent calculus 15
2.2 The matrix-parameter fixed point 18
2.3 Holomorphic dependence 21
2.4 Scalar spectral parameters and Stieltjes transforms 22
2.5 Support of the deterministic equivalent measure 24

3 Probabilistic estimates **26**
3.1 Common probabilistic estimates and auxiliary pivots 26
3.1.1 Resolvent and leave-one-out estimates 26
3.1.2 Auxiliary resolvents and inverse defects 29
3.1.3 Choice of the auxiliary pivot 29
3.1.4 Transfer from the inverse defect 31
3.2 Hilbert–Schmidt-norm route 35
3.3 Operator-norm route 39
3.3.1 One-class case 40
3.3.2 Multiple classes 45
3.4 Resolvent sandwiches 50

A Notation index **53**

B Auxiliary results **54**

References **60**

1 Main results and examples

1.1 Deterministic results

We first define the deterministic equivalent and record the properties needed for the probabilistic comparisons: existence, uniqueness, holomorphic dependence, and Stieltjes-transform representations. Columnwise self-consistent equations for heterogeneous correlation matrices appear in [WCDS12, Theorem 1] and in [KA16, Yin20]. These works formulate deterministic equivalents in asymptotic random-matrix settings; [WCDS12] also allows a positive-semidefinite additive deformation. Their equations are already defined at each matrix dimension. We use a formulation valid for every finite p, n and every real symmetric positive-semidefinite profile $\Sigma_1, \dots, \Sigma_n$, without an aspect-ratio or profile-size assumption. The construction is also related to the theory of self-consistent equations developed in [AEK19].

Theorem 1.1. *For every $z \in \mathbb{C}\backslash\mathbb{R}_+$, there are unique scalars $\tilde\Gamma_1^z, \dots, \tilde\Gamma_n^z \in \{a-b/z : a, b \geq 0\}$ and a matrix $\tilde G^z$ satisfying*

$$\tilde G^z = -\frac{1}{z}\left(I_p + \frac{1}{n}\sum_{j=1}^{n} \tilde\Gamma_j^z \Sigma_j\right)^{-1}, \qquad \tilde\Gamma_i^z = -\frac{1}{z\left(1 + \frac{1}{n}\operatorname{Tr}(\Sigma_i \tilde G^z)\right)}, \qquad i \in [n].$$

The mappings $z \mapsto \tilde\Gamma_i^z$ and $z \mapsto \tilde G^z$ are holomorphic on $\mathbb{C} \setminus \mathbb{R}_+$.

The deterministic construction and the resolvent comparison estimates extend to complex matrix parameters Z satisfying $d(\mathcal{W}(Z), \mathbb{R}_+) > 0$. Their matrix formulations, including the dependence of the constants on Z, are given in Sections 2 and 3. Set

$$\tilde g(z) := \frac{1}{p}\operatorname{Tr}(\tilde G^z), \qquad z \in \mathbb{C} \setminus \mathbb{R}_+.$$

Theorem 1.2. *For every $i \in [n]$, the mapping $z \mapsto \tilde{\Gamma}_i^z$ is the Stieltjes transform of a probability measure of compact support $\tilde{\mu}_i$ on $\mathbb{R}_+$. The mapping $z \mapsto \tilde{g}(z)$ is the Stieltjes transform of a probability measure $\tilde{\mu}$ on $\mathbb{R}_+$, with*

$$\tilde{\mu} = \left(1 - \frac{n}{p}\right)\delta_0 + \frac{1}{p}\sum_{i=1}^{n} \tilde{\mu}_i. \tag{1.1}$$

In addition, $z \mapsto \tilde{G}^z$ is the Stieltjes transform of a positive matrix-valued measure on $\mathbb{R}_+$ with total mass I_p.

The right-hand side of (1.1) is understood as a finite signed combination; the identity asserts that this combination is in fact the positive probability measure whose Stieltjes transform is $\tilde{g}$.

Section 2 establishes existence and uniqueness, the scalar and matrix-valued Stieltjes-transform representations, an explicit support bound, and the stability estimate used in the probabilistic comparisons.

1.2 Probabilistic results

Fix an index set $\Theta \subset \mathbb{N}^2$. For every $(p,n) \in \Theta$, let $X = (x_1, \dots, x_n) \in \mathcal{M}_{p,n}(\mathbb{R})$ have independent real columns with finite second moments, and retain the notation Σ_i and G^z from the introduction. The dependence on (p,n) is suppressed.

Fix a compact set $K \subset \mathbb{C} \setminus \mathbb{R}_+$. Throughout this section, $O(\cdot)$ means uniformly for $(p,n) \in \Theta$ and $z \in K$. The implicit constants may depend on K, on the fixed moment order, and on the constants appearing in the assumptions, but not on the deterministic test matrices or on the profile quantities displayed in the bounds. The assumption $p = O(n)$ in particular means that there exists a constant $C > 0$ such that $(p,n) \in \Theta \implies p \le Cn$. No uniformity is claimed as K approaches $\mathbb{R}_+$.

1.2.1 Quadratic-form moduli and directional estimates

For $q \ge 1$, define

$$M_{\mathrm{HS}}^{(q)} := \sup_{\substack{i \in [n] \\ |B|_{\mathrm{HS}} \le 1}} \left\| x_i^\top B x_i - \mathrm{Tr}(\Sigma_i B) \right\|_{L^q}, \qquad M_{\mathrm{op}}^{(q)} := \sup_{\substack{i \in [n] \\ |B|_{\mathrm{op}} \le 1}} \left\| x_i^\top B x_i - \mathrm{Tr}(\Sigma_i B) \right\|_{L^q}. \tag{1.2}$$

The suprema are over deterministic complex matrices $B \in \mathcal{M}_p$. Both moduli may depend on (p,n) and may vanish. The reduction to real symmetric tests, together with its conditional version, is recorded in Lemma B.3.

Size of the two moduli. The standing second-moment assumption on the columns ensures that $M_{\mathrm{HS}}^{(1)}$ and $M_{\mathrm{op}}^{(1)}$ are finite, by Lemma B.4; higher-order moduli may be infinite. They are nondecreasing in the moment order: for $1 \le r \le q$, $M_{\mathrm{HS}}^{(r)} \le M_{\mathrm{HS}}^{(q)}$ and $M_{\mathrm{op}}^{(r)} \le M_{\mathrm{op}}^{(q)}$.

The inclusions between the two matrix unit balls give

$$M_{\mathrm{HS}}^{(q)} \le M_{\mathrm{op}}^{(q)} \le \sqrt{p}\, M_{\mathrm{HS}}^{(q)}.$$

In particular, $M_{\mathrm{HS}}^{(q)} = o(\sqrt{p})$ implies $M_{\mathrm{op}}^{(q)} = o(p)$, but the two moduli need not have the same order.

Their dependence on the second-moment matrices Σ_i is explicit in familiar models. Suppose that $x_i = \Sigma_i^{1/2} y_i$, where the coordinates of each y_i are independent, centered, and standardized, and let $K_4 := \sup_{i,j} \|(y_i)_j\|_{L^4} < \infty$. Lemma B.9 gives

$$M_{\mathrm{HS}}^{(2)} = O\big(K_4^2 |\Sigma|_{\mathrm{op}}^{\infty}\big), \qquad M_{\mathrm{op}}^{(2)} = O\big(K_4^2 |\Sigma|_{\mathrm{HS}}^{\infty}\big),$$

with the corresponding first-moment bounds following by monotonicity. For centered Gaussian columns, these spectral scales are attained for every fixed $q \geq 2$:

$$M_{\mathrm{HS}}^{(q)} \asymp_q |\Sigma|_{\mathrm{op}}^{\infty}, \qquad M_{\mathrm{op}}^{(q)} \asymp_q |\Sigma|_{\mathrm{HS}}^{\infty},$$

by Lemma B.10. Thus the Hilbert–Schmidt normalization reflects the largest eigenvalues of the matrices Σ_i, whereas the operator normalization reflects their Hilbert–Schmidt sizes.

A large Hilbert–Schmidt size can also force a large operator-normalized modulus without coordinate independence. Indeed, Lemma 3.11 gives

$$|\Sigma|_{\mathrm{HS}}^{\infty} \leq |\Sigma|_{\mathrm{op}}^{\infty} + M_{\mathrm{op}}^{(2)}.$$

Consequences for resolvent tests. The comparison theorems bound $\| \operatorname{Tr}(A(G^z - \tilde{G}^z)) \|_{L^q}$ for deterministic A, measured in Hilbert–Schmidt or operator norm. Their constants are uniform in the test matrix; any supremum over tests is taken outside the L^q norm. Table 1 records selected specializations of Theorems 1.4 and 1.6. When both comparison theorems apply, one may take the smaller bound.

The ordering of the moduli does not order the resulting resolvent bounds: the Hilbert–Schmidt comparison uses the smaller modulus but the larger test norm $|A|_{\mathrm{HS}}$, and the two comparisons have different assumptions and coefficients involving the collection $(\Sigma_i)_{i\in[n]}$. For high-rank tests, the smaller value of $|A|_{\mathrm{op}}$ can favor the operator comparison. For directional tests, however,

$$|uu^\top|_{\mathrm{op}} = |uu^\top|_{\mathrm{HS}} = |u|^2, \qquad u \in \mathbb{R}^p,$$

so a smaller Hilbert–Schmidt modulus can improve the estimate without increasing the test norm.

For example, for centered Gaussian columns with $\Sigma_i = I_p$ and $p \asymp n$, the two moduli have orders 1 and $\sqrt{p}$, respectively. For every fixed $q \geq 2$, Theorem 1.6 therefore gives

$$\left\| u^\top (G^z - \tilde{G}^z) u \right\|_{L^q} = O\left(\frac{|u|^2}{\sqrt{n}} \right),$$

whereas the operator-test estimate in Theorem 1.4 does not yield a vanishing bound for a fixed unit direction in this regime.

1.2.2 Operator-norm route

Let $\Sigma^{(1)}, \ldots, \Sigma^{(k)}$, with $k \leq n$, be the distinct second-moment matrices. For $a \in [k]$, set

$$I_a := \{ i : \Sigma_i = \Sigma^{(a)} \}, \qquad n_a := |I_a|, \qquad \pi_a := \frac{n_a}{n} > 0.$$

The number and sizes of these classes may depend on the dimensions. Note that columns that follow different laws but admit the same second-moment matrix belong to the same class. Define

$$\tau_\Sigma := \max_{a\in[k]} \frac{1 + |\Sigma^{(a)}|_{\mathrm{op}}}{1 + \pi_a |\Sigma^{(a)}|_{\mathrm{op}}}. \tag{1.3}$$

The coefficient depends only on these matrices and their proportions, independently of z and the test. Unlike σ_Σ below, it does not measure approximation: close but unequal matrices with large operator norm belong to different classes.

Remark 1.3 (Operator size and class proportions)**.** Since $0 < \pi_a \leq 1$, each ratio in (1.3) is at least 1 and at most $\min(1 + |\Sigma^{(a)}|_{\mathrm{op}}, \pi_a^{-1})$. Thus

$$1 \leq \tau_\Sigma \leq \min\{ 1 + |\Sigma|_{\mathrm{op}}^{\infty}, (\min_a \pi_a)^{-1} \}.$$

Table 1: Comparison rates by profile regime and test geometry. Each entry is an $O(\cdot)$ bound. Blank assumption cells mean that no additional assumption of that type is imposed in the displayed specialization. A slash marks a cell where no separate bound in the corresponding quadratic-form modulus is recorded. The number k of classes in the last two rows may grow. All displayed operator specializations have $|\Sigma|_*^\infty = O(n)$; the general theorem retains an explicit trace factor and imposes no such restriction.

Result	**Assumptions**					**Rates for $\|\operatorname{Tr}(A(G^z - \tilde{G}^z))\|_{L^q}$**					
	Profile structure	p/n	$\lvert\Sigma\rvert_{\mathrm{op}}^\infty$	$\lvert\Sigma\rvert_{\mathrm{HS}}^\infty$	$\lvert\Sigma\rvert_*^\infty$	$\lvert A\rvert_{\mathrm{op}} \le 1$		$\lvert A\rvert_{\mathrm{HS}} \le 1$		***Stieltjes t.*** $(A = I_p/p)$	
						$q \ge 1$	$q \ge 2$	$q \ge 1$	$q \ge 2$	$q \ge 1$	$q \ge 2$
Th. 1.4 *Th. 1.6 for* $\lvert\cdot\rvert_{\mathrm{HS}}$	*Arbitrary*	$O(1)$	$O(1)$			$1 + M_{\mathrm{op}}^{(q)}$	$\frac{M_{\mathrm{op}}^{(q)}}{\sqrt{n}} + \frac{1 + (M_{\mathrm{op}}^{(2)})^2}{n}$	$M_{\mathrm{HS}}^{(q)} + \frac{\sqrt{p}}{n}$	$\frac{M_{\mathrm{HS}}^{(q)}}{\sqrt{n}} + \frac{[1 + (M_{\mathrm{HS}}^{(2)})^2]\sqrt{p}}{n}$	$\frac{1 + M_{\mathrm{op}}^{(q)}}{p}$	$\frac{M_{\mathrm{op}}^{(q)}}{p\sqrt{n}} + \frac{1 + (M_{\mathrm{op}}^{(2)})^2}{pn}$
Th. 1.4	*Arbitrary*		$O(1)$		$O(n)$			⁄	⁄		
Th. 1.4	*k well-balanced classes* $\left(\frac{1}{\min_a \pi_a} = O(k)\right)$				$O(n)$	$k(1 + M_{\mathrm{op}}^{(q)})$	$\frac{kM_{\mathrm{op}}^{(q)}}{\sqrt{n}} + \frac{k^2 + (M_{\mathrm{op}}^{(2)})^2}{n}$	⁄	⁄	$\frac{k(1 + M_{\mathrm{op}}^{(q)})}{p}$	$\frac{kM_{\mathrm{op}}^{(q)}}{p\sqrt{n}} + \frac{k^2 + (M_{\mathrm{op}}^{(2)})^2}{pn}$
Th. 1.6	*k commuting profiles* $(\Sigma^{(a)}\Sigma^{(b)} = \Sigma^{(b)}\Sigma^{(a)})$	$O(1)$		$O(\sqrt{p})$		⁄	⁄	$k\left(M_{\mathrm{HS}}^{(q)} + \frac{\sqrt{p}}{n}\right)$	$\frac{M_{\mathrm{HS}}^{(q)}}{\sqrt{n}} + \frac{k[1 + (M_{\mathrm{HS}}^{(2)})^2]\sqrt{p}}{n}$	$k\left(\frac{M_{\mathrm{HS}}^{(q)}}{\sqrt{p}} + \frac{1}{n}\right)$	$\frac{M_{\mathrm{HS}}^{(q)}}{\sqrt{np}} + \frac{k[1 + (M_{\mathrm{HS}}^{(2)})^2]}{n}$

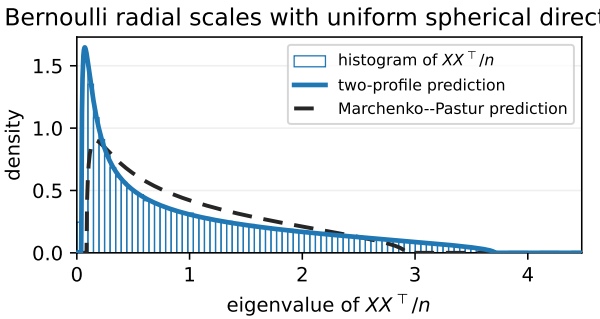


Figure 1: Empirical eigenvalue density of $\frac{1}{n}XX^\top$ for columns $x_i = \sqrt{r_i}\, y_i$, where the y_i are independent and uniform on $\{y \in \mathbb{R}^p : |y| = \sqrt{p}\}$, the r_i are independent with $\mathbb{P}(r_i = 1/5) = \mathbb{P}(r_i = 9/5) = 1/2$, and the two families are independent. Thus $|x_i|^2/p = r_i$, $\mathbb{E}[x_i x_i^\top] = I_p$, and $\mathbb{E}[x_i x_i^\top \mid r_i] = r_i I_p$. The histogram pools 3000 independent matrices with $p = 50$ and $n = 100$. The dark dashed curve is the Marchenko–Pastur prediction based on the unconditional second moment, whereas the solid blue curve is the equal-weight two-profile prediction with profile values $I_p/5$ and $9I_p/5$.

A common profile has $\tau_\Sigma = 1$, including the zero profile. Bounded operator profiles have bounded τ_Σ without a balance condition. Equally sized k classes give $\tau_\Sigma \le k$; more generally, $(\min_a \pi_a)^{-1} = O(k)$ gives $\tau_\Sigma = O(k)$, with k allowed to grow.

Theorem 1.4 (Operator comparison with explicit trace dependence). *For every deterministic $A \in \mathcal{M}_p$ and every $q \ge 1$,*

$$\| \operatorname{Tr}(A(G^z - \tilde{G}^z))\|_{L^q} = O\left(|A|_{\mathrm{op}} \left(1 + \frac{|\Sigma|_*^\infty}{n}\right)^{3/2} \tau_\Sigma \big(1 + M_{\mathrm{op}}^{(q)}\big) \right). \tag{1.4}$$

For every $q \ge 2$, one also has

$$\| \operatorname{Tr}(A(G^z - \tilde{G}^z))\|_{L^q} = O\left(|A|_{\mathrm{op}} \left(1 + \frac{|\Sigma|_*^\infty}{n}\right)^{3/2} \left[\frac{\tau_\Sigma M_{\mathrm{op}}^{(q)}}{\sqrt{n}} + \frac{\tau_\Sigma^2 + (M_{\mathrm{op}}^{(2)})^2}{n} \right] \right). \tag{1.5}$$

If $K \subset (-\infty, 0)$, the factor $\left(1 + \frac{|\Sigma|_*^\infty}{n}\right)^{3/2}$ in both bounds may be replaced by $1 + \frac{|\Sigma|_*^\infty}{n}$.

1.2.3 Hilbert–Schmidt-norm route

Define

$$\sigma_\Sigma := \inf_{\substack{T_1,\dots,T_n \in \mathcal{M}_p(\mathbb{R}) \\ T_i = T_i^\top \succeq 0 \ (i \in [n]) \\ T_i T_j = T_j T_i \ (i,j \in [n])}} (1 + \dim_{\mathbb{R}} \operatorname{span}\{T_1, \dots, T_n\}) \left(1 + \frac{1}{n} \sum_{i=1}^n |\Sigma_i - T_i|_{\mathrm{op}}\right). \tag{1.6}$$

The coefficient combines the dimension of the span of a commuting approximating family with its average operator-norm error. Any such family gives an upper bound. The matrices T_i enter only the error coefficient; the deterministic equivalent $\tilde{G}^z$ is always computed from the original profile $(\Sigma_i)_{i \in [n]}$.

Remark 1.5 (Simple bounds for the profile coefficient). The zero family gives

$$1 \le \sigma_\Sigma \le 1 + \frac{1}{n} \sum_{i=1}^n |\Sigma_i|_{\mathrm{op}} \le 1 + |\Sigma|_{\mathrm{op}}^\infty.$$

Hence bounded operator profiles have $\sigma_\Sigma = O(1)$. If the Σ_i commute pairwise, choosing $T_i = \Sigma_i$ gives

$$\sigma_\Sigma \le 1 + \dim_{\mathbb{R}} \operatorname{span}\{\Sigma_1, \dots, \Sigma_n\}.$$

Thus a profile with $k \ge 1$ distinct commuting values satisfies $\sigma_\Sigma = O(k)$. Every common profile satisfies $\sigma_\Sigma \le 2$, while the zero profile has $\sigma_\Sigma = 1$. These bounds require no class-balance condition and no bound on the norms of the commuting matrices. Examples 1.10–1.11 below illustrate these bounds and the additional flexibility obtained by using commuting approximants in (1.6).

Theorem 1.6 (Hilbert–Schmidt comparison under a profile bound). *Assume that*

$$\sqrt{p}\, |\Sigma|_{\mathrm{HS}}^\infty = O(n). \tag{1.7}$$

For every deterministic $A \in \mathcal{M}_p$ and every $q \ge 1$,

$$\left\| \operatorname{Tr}\Big(A(G^z - \tilde{G}^z)\Big) \right\|_{L^q} = O\left(\sigma_\Sigma |A|_{\mathrm{HS}} \left[M_{\mathrm{HS}}^{(q)} + \frac{|\Sigma|_{\mathrm{HS}}^\infty}{n} \right] \right). \tag{1.8}$$

For every $q \ge 2$,

$$\left\| \operatorname{Tr}\Big(A(G^z - \tilde{G}^z)\Big) \right\|_{L^q} = O\left(|A|_{\mathrm{HS}} \left[\frac{M_{\mathrm{HS}}^{(q)}}{\sqrt{n}} + \sigma_\Sigma \frac{|\Sigma|_{\mathrm{HS}}^\infty + (M_{\mathrm{HS}}^{(2)})^2 \sqrt{p}}{n} \right] \right). \tag{1.9}$$

1.2.4 Resolvent sandwiches

For $z \in \mathbb{C} \setminus \mathbb{R}_+$ and a deterministic matrix $A \in \mathcal{M}_p$, set

$$\mathcal{S}^z[A] := G^z A G^z,$$

and define $\widetilde{\mathcal{S}}^z[A]$ by

$$\widetilde{\mathcal{S}}^z[A] = \tilde{G}^z A \tilde{G}^z + \frac{1}{n^2} \sum_{i=1}^{n} \frac{\operatorname{Tr}\Big(\Sigma_i \widetilde{\mathcal{S}}^z[A]\Big)}{\Big(1 + \frac{1}{n}\operatorname{Tr}(\Sigma_i \tilde{G}^z)\Big)^2} \tilde{G}^z \Sigma_i \tilde{G}^z. \tag{1.10}$$

Proposition 3.17 shows that this equation has a unique solution, that $A \mapsto \widetilde{\mathcal{S}}^z[A]$ is linear, and that, for the matrix-parameter equivalent defined in Section 2,

$$\widetilde{\mathcal{S}}^z[A] = -\frac{d}{dt} \widetilde{G}^{zI_p - tA}\Big|_{t=0}.$$

Corollary 1.7 (Random-to-deterministic resolvent sandwiches). *Let $A, B \in \mathcal{M}_p$ be deterministic, and let $q \geq 1$. In each of the minima below, the second entry is available only when $q \geq 2$.*

(i)
$$\Big\| \operatorname{Tr}\Big(B\Big[\mathcal{S}^z[A] - \widetilde{\mathcal{S}}^z[A]\Big]\Big) \Big\|_{L^q} = O\left(|A|_{\mathrm{op}} |B|_{\mathrm{op}} \left(1 + \frac{|\Sigma|_*^\infty}{n}\right)^{3/2} \min\left\{ \tau_\Sigma\big(1 + M_{\mathrm{op}}^{(q)}\big), \frac{\tau_\Sigma M_{\mathrm{op}}^{(q)}}{\sqrt{n}} + \frac{\tau_\Sigma^2 + (M_{\mathrm{op}}^{(2)})^2}{n} \right\} \right).$$

(ii) *If $\sqrt{p}\,|\Sigma|_{\mathrm{HS}}^\infty = O(n)$, then*
$$\Big\| \operatorname{Tr}\Big(B\Big[\mathcal{S}^z[A] - \widetilde{\mathcal{S}}^z[A]\Big]\Big) \Big\|_{L^q} = O\left(|A|_{\mathrm{op}} |B|_{\mathrm{HS}} \min\left\{ \sigma_\Sigma \left(M_{\mathrm{HS}}^{(q)} + \frac{|\Sigma|_{\mathrm{HS}}^\infty}{n} \right), \frac{M_{\mathrm{HS}}^{(q)}}{\sqrt{n}} + \sigma_\Sigma \frac{|\Sigma|_{\mathrm{HS}}^\infty + (M_{\mathrm{HS}}^{(2)})^2 \sqrt{p}}{n} \right\} \right).$$

The proof, given in Subsection 3.4, follows directly from Theorems 1.4 and 1.6. The same argument extends, for each fixed k, to symmetrized sandwiches with $k+1$ resolvent factors and deterministic insertions $A_1, \ldots, A_k$ by taking successive mixed derivatives of $\widetilde{G}^{zI_p - t_1 A_1 - \cdots - t_k A_k}$ at $t_1 = \cdots = t_k = 0$.

Ridge regression. Let $y = X^\top \beta + \varepsilon$, where β is deterministic, $\mathbb{E}[\varepsilon \mid X] = 0$, and $\mathbb{E}[\varepsilon\varepsilon^\top \mid X] = \sigma_\varepsilon^2 I_n$. For fixed $\lambda > 0$, the ridge estimator $\hat{\beta}_\lambda = G^{-\lambda} X y / n$ has conditional bias $-\lambda G^{-\lambda}\beta$. If a test vector x_0, independent of (X, ε), has second moment Σ, then

$$\mathbb{E}\Big[(x_0^\top \hat{\beta}_\lambda - x_0^\top \beta)^2 \mid X\Big] = \lambda^2 \beta^\top G^{-\lambda} \Sigma G^{-\lambda} \beta + \frac{\sigma_\varepsilon^2}{n} \operatorname{Tr}\big(\Sigma\left[G^{-\lambda} - \lambda (G^{-\lambda})^2\right]\big).$$

This follows from $\hat{\beta}_\lambda - \beta = -\lambda G^{-\lambda}\beta + G^{-\lambda} X\varepsilon/n$ and $\frac{1}{n} G^{-\lambda} X X^\top G^{-\lambda} = G^{-\lambda} - \lambda (G^{-\lambda})^2$. Assume the classical conditions $p = O(n)$ and $M_{\mathrm{HS}}^{(2)} = O(1)$, which give

$$M_{\mathrm{op}}^{(2)} \leq \sqrt{p}\, M_{\mathrm{HS}}^{(2)} = O(\sqrt{p}).$$

Suppose also that $\sup_{i \in [n]} |\Sigma_i|_{\mathrm{op}} = O(1)$ and $|\Sigma|_{\mathrm{op}} = O(1)$ (the former assumption, together with $p = O(n)$, implies $\sqrt{p}\,|\Sigma|_{\mathrm{HS}}^\infty = O(n)$). We use the two bounds of Corollary 1.7 with $q = 2$: part (i) controls the squared-resolvent term in the variance with insertion I_p and test Σ, keeping the test factor $|\Sigma|_{\mathrm{op}}$ bounded even when $|\Sigma|_{\mathrm{HS}}$ is of order $\sqrt{p}$; part (ii) controls the squared-bias term with insertion Σ and test $\beta\beta^\top$, using $|\beta\beta^\top|_{\mathrm{HS}} = |\beta|^2$ and the bounded modulus $M_{\mathrm{HS}}^{(2)}$. The one-resolvent theorems control the remaining variance term. No commutation between Σ and the training profiles is required.

1.3 Rates, comparisons, and examples

1.3.1 Comparison with earlier results

The first-moment bounds give quantitative sufficient conditions within the quadratic-form approach of Bai–Zhou [BZ08] and Yaskov [Yas16, Yas14]. In the isotropic regime $\Sigma_i = I_p$ with $p \asymp n$, Theorem 1.4 gives

$$\left\| \frac{1}{p} \operatorname{Tr}(G^z - \tilde{G}^z) \right\|_{L^1} = O\left(\frac{1 + M_{\mathrm{op}}^{(1)}}{p} \right),$$

so $M_{\mathrm{op}}^{(1)} = o(p)$ yields convergence with an explicit error bound. Yaskov's universality principle already permits independent, non-identically distributed columns; our comparisons add explicit rates and L^q control of deterministic matrix tests.

For heterogeneous second-moment matrices, the constructions in [WCDS12, Yin20] use linear transforms of independent coordinates; our assumptions instead concern quadratic forms of the columns themselves. Under the size and moment assumptions of our comparison theorems, we do not require the Gaussian-column and fixed-class assumptions of [BGC16], the asymptotic simultaneous diagonalization assumed in [MWY23], or the average-isotropy assumptions of [DKL22]. The coefficients τ_Σ and σ_Σ describe how relations among the second-moment matrices enter the bounds. In particular, both coefficients remain bounded when $\max_i |\Sigma_i|_{\mathrm{op}} = O(1)$, even if all n second-moment matrices differ.

For directional tests, Hachem–Loubaton–Najim–Vallet [HLNV13] obtain quantitative bilinear-form estimates in a noncentered independent-entry model with a separable variance profile. In the real, centered version of that model with bounded variance profile and $p \asymp n$, Theorem 1.6 gives the same $n^{-1/2}$ scale for the L^2 error using uniformly bounded fourth entry moments, whereas their Theorem 1.1 assumes eighth moments for this bound. More generally, our theorem uses $M_{\mathrm{HS}}^{(2)}$ directly and does not require coordinate independence.

Unlike the proportional-dimensional setting of [HLNV13], the operator estimate is formulated without a restriction on p/n; whether it gives convergence is specified by its trace and quadratic-form factors. Existing results for $p \gg n$ include the eigenvalue-statistic CLTs of Ding–Wang [DW25], for independent entries and a common diagonal covariance, and the dimension-free ridge-regression equivalents of Cheng–Montanari [CM24] and Misiakiewicz–Saeed [MS24], under i.i.d. sampling and spectral and concentration assumptions. In the common-covariance case at $z < 0$, our fixed-point system reduces to the same effective-regularization equation as in the ridge-regression results. Theorem 1.4 instead allows different matrices Σ_i and measures the error through their traces and quadratic-form moments. Example 1.9 takes $p = n^4$ and centered Gaussian columns with covariance eigenvalues $j^{-3/4}$ and growing traces $4n + O(1)$, and yields

$$\| \operatorname{Tr}(A(G^z - \tilde{G}^z)) \|_{L^q} = O\left(\frac{|A|_{\mathrm{op}}}{\sqrt{n}} \right), \qquad q \geq 2,$$

uniformly over the choices of covariance eigenbases. This is an additive fixed-test comparison, not a CLT or the relative risk approximation, potentially at vanishing regularization, obtained in the cited works.

Finally, our fixed-K estimates do not provide spectral confinement as in [BGC16, ZZXS26], or local laws as in [KY17, AEK17, FMPW26]. For a common covariance in proportional dimensions, Ma–Misiakiewicz [MM26] obtain anisotropic local laws under quadratic-form concentration without an additional cumulant assumption, but with stronger tail and regularity hypotheses than those used in our fixed-moment comparisons. Approaching-axis estimates under Lipschitz concentration are also given in [Cho22].

1.3.2 Selected illustrations

For the normalized-trace estimates below, write $\lambda_1, \dots, \lambda_p$ for the eigenvalues of $\frac{1}{n}XX^\top$ and define the empirical spectral measure and its Stieltjes transform by

$$\mu := \frac{1}{p}\sum_{j=1}^{p} \delta_{\lambda_j}, \qquad g(z) := \frac{1}{p}\operatorname{Tr}(G^z) = \int_{\mathbb{R}_+} \frac{\mu(d\lambda)}{\lambda - z}, \qquad z \in \mathbb{C} \setminus \mathbb{R}_+.$$

Operator-norm route.

Example 1.8 (Balanced strong signals with Gaussian noise)**.** Let $x_i = v_a + y_i$ for $i \in I_a$, where $v_a \in \mathbb{R}^p$ are deterministic and the $y_i \sim \mathcal{N}(0, \Sigma)$ are independent. Assume that the class proportions $\pi_a = |I_a|/n$ satisfy $(\min_a \pi_a)^{-1} = O(k)$. Then

$$\Sigma^{(a)} = \Sigma + v_a v_a^\top, \qquad \tau_\Sigma = O(k).$$

Neither orthogonality of the signals nor commutation of the matrices $\Sigma^{(a)}$ is required.

For real symmetric B,

$$x_i^\top B x_i - \operatorname{Tr}(\Sigma^{(a)} B) = y_i^\top B y_i - \operatorname{Tr}(\Sigma B) + 2 v_a^\top B y_i.$$

Gaussian moment bounds and Lemma B.10 therefore give, for every fixed $q \geq 2$,

$$M_{\mathrm{op}}^{(q)} = O\Big(|\Sigma|_{\mathrm{HS}} + |\Sigma|_{\mathrm{op}}^{1/2} \max_a |v_a| \Big).$$

Under the trace condition $\operatorname{Tr}\Sigma + \max_a |v_a|^2 = O(n)$, Theorem 1.4 yields, for every deterministic $A \in \mathcal{M}_p$,

$$\begin{aligned} &\big\| \operatorname{Tr}\big(A(G^z - \tilde{G}^z)\big) \big\|_{L^q} \\ &\quad = O\Bigg(|A|_{\mathrm{op}} \left[\frac{k}{\sqrt{n}} \left(|\Sigma|_{\mathrm{HS}} + |\Sigma|_{\mathrm{op}}^{1/2} \max_a |v_a| \right) + \frac{k^2 + |\Sigma|_{\mathrm{HS}}^2 + |\Sigma|_{\mathrm{op}} \max_a |v_a|^2}{n} \right] \Bigg). \end{aligned}$$

For unit-variance noise $\Sigma = I_p$, suppose that $k = O(1)$, $\max_a |v_a| = O(\sqrt{p})$, and $p = O(n)$. The trace condition holds and $M_{\mathrm{op}}^{(q)} = O(\sqrt{p})$. Since $p/n = O(1)$, the estimates simplify to

$$\big\| \operatorname{Tr}\big(A(G^z - \tilde{G}^z)\big) \big\|_{L^q} = O\left(|A|_{\mathrm{op}} \sqrt{\frac{p}{n}} \right), \qquad \text{and} \qquad \|g(z) - \tilde{g}(z)\|_{L^q} = O\left(\frac{1}{\sqrt{np}} \right).$$

These estimates allow full-rank noise and second-moment matrices with $|\Sigma^{(a)}|_{\mathrm{op}} = 1 + |v_a|^2$ of order p.

Example 1.9 (Growing trace beyond proportional dimensions)**.** Let $p = n^4$, and let the columns be independent centered Gaussian vectors with covariance matrices

$$\Sigma_i = U_i \operatorname{Diag}\big(j^{-3/4} : 1 \leq j \leq p\big) U_i^\top, \qquad i \in [n],$$

where the U_i are arbitrary deterministic real orthogonal matrices. Then

$$|\Sigma|_{\mathrm{op}}^{\infty} = 1, \qquad \big(|\Sigma|_{\mathrm{HS}}^{\infty}\big)^2 = \sum_{j=1}^{n^4} j^{-3/2} \asymp 1, \qquad |\Sigma|_{*}^{\infty} = \sum_{j=1}^{n^4} j^{-3/4} = 4n + O(1).$$

Thus $\tau_\Sigma \leq 2$ and $1 + |\Sigma|_*^\infty / n = 5 + O(n^{-1})$. Lemma B.10 also gives $M_{\mathrm{op}}^{(q)} = O(1)$ for every fixed $q \geq 2$. Consequently, Theorem 1.4 yields

$$\Big\| \operatorname{Tr}\Big(A(G^z - \tilde{G}^z)\Big) \Big\|_{L^q} = O\left(\frac{|A|_{\mathrm{op}}}{\sqrt{n}} \right), \qquad q \geq 2, \tag{1.11}$$

for every deterministic $A \in \mathcal{M}_p$, uniformly for z in any fixed compact set $K \subset \mathbb{C} \setminus \mathbb{R}_+$. The constants are independent of the choices of U_i; no common eigenbasis or bound on the number of distinct covariance matrices is required. In particular, taking $A = I_p$ gives an $O(n^{-1/2})$ error for the unnormalized trace, although $p/n = n^3$. The Hilbert–Schmidt comparison does not apply, since

$$\frac{\sqrt{p}\,|\Sigma|_{\mathrm{HS}}^{\infty}}{n} \asymp n,$$

so its size condition (1.7) fails.

The population resolvent $(I_p + \frac{1}{n}\sum_{i=1}^n \Sigma_i)^{-1}$ does not provide the same approximation, even in the common-covariance case $U_i = I_p$ for every i. Write $\Sigma = \mathrm{Diag}(j^{-3/4} : 1 \le j \le p)$ for this common covariance. At $z = -1$, write $\tilde{G} := \tilde{G}^z\big|_{z=-1}$. The fixed-point equations in Theorem 1.1 can be written as

$$\tilde{G} = (I_p + \tilde{\Gamma}\Sigma)^{-1}, \qquad 0 < \tilde{\Gamma} \le 1, \qquad \frac{1}{\tilde{\Gamma}} = 1 + \frac{1}{n}\sum_{j=1}^{n^4} \frac{j^{-3/4}}{1 + \tilde{\Gamma} j^{-3/4}}.$$

Since

$$0 \le \sum_{j=1}^{n^4} j^{-3/4} - \sum_{j=1}^{n^4} \frac{j^{-3/4}}{1 + \tilde{\Gamma} j^{-3/4}} = \tilde{\Gamma} \sum_{j=1}^{n^4} \frac{j^{-3/2}}{1 + \tilde{\Gamma} j^{-3/4}} = O(1),$$

we obtain $\tilde{\Gamma}^{-1} = 5 + O(n^{-1})$, and hence $\tilde{\Gamma} = 1/5 + O(n^{-1})$. Moreover, uniformly for $0 \le \Gamma \le 1$,

$$\sum_{j=1}^{n^4} \frac{j^{-3/4}}{(1 + \Gamma j^{-3/4})(1 + j^{-3/4})} = \sum_{j=1}^{n^4} j^{-3/4} + O\left(\sum_{j=1}^{n^4} j^{-3/2}\right) = 4n + O(1).$$

It follows that

$$\mathrm{Tr}\Big(\tilde{G} - (I_p + \Sigma)^{-1}\Big) = (1 - \tilde{\Gamma}) \sum_{j=1}^{n^4} \frac{j^{-3/4}}{(1 + \tilde{\Gamma} j^{-3/4})(1 + j^{-3/4})} = \frac{16}{5} n + O(1).$$

Thus the fixed-point correction is of order n in unnormalized trace, whereas (1.11) controls the random-to-deterministic trace error at order $n^{-1/2}$.

Examples for the commuting-approximation coefficient. The next two examples bound σ_Σ; the profile-size and moment assumptions of Theorem 1.6 must still be checked.

Example 1.10 (A commuting family of low linear dimension)**.** The span-dimension bound in Remark 1.5 does not require a bound on the number of distinct profiles. For example, if

$$\Sigma_i = a_i A + b_i B, \qquad A, B \succeq 0, \quad AB = BA, \quad a_i, b_i \ge 0,$$

then $\sigma_\Sigma \le 3$. This remains true when all the profiles are distinct or their norms grow. More generally, if the matrices Σ_i are positive semidefinite, pairwise commuting, and span an r-dimensional real vector space, then $\sigma_\Sigma \le 1 + r$; no representation by nonnegative coordinates is required.

Example 1.11 (Perturbations of commuting matrices)**.** Let $T^{(1)}, \dots, T^{(k)}$ be pairwise commuting real positive semidefinite matrices, and suppose that

$$\Sigma_i = T^{(a(i))} + E_i \succeq 0, \qquad a(i) \in [k].$$

Choosing $T_i = T^{(a(i))}$ in (1.6) gives

$$\sigma_\Sigma \le \Big(1 + \dim_{\mathbb{R}} \mathrm{span}\{T^{(1)}, \dots, T^{(k)}\}\Big)\left(1 + \frac{1}{n}\sum_{i=1}^n |E_i|_{\mathrm{op}}\right) = O\left(k\left(1 + \frac{1}{n}\sum_{i=1}^n |E_i|_{\mathrm{op}}\right)\right).$$

The matrices E_i may be indefinite and need not commute. No bound on the operator norms $|T^{(a)}|_{\mathrm{op}}$ or balance among the classes is required.

For $k = 1$, write $T = T^{(1)}$, so that $\Sigma_i = T + E_i$. The bound becomes

$$\sigma_\Sigma \le 2\left(1 + \frac{1}{n}\sum_{i=1}^{n} |E_i|_{\mathrm{op}}\right).$$

Choosing $T = \overline{\Sigma} := \frac{1}{n}\sum_j \Sigma_j$ gives the explicit average-deviation estimate

$$\sigma_\Sigma \le 2\left(1 + \frac{1}{n}\sum_{i=1}^{n} |\Sigma_i - \overline{\Sigma}|_{\mathrm{op}}\right).$$

If at most m perturbations E_i are nonzero and $|E_i|_{\mathrm{op}} = O(1)$, the average-error factor in the general bound is $1 + O(m/n)$.

Why second moments alone do not suffice. The following example shows that a common unconditional second moment does not determine the limiting empirical spectral distribution when quadratic forms fail to concentrate. It also shows how conditioning on latent radii recovers the appropriate heterogeneous-profile deterministic equivalent. For a broader treatment of spectral limits for elliptical distributions, see El Karoui [EK09].

Example 1.12 (An isotropic radial mixture)**.** Let $x_i = \sqrt{r_i}\, y_i$, where the y_i are independent and uniform on the sphere of radius $\sqrt{p}$, and the r_i are independent of the directions and of each other, with $\mathbb{P}(r_i = 1/5) = \mathbb{P}(r_i = 9/5) = 1/2$. Since $\mathbb{E}[r_i] = 1$ and $\mathbb{E}[y_i y_i^\top] = I_p$, independence gives $\Sigma_i = \mathbb{E}[x_i x_i^\top] = I_p$. However, $|x_i|^2 = p r_i$ and $|r_i - 1| = 4/5$, so the tests $B = I_p$ and $B = I_p/\sqrt{p}$ give

$$\frac{M_{\mathrm{op}}^{(q)}}{p} \ge \frac{4}{5}, \qquad \frac{M_{\mathrm{HS}}^{(q)}}{\sqrt{p}} \ge \frac{4}{5}, \qquad q \ge 1.$$

Suppose that $p, n \to \infty$ with $p/n \to \gamma \in (0, \infty)$, and condition on the radii $r = (r_1, \dots, r_n)$. The columns remain independent, with conditional second moments $r_i I_p$. For every real symmetric B, the spherical variance bound

$$\mathbb{V}(y_i^\top B y_i) \le 2|B|_{\mathrm{HS}}^2$$

and $r_i \le 9/5$ give a conditional bound $M_{\mathrm{HS}}^{(2)} = O(1)$, uniformly in r, including complex tests by transpose symmetrization and real–imaginary decomposition. Theorem 1.6 therefore yields

$$\|g(z) - \tilde{g}_r(z)\|_{L^2(\cdot\,|r)} = O(n^{-1}),$$

uniformly in r, where $\tilde{g}_r$ is the deterministic equivalent for the conditional profile. Since this profile consists of scalar matrices,

$$\frac{1}{\tilde{g}_r(z)} = -z + \frac{1}{n}\sum_{i=1}^{n} \frac{r_i}{1 + (p/n) r_i \tilde{g}_r(z)}.$$

The proportions of the two radii converge in probability to $1/2$. The uniform support bound gives subsequential weak compactness for the conditional deterministic measures. Passing to the scalar equation and using the sector uniqueness argument of Proposition 2.6 then gives $\tilde{g}_r(z) \to m(z)$ in probability, where m is the unique Stieltjes transform of a probability measure on $\mathbb{R}_+$ satisfying

$$\frac{1}{m(z)} = -z + \frac{1}{2(5 + \gamma m(z))} + \frac{9}{2(5 + 9\gamma m(z))}.$$

The conditional comparison therefore gives weak convergence in probability of the empirical spectral distribution to this law.

Expansion at infinity shows that this law has first moment one and second moment

$$1 + \gamma\,\mathbb{E}[r_i^2] = 1 + \frac{41}{25}\gamma.$$

This differs from the second moment $1+\gamma$ of the Marchenko–Pastur law predicted by the unconditional profile I_p. The discrepancy is illustrated in Figure 1.

1.3.3 Independent entries

For independent entries, the first-moment Hilbert–Schmidt comparison gives qualitative convergence when the squared entries are uniformly integrable, including every fixed finite-variance entry law. Higher moments yield quantitative bounds. For $c \in [0,\infty)$, write $\mu_{\mathrm{MP},c}$ for the Marchenko–Pastur law with parameter c, with $\mu_{\mathrm{MP},0} := \delta_1$.

Corollary 1.13 (Marchenko–Pastur convergence under finite variance)**.** *For every $(p,n) \in \Theta$, suppose that the entries of X are independent copies of a centered unit-variance real random variable $\xi_{p,n}$. Assume that the family $(\xi_{p,n}^2)_{(p,n)\in\Theta}$ is uniformly integrable and that $p = O(n)$. Then, along every sequence in Θ with $p \to \infty$, for every compact $K \subset \mathbb{C} \setminus \mathbb{R}_+$,*

$$\sup_{z\in K} \|g(z) - \tilde{g}(z)\|_{L^1} \longrightarrow 0.$$

If, along the same sequence, $p/n \to c \in [0,\infty)$, the empirical spectral distribution of $\frac{1}{n}XX^\top$ converges weakly in probability to $\mu_{\mathrm{MP},c}$.

Its proof appears immediately after the proof of Theorem 1.6.

Remark 1.14 (Rates under higher moments)**.** Suppose that $p = O(n)$ and that, within each matrix, the entries are centered and i.i.d., with a law that may depend on (p,n). For any fixed $r \in [1,2]$ such that $\|X_{11}\|_{L^{2r}} = O(1)$, for every deterministic $A \in \mathcal{M}_p$,

$$\left\|\mathrm{Tr}\left(A(G^z - \tilde{G}^z)\right)\right\|_{L^1} = O\left(|A|_{\mathrm{HS}} p^{\frac{1}{r}-\frac{1}{2}}\right), \qquad \|g(z) - \tilde{g}(z)\|_{L^1} = O\left(p^{\frac{1}{r}-1}\right).$$

Indeed, since $\mathbb{E}[X_{11}^2] = O(1)$ and $\Sigma_i = \mathbb{E}[X_{11}^2] I_p$, $|\Sigma|_{\mathrm{HS}}^\infty = O(\sqrt{p})$. The common scalar profile satisfies $\sigma_\Sigma \le 2$, and Proposition B.6 gives $M_{\mathrm{HS}}^{(1)} = O(p^{\frac{1}{r}-\frac{1}{2}})$. Since $|\Sigma|_{\mathrm{HS}}^\infty/n = O(\sqrt{p}/n) = O(1)$, the first-moment estimate in Theorem 1.6 gives the first bound, and $|I_p/p|_{\mathrm{HS}} = \frac{1}{\sqrt{p}}$ gives the second.

At $r = 1$, the displayed bound does not imply convergence. Uniform integrability, as assumed in Corollary 1.13, provides a sufficient additional condition.

If $\|X_{11}\|_{L^4} = O(1)$, the variance calculation in Lemma B.9 gives directly

$$M_{\mathrm{HS}}^{(2)} = O(\|X_{11}\|_{L^4}^2) = O(1).$$

The same comparison theorem with $q = 2$ then yields

$$\left\|\mathrm{Tr}\left(A(G^z - \tilde{G}^z)\right)\right\|_{L^1} \le \left\|\mathrm{Tr}\left(A(G^z - \tilde{G}^z)\right)\right\|_{L^2} = O\left(|A|_{\mathrm{HS}}\left[\frac{1}{\sqrt{n}} + \frac{\sqrt{p}}{n}\right]\right) = O\left(\frac{|A|_{\mathrm{HS}}}{\sqrt{n}}\right),$$

and consequently

$$\|g(z) - \tilde{g}(z)\|_{L^1} \le \|g(z) - \tilde{g}(z)\|_{L^2} = O\left(\frac{1}{\sqrt{np}}\right).$$

2 Deterministic equivalents for sectorial matrix parameters

We prove the deterministic results for general complex matrix parameters. Throughout the proofs, we write $\Omega := \mathbb{C} \setminus \mathbb{R}_+$, recalling that $\mathbb{R}_+ = [0, \infty)$. For $Z \in \mathcal{M}_p$, let $\mathcal{W}(Z) := \{u^* Z u : u \in \mathbb{C}^p,\ |u| = 1\}$ be its numerical range, and set

$$\eta_Z := d\big(\mathcal{W}(Z), \mathbb{R}_+\big), \qquad \eta_z := \eta_{zI_p} = d(z, \mathbb{R}_+), \qquad z \in \Omega.$$

We call Z admissible when $\eta_Z > 0$, and associate with it the closed convex sector

$$\mathcal{C}_Z := \operatorname{cone}\big(\mathbb{R}_+ \cup -\mathcal{W}(Z)\big). \tag{2.1}$$

The sector $\mathcal{C}_Z$ specifies the admissible values of the fixed-point coefficients. It allows a common treatment of nonreal scalar parameters and negative real parameters. General matrix parameters are needed in the classwise replacement argument of the operator norm route, and in the additive deformations used to obtain resolvent sandwiches; see Subsections 3.3 and 3.4. The supporting phase introduced below gives the real-part bounds used to control the resolvents and the fixed-point denominators.

2.1 Matrix sectors and two-sided resolvent calculus

For an admissible matrix Z, $\mathcal{C}_Z$ is convex and contains $\mathbb{R}_+$. Its closedness and aperture bound are proved below. Figure 2 illustrates the geometry in the complex plane. For an

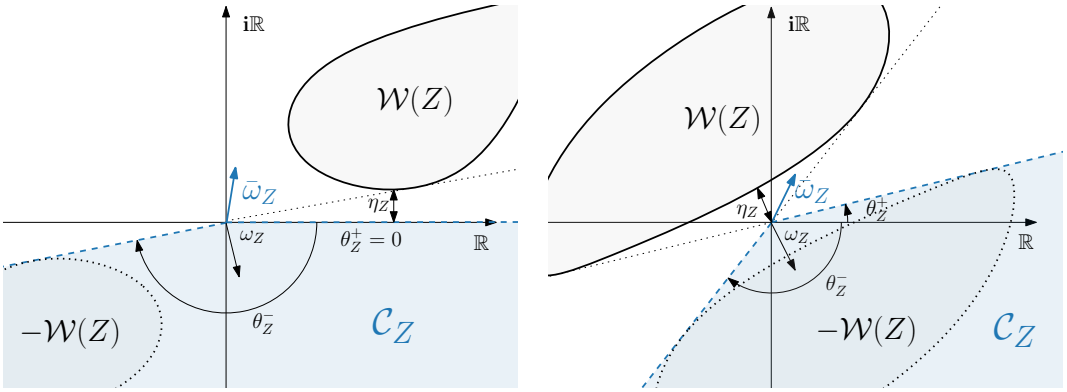


Figure 2: Geometry of the matrix sector $\mathcal{C}_Z$ for two numerical-range configurations.

admissible parameter Z, define

$$\theta_Z^+ := \sup\left\{\theta \in (-\pi, \pi) : e^{\mathbf{i}\theta} \in \mathcal{C}_Z\right\}, \qquad \theta_Z^- := \inf\left\{\theta \in (-\pi, \pi) : e^{\mathbf{i}\theta} \in \mathcal{C}_Z\right\}.$$

Lemma 2.1 (Aperture of an admissible matrix sector). *For every admissible matrix Z, the cone $\mathcal{C}_Z$ is closed, and the principal arguments of its nonzero elements form the closed interval $[\theta_Z^-, \theta_Z^+]$, with $\theta_Z^- \le 0 \le \theta_Z^+$ and*

$$0 \le \theta_Z^+ - \theta_Z^- \le 2 \arccos\left(\frac{\eta_Z}{2|Z|_{\mathrm{op}}}\right) < \pi. \tag{2.2}$$

Proof. The set $B := \operatorname{conv}(\{1\} \cup -\mathcal{W}(Z))$ is compact and convex and does not contain zero. Indeed, an identity $0 = t - (1-t)w$, with $t \in [0,1]$ and $w \in \mathcal{W}(Z)$, would force $t < 1$ and $w = t/(1-t) \in \mathbb{R}_+$, contrary to admissibility. Choose $q \in B$ of minimum modulus. Every element of $\mathcal{C}_Z$ is a nonnegative multiple of a point of B. If $r_m v_m$ converges, with $r_m \ge 0$ and $v_m \in B$, then $r_m \le |r_m v_m|/|q|$ is bounded. Compactness of B allows us to pass to a subsequence with $r_m \to r \ge 0$ and $v_m \to v \in B$, so the limit rv belongs to $\mathcal{C}_Z$. This proves closedness.

For $v \in B$ and $0 < t \le 1$, convexity and minimality give

$$0 \le \frac{|q + t(v-q)|^2 - |q|^2}{t} = 2\Re\big(\overline{q}(v-q)\big) + t|v-q|^2.$$

Letting $t \downarrow 0$ yields $\Re(\overline{q}v) \ge |q|^2 > 0$. With $M := \max_{v\in B} |v|$, it follows that

$$\Re\left(\frac{\overline{q}}{|q|}v\right) \ge \frac{|q|}{M}|v|, \qquad v \in B.$$

By addition and homogeneity, the same inequality holds on $\mathcal{C}_Z$. Thus this cone has aperture strictly smaller than π. Since it contains 1, it excludes the negative real axis; its principal arguments therefore form a closed interval $[\theta_Z^-, \theta_Z^+]$ containing zero.

Put $\delta := \theta_Z^+ - \theta_Z^-$. Compactness of $\mathcal{W}(Z)$ and $|w| \ge \eta_Z$ for $w \in \mathcal{W}(Z)$ ensure that each boundary ray is either $\mathbb{R}_+$ or contains a point of $-\mathcal{W}(Z)$. If both boundary rays contain points $-w_\pm = r_\pm e^{\mathbf{i}\theta_Z^\pm}$, then convexity gives

$$w_0 := \frac{r_- w_+ + r_+ w_-}{r_+ + r_-} \in \mathcal{W}(Z), \qquad \eta_Z \le |w_0| = \frac{2r_+r_-}{r_+ + r_-}\cos(\delta/2) \le |Z|_{\mathrm{op}}\cos(\delta/2).$$

If one boundary ray is $\mathbb{R}_+$, the other contains a point $-w = re^{\pm\mathbf{i}\delta}$ for $w \in \mathcal{W}(Z)$. Taking the point $r \in \mathbb{R}_+$ as a competitor in the definition of η_Z, we obtain

$$\eta_Z \le |w - r| = \left|-re^{\pm\mathbf{i}\delta} - r\right| = r\left|1 + e^{\pm\mathbf{i}\delta}\right| = 2r\cos(\delta/2) \le 2|Z|_{\mathrm{op}}\cos(\delta/2).$$

For $\delta = 0$, the same conclusion follows from $\eta_Z \le |Z|_{\mathrm{op}}$. These bounds prove (2.2). □

Define the bisecting phase and its angular constant by

$$\theta_Z := \frac{\theta_Z^+ + \theta_Z^-}{2}, \qquad \omega_Z := e^{\mathbf{i}\theta_Z}, \qquad c_Z := \cos\left(\frac{\theta_Z^+ - \theta_Z^-}{2}\right) > 0. \tag{2.3}$$

Lemma 2.2 (Supporting phase for an admissible parameter). *For every admissible matrix Z, the phase and constant in* (2.3) *satisfy*

$$\Re(\overline{\omega_Z}u) \ge c_Z|u|, \qquad u \in \mathcal{C}_Z, \tag{2.4}$$

and

$$\Re(\overline{\omega_Z}(-Z)) \succeq c_Z\eta_Z I_p \succ 0. \tag{2.5}$$

Proof. For $u \ne 0$ in $\mathcal{C}_Z$, write $u = |u|e^{\mathbf{i}\theta}$. Lemma 2.1 gives

$$|\theta - \theta_Z| \le \frac{\theta_Z^+ - \theta_Z^-}{2},$$

so $\Re(\overline{\omega_Z}u) = |u|\cos(\theta - \theta_Z) \ge c_Z|u|$. The assertion also holds for $u = 0$. For every unit vector v, one has $-v^*Zv \in \mathcal{C}_Z$ and $|v^*Zv| \ge \eta_Z$. Hence

$$v^*\Re(\overline{\omega_Z}(-Z))v = \Re\big(\overline{\omega_Z}(-v^*Zv)\big) \ge c_Z|v^*Zv| \ge c_Z\eta_Z,$$

which proves (2.5). □

We write

$$\overline{\mathcal{C}_Z} := \{\bar{w} : w \in \mathcal{C}_Z\}$$

for the conjugate cone. The next lemma applies to any supporting phase satisfying its hypotheses.

Lemma 2.3 (Sectorial resolvent calculus). *Let $Z \in \mathcal{M}_p$, let $\mathcal{C}_Z$ be given by* (2.1)*, and suppose that a unit phase ω and constants $c, \kappa > 0$ satisfy*

$$\forall v \in \mathcal{C}_Z : \quad \Re(\overline{\omega} v) \geq c|v| \qquad \text{and} \qquad \Re(\overline{\omega}(-Z)) \succeq \kappa I_p. \tag{2.6}$$

Given $D \in \mathcal{D}_n(\mathcal{C}_Z)$ and positive semidefinite matrices $S_1, \ldots, S_n \in \mathcal{M}_p$, set

$$S := \frac{1}{n} \sum_{j=1}^{n} D_j S_j \qquad \text{and} \qquad H = (S - Z)^{-1}.$$

Then H is well defined and satisfies

$$|H|_{\mathrm{op}} \leq \frac{1}{\kappa}.$$

Moreover, $\Re(\omega H) = H^* \Re(\overline{\omega}(S - Z))\, H = H \Re(\overline{\omega}(S - Z))\, H^*$, and the following Ward inequalities *hold:*

$$H^* H, \; H H^* \preceq \frac{1}{\kappa} \Re(\omega H).$$

For $x \in \mathbb{C}^p$ and $\Sigma \succeq 0$,

$$x^* H x \in \overline{\mathcal{C}_Z}, \qquad \mathrm{Tr}(\Sigma H) \in \overline{\mathcal{C}_Z}.$$

Finally, for every $w \in \overline{\mathcal{C}_Z}$,

$$|1 + w| \geq c(1 + |w|), \qquad (1 + w)^{-1} \in \mathcal{C}_Z.$$

Proof. For every unit vector u, the definition of $\mathcal{C}_Z$ gives $-u^* Z u \in \mathcal{C}_Z$. Since $u^* S_j u \geq 0$ and $D_j \in \mathcal{C}_Z$, one also has

$$u^* S u = \frac{1}{n} \sum_j D_j (u^* S_j u) \in \mathcal{C}_Z, \qquad u^*(S - Z)u \in \mathcal{C}_Z.$$

These inclusions extend to every u by homogeneity. Since $1 \in \mathcal{C}_Z$, the supporting phase satisfies $\Re(\omega) \geq c$. In particular, $\Re(\overline{\omega}(s - w)) \geq cs + \kappa$ for $s \geq 0$ and $w \in \mathcal{W}(Z)$, so these hypotheses imply $\eta_Z \geq \kappa > 0$. The supporting-phase and coercivity bounds give

$$\Re(\overline{\omega}(S - Z)) \succeq \frac{c}{n} \sum_j |D_j| S_j + \kappa I_p \succeq \kappa I_p.$$

Thus $|(S - Z)u| \geq \kappa |u|$, proving invertibility and the inverse norm bound. Multiplying the identity $H(S - Z) = (S - Z)H = I_p$ and its adjoint in the two orders gives the displayed inverse identities. Coercivity then yields both Ward inequalities.

For the sector inclusions, set $y = Hx$. Then

$$x^* H x = ((S - Z)y)^* y = \overline{y^*(S - Z)y} \in \overline{\mathcal{C}_Z}.$$

Diagonalizing $\Sigma \succeq 0$ and summing its eigenvalue-weighted quadratic forms proves the trace assertion.

Since $1 \in \mathcal{C}_Z$, the supporting phase gives $\Re(\omega) \geq c$. For $w \in \overline{\mathcal{C}_Z}$, it also gives $\Re(\omega w) \geq c|w|$, so

$$|1 + w| \geq \Re(\omega(1 + w)) \geq c(1 + |w|).$$

Moreover,

$$(1 + w)^{-1} = \frac{\overline{1 + w}}{|1 + w|^2} \in \mathcal{C}_Z.$$

□

Expectations of integrable random variables taking values in either sector, and line segments joining points in either sector, remain in that sector. All assertions of the lemma remain valid when $\mathcal{C}_Z$ is replaced throughout by a larger closed convex sector containing $\mathcal{C}_Z$ and satisfying the same supporting-phase bound. The expectation and line-segment statements follow from closedness and convexity. The extension to larger sectors follows by the same proof, retaining the coercivity bound in (2.6) and allowing the coefficients D_j in the larger sector.

For every admissible Z, Lemma 2.2 lets us apply the lemma with $\omega = \omega_Z$, $c = c_Z$, and $\kappa = c_Z\eta_Z$.

2.2 The matrix-parameter fixed point

Let $\Sigma_1, \dots, \Sigma_n \in \mathcal{M}_p(\mathbb{R})$ be symmetric positive-semidefinite matrices. For $D \in \mathcal{D}_n(\mathcal{C}_Z)$, define

$$G_D^Z := \left(\frac{1}{n}\sum_{j=1}^{n} D_j\Sigma_j - Z\right)^{-1}, \qquad \Phi^Z(D)_i := \left(1 + \frac{1}{n}\operatorname{Tr}(\Sigma_i G_D^Z)\right)^{-1}, \qquad i \in [n]. \quad (2.7)$$

By Lemma 2.3,

$$\frac{1}{n}\operatorname{Tr}(\Sigma_i G_D^Z) \in \overline{\mathcal{C}_Z}.$$

The reciprocal bounds in the same lemma give

$$\Phi^Z(D)_i \in \mathcal{C}_Z, \qquad |\Phi^Z(D)_i| \le c_Z^{-1}.$$

Thus Φ^Z maps the nonempty compact convex set

$$\left\{D \in \mathcal{D}_n(\mathcal{C}_Z) : |D|_\infty \le c_Z^{-1}\right\}$$

continuously into itself. Brouwer's fixed-point theorem therefore gives the following result.

Lemma 2.4 (Sector invariance and existence). *For every admissible Z, the mapping Φ^Z has a fixed point $\widetilde{\Delta}^Z \in \mathcal{D}_n(\mathcal{C}_Z)$, and*

$$|\widetilde{\Delta}^Z|_\infty \le c_Z^{-1}.$$

For a fixed point supplied by the lemma, define

$$\widetilde{G}^Z := G^Z_{\widetilde{\Delta}^Z}.$$

This matrix satisfies

$$\widetilde{G}^Z = \left(-Z + \frac{1}{n}\sum_{i=1}^{n} \frac{\Sigma_i}{1 + \frac{1}{n}\operatorname{Tr}(\Sigma_i \widetilde{G}^Z)}\right)^{-1}. \quad (2.8)$$

We now prove uniqueness using an estimate that will also be used for stability. With the phase ω_Z defined in (2.3), write, for a scalar or square matrix A,

$$\mathfrak{C}_Z(A) := \Re(\overline{\omega_Z} A), \qquad \mathfrak{C}_{\bar{Z}}(A) = \Re(\omega_Z A). \quad (2.9)$$

Here $\Re$ denotes the real part for scalars and the Hermitian part for matrices. The second identity follows from $\omega_{\bar{Z}} = \overline{\omega_Z}$, since complex conjugation reflects the sector and its bisecting phase. In particular, $\mathfrak{C}_Z(1) = \mathfrak{C}_{\bar{Z}}(1) \ge c_Z > 0$. For Hermitian Σ, taking the trace commutes with this operation:

$$\operatorname{Tr}(\Sigma \mathfrak{C}_{\bar{Z}}(A)) = \mathfrak{C}_{\bar{Z}}(\operatorname{Tr}(\Sigma A))\,.$$

For profiles D, D' for which the two resolvents are defined, set

$$\Psi^Z(D, D')_{ij} := \frac{D_i D'_i}{n^2}\operatorname{Tr}\big(\Sigma_i G_D^Z \Sigma_j G_{D'}^Z\big)\,.$$

For $D, D' \in \mathcal{D}_n(\mathcal{C}_Z \setminus \{0\})$, define the inverse defect

$$r_i(D) := \frac{1}{D_i} - 1 - \frac{1}{n}\operatorname{Tr}(\Sigma_i G_D^Z). \tag{2.10}$$

Thus $r(D) = 0$ precisely when D is a fixed point of Φ^Z. The resolvent identity gives

$$G_D^Z - G_{D'}^Z = -\frac{1}{n} G_D^Z \left(\sum_{j=1}^n (D_j - D'_j)\Sigma_j \right) G_{D'}^Z. \tag{2.11}$$

Subtracting (2.10) evaluated at D and D', using $D_i^{-1} - (D'_i)^{-1} = -(D_i - D'_i)/(D_i D'_i)$, and substituting the resolvent identity (2.11) gives, for each $i \in [n]$,

$$r_i(D) - r_i(D') = -\frac{D_i - D'_i}{D_i D'_i} + \frac{1}{n^2} \sum_{j=1}^n (D_j - D'_j) \operatorname{Tr}\big(\Sigma_i G_D^Z \Sigma_j G_{D'}^Z\big).$$

Multiplying the i-th identity by $-D_i D'_i$ and using the definition of $\Psi^Z(D, D')$ yields the exact secant equation

$$\big(I_n - \Psi^Z(D, D')\big)(D - D') = -DD'\big(r(D) - r(D')\big). \tag{2.12}$$

Here $D - D'$ and $r(D) - r(D')$ are viewed as vectors, and the diagonal matrix DD' acts coordinatewise. For two fixed points, this reduces to $D - D' = \Psi^Z(D, D')(D - D')$; a strict contraction of $\Psi^Z(D, D')$ therefore proves uniqueness. Taking $D' = \widetilde{\Delta}^Z$ instead leaves only $r(D)$ on the right, giving (3.17) below. Bounds on this defect then control $D - \widetilde{\Delta}^Z$, and (2.11) transfers them to $G_D^Z - \widetilde{G}^Z$.

For such profiles, equip $\mathbb{C}^n$ with

$$|u|_{D,D'} := \max_{i \in [n]} \frac{|u_i|}{\sqrt{\mathfrak{C}_Z(D_i)\mathfrak{C}_Z(D'_i)}}.$$

For a linear map $T : \mathbb{C}^n \to \mathbb{C}^n$, denote its induced operator norm by

$$|T|_{D,D'} := \sup_{u \neq 0} \frac{|Tu|_{D,D'}}{|u|_{D,D'}}.$$

Lemma 2.5 (Weighted secant estimate). *For $D, D' \in \mathcal{D}_n(\mathcal{C}_Z \setminus \{0\})$ and every $u \in \mathbb{C}^n$,*

$$|\Psi^Z(D, D')u|_{D,D'} \;\le\; \max_{i \in [n]} \sqrt{\frac{|D_i|^2 \frac{1}{n}\mathfrak{C}_{\bar{Z}}\big(\operatorname{Tr}(\Sigma_i G_D^Z)\big)}{\mathfrak{C}_Z(D_i)} \frac{|D'_i|^2 \frac{1}{n}\mathfrak{C}_{\bar{Z}}\big(\operatorname{Tr}(\Sigma_i G_{D'}^Z)\big)}{\mathfrak{C}_Z(D'_i)}}\, |u|_{D,D'}. \tag{2.13}$$

The quantities $\frac{1}{n}\mathfrak{C}_{\bar{Z}}\big(\operatorname{Tr}(\Sigma_i G_D^Z)\big)$ are nonnegative.

Proof. For each i, j, the Hilbert–Schmidt Cauchy–Schwarz inequality gives

$$\begin{aligned} |\Psi^Z(D, D')_{ij}| &= \frac{|D_i D'_i|}{n^2} \left|\operatorname{Tr}(\Sigma_i G_D^Z \Sigma_j G_{D'}^Z)\right| \\ &\le \frac{|D_i D'_i|}{n^2} \operatorname{Tr}\big(\Sigma_i G_D^Z \Sigma_j (G_D^Z)^*\big)^{1/2} \operatorname{Tr}\big(\Sigma_i (G_{D'}^Z)^* \Sigma_j G_{D'}^Z\big)^{1/2}. \end{aligned}$$

Since $-\mathcal{W}(Z) \subset \mathcal{C}_Z$, Lemma 2.2 gives $\mathfrak{C}_Z(-Z) \succeq 0$. Then the two inverse identities in Lemma 2.3 give

$$\frac{1}{n} \sum_j \mathfrak{C}_Z(D_j) G_D^Z \Sigma_j (G_D^Z)^* = \mathfrak{C}_{\bar{Z}}(G_D^Z) - G_D^Z \mathfrak{C}_Z(-Z)(G_D^Z)^* \preceq \mathfrak{C}_{\bar{Z}}(G_D^Z).$$

Interchanging the two outer resolvent factors on the left gives the same upper bound $\mathfrak{C}_{\bar{Z}}(G_D^Z)$. Multiplying each absolute kernel entry by $\sqrt{\mathfrak{C}_Z(D_j)\mathfrak{C}_Z(D'_j)}$ and applying Cauchy–Schwarz in j therefore gives

$$\sum_{j=1}^{n} |\Psi^Z(D,D')_{ij}|\sqrt{\mathfrak{C}_Z(D_j)\mathfrak{C}_Z(D'_j)} \le \frac{|D_iD'_i|}{n}\sqrt{\operatorname{Tr}(\Sigma_i\mathfrak{C}_{\bar{Z}}(G_D^Z))\operatorname{Tr}(\Sigma_i\mathfrak{C}_{\bar{Z}}(G_{D'}^Z))}.$$

Bounding $|(\Psi^Z(D,D')u)_i|$ by this row sum times $|u|_{D,D'}$ and dividing by $\sqrt{\mathfrak{C}_Z(D_i)\mathfrak{C}_Z(D'_i)}$ proves (2.13). □

The same estimate holds if the canonical phase is replaced by any unit phase satisfying the hypotheses of Lemma 2.3, provided both rotated real-part maps and the weights in $|\cdot|_{D,D'}$ are formed with that phase.

Proposition 2.6 (Fixed-point secant contraction and uniqueness). *If $D, D' \in \mathcal{D}_n(\mathcal{C}_Z)$ are fixed points of Φ^Z, then*

$$|\Psi^Z(D,D')|_{D,D'} \le \max_{i\in[n]} \sqrt{\frac{\frac{1}{n}\mathfrak{C}_{\bar{Z}}\big(\operatorname{Tr}(\Sigma_i G_D^Z)\big)}{\mathfrak{C}_Z(1)+\frac{1}{n}\mathfrak{C}_{\bar{Z}}\big(\operatorname{Tr}(\Sigma_i G_D^Z)\big)}\,\frac{\frac{1}{n}\mathfrak{C}_{\bar{Z}}\big(\operatorname{Tr}(\Sigma_i G_{D'}^Z)\big)}{\mathfrak{C}_Z(1)+\frac{1}{n}\mathfrak{C}_{\bar{Z}}\big(\operatorname{Tr}(\Sigma_i G_{D'}^Z)\big)}} < 1. \quad (2.14)$$

Consequently, the fixed point is unique in $\mathcal{D}_n(\mathcal{C}_Z)$, and $I_n - \Psi^Z(D,D)$ is invertible.

Proof. By (2.7), each component of a fixed point is the inverse of a finite nonzero scalar. Consequently, $\mathfrak{C}_Z(D_i) \ge c_Z|D_i| > 0$, and similarly for D', so the weighted norms are well defined. At a fixed point,

$$\frac{1}{D_i} = 1 + \frac{1}{n}\operatorname{Tr}(\Sigma_i G_D^Z).$$

Applying $\mathfrak{C}_{\bar{Z}}$ gives the exact identity

$$\frac{\mathfrak{C}_Z(D_i)}{|D_i|^2} = \mathfrak{C}_Z(1) + \frac{1}{n}\mathfrak{C}_{\bar{Z}}\big(\operatorname{Tr}(\Sigma_i G_D^Z)\big).$$

Substitution in Lemma 2.5 proves (2.14); it is strict because $\mathfrak{C}_Z(1) \ge c_Z > 0$.

If $U := D - D'$, equation (2.12) with $r(D) = r(D') = 0$ gives

$$U = \Psi^Z(D,D')U.$$

The strict contraction gives $U = 0$, hence $D = D'$. Taking $D' = D$, the same norm bound gives the invertibility of $I_n - \Psi^Z(D,D)$ by the Neumann series. □

Proposition 2.7 (Sectorial matrix-parameter deterministic equivalent). *For every admissible matrix Z, there is a unique $\widetilde{\Delta}^Z \in \mathcal{D}_n(\mathcal{C}_Z)$ satisfying*

$$\widetilde{\Delta}^Z = \Phi^Z(\widetilde{\Delta}^Z).$$

The associated matrix $\widetilde{G}^Z = G^Z_{\widetilde{\Delta}^Z}$ satisfies (2.8), with

$$|\widetilde{\Delta}^Z|_\infty \le \frac{1}{c_Z}, \qquad |\widetilde{G}^Z|_{\mathrm{op}} \le \frac{1}{c_Z\eta_Z}.$$

Proof. Existence follows from Lemma 2.4, and uniqueness from Proposition 2.6. The bounds follow from the existence lemma and Lemma 2.3. □

Lemma 2.8 (Trace identity for the deterministic equivalent). *At the matrix-parameter fixed point,*

$$\operatorname{Tr}(Z\widetilde{G}^Z) = n - p - \sum_{i=1}^{n} \widetilde{\Delta}_i^Z.$$

Proof. The identity

$$\left(-Z + \frac{1}{n}\sum_i \widetilde{\Delta}_i^Z \Sigma_i\right) \widetilde{G}^Z = I_p$$

and the fixed-point equations give

$$p = -\operatorname{Tr}(Z\widetilde{G}^Z) + \frac{1}{n}\sum_i \widetilde{\Delta}_i^Z \operatorname{Tr}(\Sigma_i \widetilde{G}^Z) = -\operatorname{Tr}(Z\widetilde{G}^Z) + \sum_i (1 - \widetilde{\Delta}_i^Z).$$

This proves the identity. □

2.3 Holomorphic dependence

Proposition 2.9 (Holomorphic dependence on the matrix parameter). *Let $U \subset \mathbb{C}$ be open, and let $t \mapsto Z_t \in \mathcal{M}_p$ have holomorphic entries. Assume that*

$$d\big(\mathcal{W}(Z_t), \mathbb{R}_+\big) > 0, \qquad t \in U.$$

Then the mappings

$$t \longmapsto \widetilde{\Delta}^{Z_t}, \qquad t \longmapsto \widetilde{G}^{Z_t}$$

are holomorphic on U.

Proof. Fix $t_0 \in U$, and choose $r > 0$ so that the closed disk $\overline{B}(t_0, r)$ is contained in U. For s, t in this disk,

$$|\eta_{Z_s} - \eta_{Z_t}| \le |Z_s - Z_t|_{\mathrm{op}},$$

because $|v^*(Z_s - Z_t)v| \le |Z_s - Z_t|_{\mathrm{op}}$ for every unit vector v. Consequently, continuity and compactness give

$$\eta_* := \inf_{t \in \overline{B}(t_0, r)} \eta_{Z_t} > 0.$$

Lemma 2.1 also gives

$$c_* := \inf_{t \in \overline{B}(t_0, r)} c_{Z_t} \ge \frac{\eta_*}{2 \sup_{t \in \overline{B}(t_0, r)} |Z_t|_{\mathrm{op}}} > 0.$$

To identify the solution given by the implicit-function theorem, we first prove continuity at t_0. Let $t_m \to t_0$ in this disk, and write $Z_m := Z_{t_m}$ and $Z_0 := Z_{t_0}$.

We need closedness when the matrix parameter varies. Let $u_m \in \mathcal{C}_{Z_m}$ satisfy $u_m \to u$. By the definition of $\mathcal{C}_{Z_m}$ and convexity of $\mathcal{W}(Z_m)$, choose $a_m, b_m \ge 0$ and $w_m \in \mathcal{W}(Z_m)$ such that

$$u_m = a_m - b_m w_m.$$

The uniform supporting and separation bounds give $b_m \le |u_m|/(c_*\eta_*) = O(1)$. Since $d(w_m, \mathcal{W}(Z_0)) \le |Z_m - Z_0|_{\mathrm{op}}$, we have

$$d(u_m, \mathcal{C}_{Z_0}) \le b_m |Z_m - Z_0|_{\mathrm{op}} \longrightarrow 0,$$

and closedness of $\mathcal{C}_{Z_0}$ yields $u \in \mathcal{C}_{Z_0}$.

The bound $|\widetilde{\Delta}^{Z_m}|_\infty \le c_*^{-1}$, supplied by Lemma 2.4, makes $(\widetilde{\Delta}^{Z_m})_m$ bounded. Every convergent subsequence therefore has a limit in $\mathcal{D}_n(\mathcal{C}_{Z_0})$, which satisfies the fixed-point equation by the resolvent bounds and continuity of matrix inversion. Uniqueness identifies this limit with $\widetilde{\Delta}^{Z_0}$. Consequently,

$$\widetilde{\Delta}^{Z_m} \longrightarrow \widetilde{\Delta}^{Z_0}.$$

Now $\Phi^{Z_t}(D)$ denotes the rational expression in (2.7), defined on the open subset of $U \times \mathcal{D}_n(\mathbb{C})$ where the matrix defining $G_D^{Z_t}$ is invertible and all scalar denominators are nonzero. This open set contains $(t_0, \widetilde{\Delta}^{Z_{t_0}})$. The mapping

$$F(t, D) := D - \Phi^{Z_t}(D)$$

is therefore holomorphic in a neighborhood of this point. Its derivative with respect to D is

$$\partial_D F(t_0, \widetilde{\Delta}^{Z_{t_0}}) = I_n - \Psi^{Z_{t_0}}\left(\widetilde{\Delta}^{Z_{t_0}}, \widetilde{\Delta}^{Z_{t_0}}\right),$$

which is invertible by Proposition 2.6. The holomorphic implicit-function theorem gives a holomorphic solution near t_0, unique among solutions near $\widetilde{\Delta}^{Z_{t_0}}$. The continuity just proved ensures that this solution equals $\widetilde{\Delta}^{Z_t}$.

Since t_0 is arbitrary, $t \mapsto \widetilde{\Delta}^{Z_t}$ is holomorphic on U. Holomorphy of $t \mapsto \widetilde{G}^{Z_t}$ follows from (2.7). □

2.4 Scalar spectral parameters and Stieltjes transforms

The scalar quantities of Theorem 1.1 are related to the matrix fixed point by

$$\tilde{\Gamma}_i^z = -\frac{1}{z}\widetilde{\Delta}_i^{zI_p}, \qquad \tilde{G}^z = \widetilde{G}^{zI_p}, \qquad z \in \Omega.$$

Write $\tilde{\Gamma}^z := \mathrm{Diag}(\tilde{\Gamma}_1^z, \dots, \tilde{\Gamma}_n^z)$ and set

$$\mathcal{C}_z := \mathcal{C}_{zI_p} = \mathrm{cone}\big(\mathbb{R}_+ \cup \{-z\}\big).$$

Since $-z^{-1}\mathcal{C}_z = \overline{\mathcal{C}_z}$, the following is a direct corollary of Propositions 2.7 and 2.9.

Corollary 2.10 (Scalar fixed point and holomorphy). *For every $z \in \Omega$, the matrix $\tilde{\Gamma}^z$ is the unique $D \in \mathcal{D}_n(\overline{\mathcal{C}_z})$ satisfying*

$$D_i = -\frac{1}{z\left(1 + \frac{1}{n}\mathrm{Tr}(\Sigma_i G_{-zD}^{zI_p})\right)}, \qquad i \in [n].$$

Moreover, the mappings

$$z \longmapsto \tilde{\Gamma}^z, \qquad z \longmapsto \tilde{G}^z$$

are holomorphic on Ω.

Proof of Theorem 1.1. Corollary 2.10 gives existence, uniqueness, and holomorphy, since $\overline{\mathcal{C}_z} = \{a - b/z : a, b \geq 0\}$. Substituting $\widetilde{\Delta}_i^{zI_p} = -z\tilde{\Gamma}_i^z$ into the definition $\tilde{G}^z = G^{zI_p}_{\widetilde{\Delta}^{zI_p}}$ gives the stated formula for $\tilde{G}^z$. □

Lemma 2.8, applied with $Z = zI_p$, gives

$$z\,\mathrm{Tr}(\tilde{G}^z) = n - p + z\sum_{i=1}^n \tilde{\Gamma}_i^z,$$

and hence

$$\tilde{g}(z) = \frac{1}{z}\left(\frac{n}{p} - 1\right) + \frac{1}{p}\sum_{i=1}^n \tilde{\Gamma}_i^z. \tag{2.15}$$

Uniqueness gives the following conjugation identities; the last equality also uses $(\tilde{G}^z)^\top = \tilde{G}^z$.

$$\tilde{\Gamma}_i^{\bar{z}} = \overline{\tilde{\Gamma}_i^z}, \qquad \tilde{G}^{\bar{z}} = \overline{\tilde{G}^z} = (\tilde{G}^z)^*. \tag{2.16}$$

We next prove that each mapping $z \mapsto \tilde{\Gamma}_i^z$ is a Stieltjes transform. We use the characterization and inversion formula in Theorem B.11 of Appendix B.

Proposition 2.11. *For every $i \in [n]$, there is a probability measure $\tilde{\mu}_i$ supported on $\mathbb{R}_+$ such that*

$$\tilde{\Gamma}_i^z = \int_{\mathbb{R}_+} \frac{\tilde{\mu}_i(dt)}{t - z}, \qquad z \in \Omega.$$

Proof. By Corollary 2.10, the mappings $z \mapsto \tilde{\Gamma}_i^z$ are holomorphic on Ω. For $\Im z > 0$, their values lie in $\overline{\mathcal{C}_z}$, so $\Im(\tilde{\Gamma}_i^z) \geq 0$ and $\Im(z\tilde{\Gamma}_i^z) \geq 0$. The inverse identity gives

$$\Im(z\tilde{G}^z) = \frac{|z|^2}{n} \sum_{j=1}^{n} \Im(\tilde{\Gamma}_j^z) \tilde{G}^z \Sigma_j \tilde{G}^{\bar{z}} \succeq 0. \tag{2.17}$$

The fixed-point equation therefore yields

$$\Im\left(-\frac{1}{\tilde{\Gamma}_i^z}\right) = \Im z + \frac{1}{n} \operatorname{Tr}\Big(\Sigma_i \Im(z\tilde{G}^z)\Big) > 0,$$

hence $\Im(\tilde{\Gamma}_i^z) > 0$. Since $\Im(-1/\tilde{\Gamma}_i^z) \leq |\tilde{\Gamma}_i^z|^{-1}$, the same identity gives $|\tilde{\Gamma}_i^z| \leq (\Im z)^{-1}$, and in particular

$$|\tilde{\Gamma}_i^{\mathbf{i}y}| \leq y^{-1}.$$

For fixed p, n, letting $y \to \infty$ gives

$$\frac{1}{n} \sum_j \tilde{\Gamma}_j^{\mathbf{i}y} \Sigma_j \longrightarrow 0, \qquad \mathbf{i}y\, \tilde{G}^{\mathbf{i}y} \longrightarrow -I_p.$$

The fixed-point equation therefore gives

$$\mathbf{i}y\, \tilde{\Gamma}_i^{\mathbf{i}y} \longrightarrow -1.$$

Theorem B.11 produces a probability measure on $\mathbb{R}$. Since

$$\Im(z\tilde{\Gamma}_i^z) \geq 0, \qquad \Im z > 0,$$

the same theorem places its support in $\mathbb{R}_+$. Its Stieltjes transform agrees with $z \mapsto \tilde{\Gamma}_i^z$ on the upper half-plane. Since both functions are holomorphic on Ω, they agree there by the identity theorem. □

Proposition 2.12. *The mapping $z \mapsto \tilde{G}^z$ is the Stieltjes transform of a positive matrix-valued measure $\tilde{M}$ on $\mathbb{R}_+$ with total mass I_p. Consequently,*

$$|\tilde{G}^z|_{\mathrm{op}} \leq \frac{1}{\eta_z}, \qquad z \in \Omega.$$

Proof. For $\Im z > 0$,

$$\tilde{G}^z = -\left(zI_p + \frac{z}{n} \sum_j \tilde{\Gamma}_j^z \Sigma_j\right)^{-1}.$$

Since $\Im(z\tilde{\Gamma}_j^z) \geq 0$, the inverse identity gives

$$\Im(\tilde{G}^z) = \tilde{G}^{\bar{z}} \left(\Im(z) I_p + \frac{1}{n} \sum_j \Im(z\tilde{\Gamma}_j^z) \Sigma_j\right) \tilde{G}^z \succ 0.$$

Furthermore, (2.17) and the preceding proof give

$$\Im(z\tilde{G}^z) \succeq 0, \qquad \mathbf{i}y\, \tilde{G}^{\mathbf{i}y} \longrightarrow -I_p.$$

Fix $u \in \mathbb{C}^p$. For $u = 0$, the corresponding zero measure gives the assertion trivially. If $u \neq 0$, apply Theorem B.11 to the normalized scalar mapping

$$z \longmapsto |u|^{-2} u^* \tilde{G}^z u.$$

It has positive imaginary part when $\Im z > 0$; its product with z has nonnegative imaginary part there, and $\mathbf{i}y|u|^{-2}u^*\tilde{G}^{\mathbf{i}y}u \to -1$. Consequently, $z \mapsto u^*\tilde{G}^z u$ is the Stieltjes transform of a positive measure on $\mathbb{R}_+$ of total mass $|u|^2$. Polarization and uniqueness of scalar Stieltjes transforms give a positive matrix-valued measure $\tilde{M}$ with total mass I_p. The identity theorem, applied entrywise, extends this representation to Ω:

$$\tilde{G}^z = \int_{\mathbb{R}_+} \frac{\tilde{M}(dt)}{t - z}, \qquad z \in \Omega.$$

Positivity of the matrix-valued measure and Cauchy–Schwarz for its scalar polarizations give, for $u, v \in \mathbb{C}^p$,

$$|u^*\tilde{G}^z v| \le \frac{|u|\,|v|}{\eta_z}.$$

Taking the supremum over unit vectors u, v gives the stated operator-norm bound. □

2.5 Support of the deterministic equivalent measure

To complete the support assertion in Theorem 1.2, set

$$x_\Sigma := \max\left\{\frac{8p}{n}, 4\right\} |\Sigma|_{\mathrm{op}}^{\infty}.$$

Lemma 2.13. *Assume that $|\Sigma|_{\mathrm{op}}^{\infty} > 0$. For every $z \in \mathbb{C}$ with $\Im z > 0$ and $\Re z \ge x_\Sigma$,*

$$\Re\left(-\frac{1}{\tilde{\Gamma}_i^z}\right) \ge \frac{\Re z}{2}, \qquad i \in [n],$$

and

$$|z\tilde{G}^z|_{\mathrm{op}} \le 2.$$

Proof. Put $x := \Re z \ge x_\Sigma$. Consider the compact convex set

$$\left\{D \in \mathcal{D}_n(\mathcal{C}_z) : |D|_\infty \le \frac{2|z|}{x}\right\}.$$

It contains I_n. If D belongs to this set, then

$$\left|\frac{1}{n}\sum_j \left(-\frac{D_j}{z}\right)\Sigma_j\right|_{\mathrm{op}} \le \frac{2|\Sigma|_{\mathrm{op}}^{\infty}}{x} \le \frac{1}{2},$$

so the Neumann series gives $|zG_D^{zI_p}|_{\mathrm{op}} \le 2$. Hence

$$\Re\left(z + \frac{z}{n}\operatorname{Tr}(\Sigma_i G_D^{zI_p})\right) \ge x - \frac{2p}{n}|\Sigma|_{\mathrm{op}}^{\infty} \ge \frac{x}{2}.$$

The map Φ^{zI_p} maps $\mathcal{D}_n(\mathcal{C}_z)$ into itself. Its image satisfies

$$-\frac{1}{z}\Phi^{zI_p}(D)_i = -\frac{1}{z + \frac{z}{n}\operatorname{Tr}(\Sigma_i G_D^{zI_p})},$$

and therefore

$$|\Phi^{zI_p}(D)_i| \le \frac{2|z|}{x}.$$

Thus Φ^{zI_p} maps the displayed set into itself. Brouwer's fixed-point theorem gives a fixed point in this set. Proposition 2.7 identifies this fixed point with $\widetilde{\Delta}^{zI_p}$. Since $\tilde{\Gamma}^z = -z^{-1}\widetilde{\Delta}^{zI_p}$, the two asserted bounds follow. □

Proof of Theorem 1.2. Proposition 2.11 provides the probability measures $\tilde{\mu}_i$. By Proposition 2.12, $\tilde{G}^z$ is the Stieltjes transform of a positive matrix-valued measure $\tilde{M}$ on $\mathbb{R}_+$ with total mass I_p. Taking its normalized trace defines the probability measure $\tilde{\mu} := \frac{1}{p}\operatorname{Tr}\tilde{M}$, whose Stieltjes transform is $\tilde{g}$. Equation (2.15) shows that this transform also equals that of the finite signed measure $(1-n/p)\delta_0 + \frac{1}{p}\sum_i \tilde{\mu}_i$. Uniqueness of Stieltjes transforms therefore proves the stated measure identity.

If $|\Sigma|_{\mathrm{op}}^{\infty} = 0$, every Σ_i vanishes, so $\tilde{\Gamma}_i^z = -1/z$ and $\tilde{G}^z = -I_p/z$. Thus $\tilde{\mu}_i = \tilde{\mu} = \delta_0$, and the support assertion follows. Assume henceforth that $|\Sigma|_{\mathrm{op}}^{\infty} > 0$, and define

$$M(z) := \max_{j\in[n]} \Im(\tilde{\Gamma}_j^z).$$

For $z = x + \mathbf{i}y$ with $x \in \mathbb{R}$ and $y > 0$, (2.17) gives

$$\Im\left(-\frac{1}{\tilde{\Gamma}_i^z}\right) = y + \frac{|z|^2}{n^2}\sum_{j=1}^{n} \Im(\tilde{\Gamma}_j^z)\operatorname{Tr}\left(\Sigma_i \tilde{G}^z \Sigma_j \tilde{G}^{\bar{z}}\right). \tag{2.18}$$

For $x \geq x_\Sigma$, Lemma 2.13 gives $|z\tilde{G}^z|_{\mathrm{op}} \leq 2$, and hence

$$|z|^2 \operatorname{Tr}\left(\Sigma_i \tilde{G}^z \Sigma_j \tilde{G}^{\bar{z}}\right) = \left|\Sigma_i^{1/2} z \tilde{G}^z \Sigma_j^{1/2}\right|_{\mathrm{HS}}^2 \leq 4p(|\Sigma|_{\mathrm{op}}^{\infty})^2.$$

Thus

$$\Im\left(-\frac{1}{\tilde{\Gamma}_i^z}\right) \leq y + \frac{4p(|\Sigma|_{\mathrm{op}}^{\infty})^2}{n} M(z).$$

The real-part estimate in Lemma 2.13 yields

$$\Im(\tilde{\Gamma}_i^z) \leq \frac{4}{x^2}\Im\left(-\frac{1}{\tilde{\Gamma}_i^z}\right).$$

Taking the maximum over i gives

$$M(z) \leq \frac{4y}{x^2} + \frac{16p(|\Sigma|_{\mathrm{op}}^{\infty})^2}{nx^2} M(z).$$

The last coefficient is at most $1/2$ for $x \geq x_\Sigma$: if $p \leq n/2$, use $x_\Sigma \geq 4|\Sigma|_{\mathrm{op}}^{\infty}$, while if $p \geq n/2$, use $x_\Sigma \geq 8p|\Sigma|_{\mathrm{op}}^{\infty}/n$. Therefore

$$M(x + \mathbf{i}y) \leq \frac{8y}{x^2}, \qquad x \geq x_\Sigma.$$

Fix $i \in [n]$, and let $x_\Sigma < a < b$ satisfy $\tilde{\mu}_i(\{a\}) = \tilde{\mu}_i(\{b\}) = 0$. Uniformly for $x \in [a, b]$,

$$0 \leq \Im(\tilde{\Gamma}_i^{x+\mathbf{i}y}) \leq \frac{8y}{x^2}.$$

Letting $y \downarrow 0$ in the Stieltjes inversion formula gives $\tilde{\mu}_i([a, b]) = 0$. Exhausting (x_Σ, ∞) by such intervals proves

$$\operatorname{supp}(\tilde{\mu}_i) \subset [0, x_\Sigma].$$

The measure identity proved above now gives

$$\operatorname{supp}(\tilde{\mu}) \subset \{0\} \cup \bigcup_{i=1}^{n} \operatorname{supp}(\tilde{\mu}_i) \subset [0, x_\Sigma] = \left[0,\, \max\left\{\frac{8p}{n}, 4\right\}|\Sigma|_{\mathrm{op}}^{\infty}\right].$$

□

3 Probabilistic estimates

We separate the resolvent error into its fluctuation and its bias. The common estimates first control $G^Z - \mathbb{E}[G^Z]$ by independent-copy replacement. For the bias, each route introduces a deterministic diagonal matrix D and estimates both $\mathbb{E}[G^Z] - G_D^Z$ and the inverse defect $r(D)$ defined in (2.10). The Hilbert–Schmidt route uses coordinatewise defect bounds, whereas the operator route uses a weighted sum of absolute defects. The two transfer estimates below then control $G_D^Z - \widetilde{G}^Z$. The final subsection uses additive deformation to obtain the corresponding sandwich estimates.

3.1 Common probabilistic estimates and auxiliary pivots

Fix a nonempty compact set $K \subset \mathbb{C}$ such that[1]

$$\eta_K := 1 \wedge d(\operatorname{conv} K, \mathbb{R}_+) > 0. \tag{3.1}$$

The convex-hull restriction entails no loss of generality for the scalar results: every compact subset of Ω admits a finite cover by compact sets satisfying it. Uniform estimates on the original compact set follow by taking the maximum of the resulting constants.

Define

$$\mathcal{C}_K := \operatorname{cone}\big(\mathbb{R}_+ \cup (-K)\big), \tag{3.2}$$

where cone denotes the convex conic hull. The geometric proof of Lemma 2.1, applied to $\operatorname{conv} K$, shows that $\mathcal{C}_K$ is a closed sector of aperture less than π. Let ω_K be its bisecting unit phase and set

$$c_K := \inf_{w \in \mathcal{C}_K \setminus \{0\}} \frac{\Re(\overline{\omega_K} w)}{|w|} > 0. \tag{3.3}$$

For a scalar or square matrix A, put

$$\mathfrak{C}_K(A) := \Re(\overline{\omega_K} A), \qquad \mathfrak{C}_{\overline{K}}(A) := \Re(\omega_K A). \tag{3.4}$$

For matrices, $\Re$ denotes the Hermitian part. The second map corresponds to the conjugate sector $\overline{\mathcal{C}_K}$. The sector and phase remain fixed throughout the section, including conditional arguments.

For every $p \geq 1$, define

$$\mathcal{K}_p := \{Z \in \mathcal{M}_p : \mathcal{C}_Z \subseteq \mathcal{C}_K, \quad \mathfrak{C}_K(-Z) \succeq c_K \eta_K I_p\}. \tag{3.5}$$

Every $z \in K$ gives $zI_p \in \mathcal{K}_p$. For any $Z \in \mathcal{K}_p$, the defining conditions imply

$$\eta_Z \geq c_K \eta_K > 0, \qquad c_Z \geq c_K. \tag{3.6}$$

For a parameter family $Z = (Z_p)_{(p,n)\in\Theta}$, write $Z \in \mathcal{K}$ when $Z_p \in \mathcal{K}_p$ at every index; in matrix formulas, Z denotes the current component, with its indices suppressed. We use the $O(\cdot)$ convention of Section 1.2, with $Z \in \mathcal{K}$ in place of $z \in K$ and with K fixed. In particular, $|Z|_{\mathrm{op}} = O(1)$ means $\sup_{(p,n)\in\Theta} |Z_p|_{\mathrm{op}} < \infty$.

3.1.1 Resolvent and leave-one-out estimates

Given $Z \in \mathcal{K}$, put

$$G := G^Z = \left(\frac{1}{n} X X^\top - Z\right)^{-1}.$$

For $i \in [n]$, let

$$X_{-i} := (x_1, \ldots, x_{i-1}, 0, x_{i+1}, \ldots, x_n), \qquad G_{-i} := G_{-i}^Z := \left(\frac{1}{n} X_{-i} X_{-i}^\top - Z\right)^{-1}.$$

[1] The inequality $\eta_K \leq 1$ will be used in particular in (3.56).

Resolvent, sector, and pivot identities. Define the scaled Schur pivots by

$$\Delta_i^Z := \left(1 + \frac{1}{n} x_i^\top G_{-i}^Z x_i\right)^{-1}, \qquad i \in [n]. \tag{3.7}$$

We suppress the superscript Z below and put $\Delta := \mathrm{Diag}(\Delta_1, \dots, \Delta_n)$.

Apply Lemma 2.3 with $\omega = \omega_K$, $c = c_K$, and $\kappa = c_K \eta_K$. Every $T \in \{G, G_{-i}\}$ is well defined and satisfies for all $u \in \mathbb{C}^p$

$$|T|_{\mathrm{op}} \le \frac{1}{c_K \eta_K}, \qquad u^* T u \in \overline{\mathcal{C}_Z}, \qquad T^* T,\ T T^* \preceq \frac{1}{c_K \eta_K} \mathfrak{C}_{\bar{K}}(T). \tag{3.8}$$

The sector conclusions and reciprocal bounds of Lemma 2.3 also give

$$x_i^\top G_{-i} x_i,\ \mathrm{Tr}(\Sigma_i G_{-i}) \in \overline{\mathcal{C}_Z}, \qquad \Delta_i \in \mathcal{C}_Z \setminus \{0\}, \qquad |\Delta_i| \le \frac{1}{c_K}.$$

Expectations and convex combinations remain in the same sectors. The lemma's reciprocal estimates therefore also apply to them.

Sherman–Morrison gives

$$G - G_{-i} = -\frac{\Delta_i}{n} G_{-i} x_i x_i^\top G_{-i}, \qquad G x_i = \Delta_i G_{-i} x_i, \qquad x_i^\top G = \Delta_i x_i^\top G_{-i}. \tag{3.9}$$

Consequently,

$$\frac{1}{n} x_i^\top G x_i = 1 - \Delta_i. \tag{3.10}$$

Concentration and leave-one-out estimates. Set

$$c_{n,q} := n^{1/(q \wedge 2) - 1}, \qquad q \ge 1. \tag{3.11}$$

Proposition 3.1 (Random resolvent fluctuation). *For every deterministic $A \in \mathcal{M}_p$ and every $q \ge 1$,*

$$\left\| \mathrm{Tr}\big(A(G - \mathbb{E}[G])\big) \right\|_{L^q} = O\left(c_{n,q} |A|_{\mathrm{op}} M_{\mathrm{op}}^{(q)}\right).$$

If $\sqrt{p}\, |\Sigma|_{\mathrm{HS}}^{\infty} = O(n)$, then also

$$\left\| \mathrm{Tr}\big(A(G - \mathbb{E}[G])\big) \right\|_{L^q} = O\left(c_{n,q} |A|_{\mathrm{HS}} M_{\mathrm{HS}}^{(q)}\right).$$

The same estimates hold with G replaced by any G_{-i}.

Proof. Let x_i' be an independent copy of x_i, independent of all columns, and write $G^{(i)}$ for the resolvent obtained by replacing x_i with x_i'. It satisfies the same bounds (3.8) by Lemma 2.3. Conditionally on X_{-i}, write

$$\begin{aligned} t &:= x_i^\top G_{-i} x_i, & s &:= x_i^\top G_{-i} A G_{-i} x_i, \\ t' &:= (x_i')^\top G_{-i} x_i', & s' &:= (x_i')^\top G_{-i} A G_{-i} x_i', \end{aligned}$$

and

$$t^\Sigma := \mathrm{Tr}(\Sigma_i G_{-i}), \qquad s^\Sigma := \mathrm{Tr}(\Sigma_i G_{-i} A G_{-i}).$$

The rank-one formula gives

$$\mathrm{Tr}\left(A(G - G^{(i)})\right) = -\frac{1}{n}\left(\frac{s}{1 + t/n} - \frac{s'}{1 + t'/n}\right). \tag{3.12}$$

Moreover,

$$\frac{s}{1+t/n}-\frac{s^\Sigma}{1+t^\Sigma/n}=\frac{s-s^\Sigma}{1+t/n}-\frac{s^\Sigma}{n(1+t^\Sigma/n)}\frac{t-t^\Sigma}{1+t/n}. \tag{3.13}$$

The term $s^\Sigma/(1+t^\Sigma/n)$ depends only on X_{-i} and cancels between the two copies in (3.12). Thus each term in the replacement difference contains a quadratic form centered conditionally on X_{-i}.

We first use operator tests. Hilbert–Schmidt Cauchy–Schwarz gives

$$|s^\Sigma|\le|A|_{\mathrm{op}}\sqrt{\mathrm{Tr}(\Sigma_iG_{-i}^*G_{-i})\,\mathrm{Tr}(\Sigma_iG_{-i}G_{-i}^*)}=O\big(|A|_{\mathrm{op}}\mathfrak{C}_{\bar K}(t^\Sigma)\big)\,,$$

thanks to the two sectorial Ward inequalities (3.8):

$$G_{-i}^*G_{-i},\ G_{-i}G_{-i}^*\preceq\frac{\mathfrak{C}_{\bar K}(G_{-i})}{c_K\eta_K}.$$

Since $t^\Sigma\in\overline{\mathcal{C}_Z}$, Lemma 2.3 gives

$$\frac{|s^\Sigma|}{n|1+t^\Sigma/n|}=O(|A|_{\mathrm{op}}). \tag{3.14}$$

Conditional quadratic-form control and (3.8) give

$$\|s-s^\Sigma\|_{L^q(x_i|X_{-i})}=O\left(M_{\mathrm{op}}^{(q)}|A|_{\mathrm{op}}\right)\qquad\text{and}\qquad\|t-t^\Sigma\|_{L^q(x_i|X_{-i})}\quad=O\left(M_{\mathrm{op}}^{(q)}\right).$$

Since G_{-i} depends only on X_{-i}, Lemma B.3 applies to the complex test matrices $G_{-i}AG_{-i}$ and G_{-i}. Combining these estimates with (3.13) and (3.14) yields

$$\left\|\frac{s}{1+t/n}-\frac{s^\Sigma}{1+t^\Sigma/n}\right\|_{L^q(x_i|X_{-i})}=O\left(M_{\mathrm{op}}^{(q)}|A|_{\mathrm{op}}\right).$$

For Hilbert–Schmidt tests, conditional quadratic-form control and the resolvent bounds give

$$\begin{aligned}&\|s-s^\Sigma\|_{L^q(x_i|X_{-i})}=O\left(M_{\mathrm{HS}}^{(q)}|A|_{\mathrm{HS}}\right),\\&\|t-t^\Sigma\|_{L^q(x_i|X_{-i})}=O\left(M_{\mathrm{HS}}^{(q)}\sqrt{p}\right),\qquad\qquad|s^\Sigma|=O\left(|\Sigma_i|_{\mathrm{HS}}|A|_{\mathrm{HS}}\right).\end{aligned}$$

Together with the uniform reciprocal bounds of Lemma 2.3, (3.13) therefore yields

$$\left\|\frac{s}{1+t/n}-\frac{s^\Sigma}{1+t^\Sigma/n}\right\|_{L^q(x_i|X_{-i})}=O\left(M_{\mathrm{HS}}^{(q)}|A|_{\mathrm{HS}}\left(1+\frac{\sqrt{p}\,|\Sigma|_{\mathrm{HS}}^\infty}{n}\right)\right)=O\left(M_{\mathrm{HS}}^{(q)}|A|_{\mathrm{HS}}\right).$$

The bounds for the operator and Hilbert–Schmidt tests also hold with x_i' in place of x_i. In particular, (3.12) gives

$$\|\,\mathrm{Tr}(A(G-G^{(i)}))\|_{L^q(x_i,x_i'|X_{-i})}=O\left(\frac{M_{\mathrm{op}}^{(q)}|A|_{\mathrm{op}}}{n}\right).$$

Under (1.7), it also gives

$$\|\,\mathrm{Tr}(A(G-G^{(i)}))\|_{L^q(x_i,x_i'|X_{-i})}=O\left(\frac{M_{\mathrm{HS}}^{(q)}|A|_{\mathrm{HS}}}{n}\right).$$

After integrating these conditional moment bounds, the replacement-moment inequality in Theorem B.1 gives both fluctuation estimates. For G_{-i}, apply the same argument as a function of $(x_j)_{j\neq i}$, replacing a column x_j with $j\neq i$; the normalization remains $1/n$. The case $n=1$ is trivial. □

Lemma 3.2 (Mean Hilbert–Schmidt leave-one-out bound). *For every $i \in [n]$,*

$$|\mathbb{E}[G - G_{-i}]|_{\mathrm{HS}} = O\left(\frac{|\Sigma_i|_{\mathrm{HS}}}{n}\right). \tag{3.15}$$

Proof. Fix a deterministic matrix A with $|A|_{\mathrm{HS}} \le 1$. The rank-one identity (3.9) and the pivot bound in Lemma 2.3 imply

$$|\mathrm{Tr}(A\mathbb{E}[G - G_{-i}])| = O\left(\frac{1}{n}\mathbb{E}\left[\left|x_i^\top G_{-i} A G_{-i} x_i\right|\right]\right).$$

Condition on X_{-i} and apply Lemma B.4 to $B = G_{-i}AG_{-i}$. Since $|G_{-i}AG_{-i}|_{\mathrm{HS}} = O(|A|_{\mathrm{HS}})$,

$$|\mathrm{Tr}(A\mathbb{E}[G - G_{-i}])| = O\left(\frac{|\Sigma_i|_{\mathrm{HS}}}{n}\right).$$

Hilbert–Schmidt duality proves (3.15). □

3.1.2 Auxiliary resolvents and inverse defects

Fix $Z \in \mathcal{K}$, and write

$$\widetilde{\Delta} := \widetilde{\Delta}^Z, \qquad \widetilde{G} := \widetilde{G}^Z = G^Z_{\widetilde{\Delta}}.$$

For a deterministic $D \in \mathcal{D}_n(\mathcal{C}_Z)$, set

$$G_D := G_D^Z := \left(\frac{1}{n}\sum_{i=1}^n D_i\Sigma_i - Z\right)^{-1}.$$

Applying Lemma 2.3 with the same sectorial constants shows that G_D is well defined and satisfies (3.8). The two probabilistic routes use the same decomposition

$$G - \widetilde{G} = (G - \mathbb{E}[G]) + (\mathbb{E}[G] - G_D) + (G_D - \widetilde{G}).$$

Proposition 3.1 controls the first term. The two routes estimate the middle term using their respective pivots.

To estimate $G_D - \widetilde{G}$, we use the inverse defect $r(D)$ defined in (2.10). For $D \in \mathcal{D}_n(\mathcal{C}_Z \setminus \{0\})$, uniqueness of the deterministic fixed point gives $r(D) = 0$ exactly when $D = \widetilde{\Delta}$. Applying (2.11) with $D' = \widetilde{\Delta}$ gives

$$G_D - \widetilde{G} = -\frac{1}{n}G_D\left(\sum_{j=1}^n (D_j - \widetilde{\Delta}_j)\Sigma_j\right)\widetilde{G}. \tag{3.16}$$

The same specialization in (2.12), using $r(\widetilde{\Delta}) = 0$, gives

$$D - \widetilde{\Delta} = -D\widetilde{\Delta}\, r(D) + \Psi^Z(D, \widetilde{\Delta})(D - \widetilde{\Delta}). \tag{3.17}$$

The secant equation (3.17) transfers a bound on $r(D)$ to $G_D - \widetilde{G}$. The appropriate measure of $r(D)$ depends on the test geometry: the Hilbert–Schmidt route uses $|r(D)|_\infty$, whereas the operator-norm route uses $\sum_i |D_i r_i(D)|$.

3.1.3 Choice of the auxiliary pivot

We choose D so that the estimate of $\mathbb{E}[G] - G_D$ also gives the defect bound needed in Proposition 3.3.

Hilbert–Schmidt-norm route. Define

$$\check{\Delta}_i := \big(\mathbb{E}[\Delta_i^{-1}]\big)^{-1} = \left(1 + \frac{1}{n}\operatorname{Tr}(\Sigma_i \mathbb{E}[G_{-i}])\right)^{-1},$$

and set

$$\check{\Delta} := \operatorname{Diag}(\check{\Delta}_1, \dots, \check{\Delta}_n), \qquad \check{G} := \check{G}^Z := G^Z_{\check{\Delta}}.$$

The sector and reciprocal bounds in Lemma 2.3 give

$$\check{\Delta}_i \in \mathcal{C}_Z \setminus \{0\}, \qquad |\check{\Delta}|_\infty = O(1),$$

so $\check{G}$ is well defined. By the definitions of $\check{\Delta}_i$ and r_i,

$$r_i(\check{\Delta}) = \frac{1}{n}\operatorname{Tr}\big(\Sigma_i(\mathbb{E}[G_{-i}] - \check{G})\big). \tag{3.18}$$

Consequently,

$$|r_i(\check{\Delta})| \le \frac{|\Sigma_i|_{\mathrm{HS}}}{n}\left(|\mathbb{E}[G] - \check{G}|_{\mathrm{HS}} + |\mathbb{E}[G_{-i} - G]|_{\mathrm{HS}}\right).$$

Thus the mean-resolvent bias and the mean leave-one-out error control $|r(\check{\Delta})|_\infty$ directly. The check pivot also satisfies

$$\check{\Delta}_i - \Delta_i = \frac{\check{\Delta}_i \Delta_i}{n}\left(x_i^\top G_{-i} x_i - \mathbb{E}[x_i^\top G_{-i} x_i]\right).$$

The uniform bounds on the pivots reduce the estimate of $\check{\Delta}_i - \Delta_i$ to the centered quantity on the right-hand side. The factor $|\Sigma_i|_{\mathrm{HS}}/n$ in the bound for $r_i(\check{\Delta})$ comes from the Hilbert–Schmidt trace inequality.

Operator-norm route. For tests A measured in operator norm, we bound $G_D - \widetilde{G}$ in nuclear norm. The transfer uses the weighted defect

$$\sum_{i=1}^n |D_i r_i(D)|.$$

For this bound, we use expectations of the Schur pivots. First suppose that $\Sigma_i = \Sigma$ for every i, and set

$$\widehat{\delta} := \frac{1}{n}\sum_{i=1}^n \mathbb{E}[\Delta_i], \qquad \widehat{G} := (\widehat{\delta}\Sigma - Z)^{-1}.$$

The two resolvent equations are

$$\frac{1}{n}XX^\top G = I_p + ZG, \qquad\qquad \widehat{\delta}\Sigma\widehat{G} = I_p + Z\widehat{G}. \tag{3.19}$$

Since

$$\mathbb{E}\left[\operatorname{Tr}\left(\frac{1}{n}XX^\top G\right)\right] = n(1 - \widehat{\delta}),$$

the common inverse defect satisfies

$$n\widehat{\delta}\, r_i(\widehat{\delta} I_n) = \operatorname{Tr}\Big(Z(\mathbb{E}[G] - \widehat{G})\Big).$$

This identity controls the weighted defect by testing $\mathbb{E}[G] - \widehat{G}$ against Z.

For several classes, the same identity is applied at each replacement stage using the expected pivots defined in (3.45). If the final deterministic pivot is constant on I_a, with $D_i = \widehat{\delta}^{(a)}$ for $i \in I_a$, then

$$n_a(1 - D_i) - \pi_a D_i \operatorname{Tr}(\Sigma^{(a)} G_D) = n_a D_i r_i(D), \qquad i \in I_a.$$

The successive replacement argument estimates the change from the expected random contribution $n_a(1-D_i)$ to the final deterministic contribution $\pi_a D_i \operatorname{Tr}(\Sigma^{(a)} G_D)$. Summing the absolute error bounds over the classes controls $\sum_i |D_i r_i(D)|$. Replacing the classwise expected pivots by the check pivots would introduce

$$\big(\mathbb{E}[\Delta_i^{-1}]\big)^{-1} - \mathbb{E}[\Delta_i],$$

which would require a separate estimate in this argument.

Accordingly, the two routes use different pivots for a precise algebraic reason:

$$\check{\Delta} \quad \longrightarrow \quad |r(\check{\Delta})|_\infty \quad \longrightarrow \quad |\check{G} - \widetilde{G}|_{\mathrm{HS}},$$

whereas

$$D_i = \widehat{\delta}^{(a)} \text{ on } I_a \quad \longrightarrow \quad \sum_i |D_i r_i(D)| \quad \longrightarrow \quad |G_D - \widetilde{G}|_*.$$

3.1.4 Transfer from the inverse defect

The following proposition bounds $G_D^Z - \widetilde{G}^Z$ in Hilbert–Schmidt and nuclear norm using the secant equation (3.17). Part (i) uses a weighted supremum norm, while part (ii) uses weighted column sums.

For the weighted secant estimates below, use

$$|u|_{D,D'} := \max_{i\in[n]} \frac{|u_i|}{\sqrt{\mathfrak{C}_K(D_i)\mathfrak{C}_K(D_i')}},$$

and the corresponding induced operator norm.

Proposition 3.3 (Two transfers from the inverse defect). *Let $Z \in \mathcal{K}$ and*

$$D \in \mathcal{D}_n(\mathcal{C}_Z \setminus \{0\}).$$

Write $\widetilde{\Delta} = \widetilde{\Delta}^Z$ and $\widetilde{G} = \widetilde{G}^Z$.

(i) *Assume that*

$$|\Sigma|_*^\infty = O(n), \qquad |D|_\infty = O(1).$$

Then

$$|G_D^Z - \widetilde{G}|_{\mathrm{HS}} = O(\min\{|\Sigma|_{\mathrm{HS}}^\infty, \sqrt{p}\}|r(D)|_\infty)\,. \tag{3.20}$$

(ii) *For every $D \in \mathcal{D}_n(\mathcal{C}_Z \setminus \{0\})$ such that $|D|_\infty = O(1)$, assume there exists $\rho > 1$ satisfying $|D_i|^{-1} = O(\rho)$ and $|\widetilde{\Delta}_i|^{-1} = O(\rho)$. Then*

$$|G_D^Z - \widetilde{G}|_* = O\left(\sqrt{\rho}\sum_{i=1}^n |D_i r_i(D)|\right). \tag{3.21}$$

Proof. Since $r(\widetilde{\Delta}) = 0$, the secant equation reads

$$D - \widetilde{\Delta} = -D\widetilde{\Delta} r(D) + \Psi^Z(D, \widetilde{\Delta})(D - \widetilde{\Delta}). \tag{3.22}$$

Proof of (i). The argument is the same weighted contraction as in the uniqueness proof, with the defect $r(D)$ acting as a forcing term.

The assumption $|\Sigma|_*^\infty = O(n)$, together with the sectorial resolvent bound, gives

$$\frac{1}{n}\mathfrak{C}_{\bar{K}}(\operatorname{Tr}(\Sigma_i G_D^Z)) + \frac{1}{n}\mathfrak{C}_{\bar{K}}(\operatorname{Tr}(\Sigma_i \widetilde{G}^Z)) = O(1)$$

uniformly in i. Suppose first that

$$|r(D)|_\infty \le \frac{c_K}{2}.$$

Applying $\mathfrak{C}_{\bar{K}}$ to the definition of $r_i(D)$, and using the fixed-point equation for $\widetilde{\Delta}$, gives

$$\frac{\mathfrak{C}_K(D_i)}{|D_i|^2} = \mathfrak{C}_K(1) + \frac{1}{n}\mathfrak{C}_{\bar{K}}(\mathrm{Tr}(\Sigma_i G_D^Z)) + \mathfrak{C}_{\bar{K}}(r_i(D)) \ge \frac{c_K}{2} + \frac{1}{n}\mathfrak{C}_{\bar{K}}(\mathrm{Tr}(\Sigma_i G_D^Z)),$$
$$\frac{\mathfrak{C}_K(\widetilde{\Delta}_i)}{|\widetilde{\Delta}_i|^2} = \mathfrak{C}_K(1) + \frac{1}{n}\mathfrak{C}_{\bar{K}}(\mathrm{Tr}(\Sigma_i \widetilde{G}^Z)) \ge c_K + \frac{1}{n}\mathfrak{C}_{\bar{K}}(\mathrm{Tr}(\Sigma_i \widetilde{G}^Z)).$$

Hence Lemma 2.5 gives, uniformly in $Z \in \mathcal{K}$,

$$|\Psi^Z(D, \widetilde{\Delta})|_{D,\widetilde{\Delta}} \le 1 - c$$

for some $c > 0$. Applying this to (3.22) therefore yields

$$|D - \widetilde{\Delta}|_{D,\widetilde{\Delta}} = O\left(\max_{i\in[n]} \frac{|D_i \widetilde{\Delta}_i r_i(D)|}{\sqrt{\mathfrak{C}_K(D_i)\mathfrak{C}_K(\widetilde{\Delta}_i)}}\right) = O(|r(D)|_\infty).$$

Since D is bounded and $|\widetilde{\Delta}|_\infty = O(1)$ by Proposition 2.7, this also gives

$$|D - \widetilde{\Delta}|_\infty = O(|r(D)|_\infty). \tag{3.23}$$

The resolvent identity now gives directly

$$|G_D^Z - \widetilde{G}|_{\mathrm{HS}} \le \frac{1}{n}\sum_i |D_i - \widetilde{\Delta}_i|\, |G_D^Z \Sigma_i \widetilde{G}|_{\mathrm{HS}} = O(|\Sigma|_{\mathrm{HS}}^\infty |r(D)|_\infty)\,.$$

We also obtain a bound with $\sqrt{p}$ in place of $|\Sigma|_{\mathrm{HS}}^\infty$. Using

$$|D_i - \widetilde{\Delta}_i| \le |D - \widetilde{\Delta}|_{D,\widetilde{\Delta}} \sqrt{\mathfrak{C}_K(D_i)\mathfrak{C}_K(\widetilde{\Delta}_i)}$$

in the same resolvent identity, followed by Cauchy–Schwarz in i and the two Ward inequalities from Lemma 2.3, gives

$$|G_D^Z - \widetilde{G}|_{\mathrm{op}} = O\Big(|D - \widetilde{\Delta}|_{D,\widetilde{\Delta}}\Big) = O(|r(D)|_\infty).$$

Consequently,

$$|G_D^Z - \widetilde{G}|_{\mathrm{HS}} = O(\sqrt{p}\,|r(D)|_\infty).$$

Taking the better of the two estimates proves (3.20) when $|r(D)|_\infty \le c_K/2$. If instead $|r(D)|_\infty > c_K/2$, the resolvent identity and the uniform bounds on $D, \widetilde{\Delta}, G_D^Z, \widetilde{G}$ give

$$|G_D^Z - \widetilde{G}|_{\mathrm{HS}} = O(\min\{|\Sigma|_{\mathrm{HS}}^\infty, \sqrt{p}\})\,,$$

and the factor $|r(D)|_\infty$ may be inserted because it is bounded below by a positive constant. This proves (i).

Proof of (ii). Here we retain the term involving $\mathfrak{C}_K(-Z)$ in the Ward identities. Since $Z \in \mathcal{K}$, we have

$$\mathfrak{C}_K(-Z) \succeq c_K \eta_K I_p.$$

This term will play two roles: it controls the nuclear norm of the resolvent difference, and it appears as the deficit in the weighted column sums of the secant kernel. For $i \in [n]$, set

$$h_i := \frac{1}{n}\,\mathrm{Tr}\big(\Sigma_i (G_D^Z)^* \mathfrak{C}_K(-Z) G_D^Z\big)\,, \qquad \widetilde{h}_i := \frac{1}{n}\,\mathrm{Tr}\Big(\Sigma_i \widetilde{G}\mathfrak{C}_K(-Z)\widetilde{G}^*\Big)\,.$$

By coercivity of $\mathfrak{C}_K(-Z)$ and the resolvent identity,

$$|G_D^Z - \widetilde{G}|_* \le \frac{1}{n}\sum_i |D_i - \widetilde{\Delta}_i|\,|G_D^Z \Sigma_i \widetilde{G}|_* = O\left(\sum_i |D_i - \widetilde{\Delta}_i|\sqrt{h_i \widetilde{h}_i}\right). \tag{3.24}$$

Thus it remains to control the sum on the right. Use the column weights

$$w_i := \frac{\sqrt{\mathfrak{C}_K(D_i)\mathfrak{C}_K(\widetilde{\Delta}_i)}}{|D_i \widetilde{\Delta}_i|}.$$

These weights cancel $|D_i\widetilde{\Delta}_i|$ in the i-th row of the secant matrix. Cauchy–Schwarz and the Ward identities then give

$$\begin{aligned}\sum_i w_i |\Psi^Z(D,\widetilde{\Delta})_{ij}| &\le \frac{1}{n^2}\sqrt{\sum_i \mathfrak{C}_K(D_i)\,\mathrm{Tr}\left(\Sigma_i G_D^Z \Sigma_j (G_D^Z)^*\right)}\sqrt{\sum_i \mathfrak{C}_K(\widetilde{\Delta}_i)\,\mathrm{Tr}\left(\Sigma_i \widetilde{G}^* \Sigma_j \widetilde{G}\right)} \\ &= \sqrt{\frac{1}{n}\mathfrak{C}_{\bar{K}}(\mathrm{Tr}(\Sigma_j G_D^Z)) - h_j}\sqrt{\frac{1}{n}\mathfrak{C}_{\bar{K}}(\mathrm{Tr}(\Sigma_j \widetilde{G}^Z)) - \widetilde{h}_j}. \end{aligned} \tag{3.25}$$

Applying the two-term Cauchy–Schwarz inequality to (3.25) gives

$$\sum_i w_i |\Psi^Z(D,\widetilde{\Delta})_{ij}| + \sqrt{h_j \widetilde{h}_j} \le \sqrt{\frac{1}{n}\mathfrak{C}_{\bar{K}}(\mathrm{Tr}(\Sigma_j G_D^Z))}\sqrt{\frac{1}{n}\mathfrak{C}_{\bar{K}}(\mathrm{Tr}(\Sigma_j \widetilde{G}^Z))}. \tag{3.26}$$

The inverse-defect equation (2.10) and the fixed-point equation in Proposition 2.7 imply

$$\frac{1}{n}\mathfrak{C}_{\bar{K}}(\mathrm{Tr}(\Sigma_j G_D^Z)) \le \frac{\mathfrak{C}_K(D_j)}{|D_j|^2} + |r_j(D)|, \qquad \frac{1}{n}\mathfrak{C}_{\bar{K}}(\mathrm{Tr}(\Sigma_j \widetilde{G}^Z)) \le \frac{\mathfrak{C}_K(\widetilde{\Delta}_j)}{|\widetilde{\Delta}_j|^2}.$$

All quantities under the square roots are nonnegative, and $\mathfrak{C}_K(D_j) > 0$ by the sector assumptions. Using $\sqrt{1+t} \le 1 + t/2$ for $t \ge 0$ therefore yields

$$\sum_i w_i |\Psi^Z(D,\widetilde{\Delta})_{ij}| + \sqrt{h_j \widetilde{h}_j} \le w_j + \frac{|r_j(D)|}{2}\sqrt{\frac{\mathfrak{C}_K(\widetilde{\Delta}_j)/|\widetilde{\Delta}_j|^2}{\mathfrak{C}_K(D_j)/|D_j|^2}}. \tag{3.27}$$

We can now return to the secant equation (3.22). Taking absolute values, multiplying the i-th coordinate by w_i, and summing over i, gives

$$\sum_i w_i |D_i - \widetilde{\Delta}_i| \le \sum_i |D_i \widetilde{\Delta}_i r_i(D)| w_i + \sum_j |D_j - \widetilde{\Delta}_j| \sum_i w_i |\Psi^Z(D,\widetilde{\Delta})_{ij}|.$$

Insert (3.27) and cancel $\sum_i w_i |D_i - \widetilde{\Delta}_i|$ from the two sides to obtain

$$\sum_i |D_i - \widetilde{\Delta}_i|\sqrt{h_i \widetilde{h}_i} \le \sum_i |r_i(D)| \left[\sqrt{\mathfrak{C}_K(D_i)\mathfrak{C}_K(\widetilde{\Delta}_i)} + \frac{|D_i - \widetilde{\Delta}_i|}{2}\sqrt{\frac{\mathfrak{C}_K(\widetilde{\Delta}_i)/|\widetilde{\Delta}_i|^2}{\mathfrak{C}_K(D_i)/|D_i|^2}}\right]. \tag{3.28}$$

Since $c_K|u| \le \mathfrak{C}_K(u) \le |u|$ for every $u \in \mathcal{C}_Z$, we have

$$\sqrt{\mathfrak{C}_K(D_i)\mathfrak{C}_K(\widetilde{\Delta}_i)} = O\left(\sqrt{|D_i\widetilde{\Delta}_i|}\right) \qquad \text{and} \qquad \sqrt{\frac{\mathfrak{C}_K(\widetilde{\Delta}_i)/|\widetilde{\Delta}_i|^2}{\mathfrak{C}_K(D_i)/|D_i|^2}} = O\left(\sqrt{\frac{|D_i|}{|\widetilde{\Delta}_i|}}\right).$$

Using also $|D_i - \widetilde{\Delta}_i| \le |D_i| + |\widetilde{\Delta}_i|$, we can deduce from (3.24) and (3.28):

$$|G_D^Z - \widetilde{G}|_* \le \frac{1}{c_K \eta_K} \sum_i |D_i - \widetilde{\Delta}_i| \sqrt{h_i \widetilde{h}_i} \le O\left(\sum_i |D_i r_i(D)| \left[\sqrt{\frac{|D_i|}{|\widetilde{\Delta}_i|}} + \sqrt{\frac{|\widetilde{\Delta}_i|}{|D_i|}} \right] \right). \quad (3.29)$$

Finally, under the stated bounded-pivot assumptions,

$$\frac{|D_i|}{|\widetilde{\Delta}_i|} + \frac{|\widetilde{\Delta}_i|}{|D_i|} = O(\rho),$$

uniformly in i, and hence each square-root ratio in (3.29) is $O(\sqrt{\rho})$. This proves (3.21). □

Corollary 3.4 (Inverse-defect transfer for negative Hermitian parameters). *Let $Z \in \mathcal{K}$ satisfy $Z = Z^* \prec 0$, and let $D \in \mathcal{D}_n((0,\infty))$. Then*

$$|G_D^Z - \widetilde{G}^Z|_* \le \frac{1}{\eta_Z} \sum_{i=1}^n |D_i r_i(D)|. \quad (3.30)$$

The coefficient η_Z^{-1} is uniformly bounded on $\mathcal{K}$ by (3.6).

Proof. Set

$$\alpha_i := \frac{D_i}{\Phi^Z(D)_i} = 1 - D_i r_i(D) > 0.$$

For nonnegative diagonal coefficients, increasing one entry decreases the resolvent in the positive-semidefinite order. Hence Φ^Z is coordinatewise increasing. Consequently, the continuous map

$$E \longmapsto \mathrm{Diag}\big(\min\{\alpha_i, 1\} \Phi^Z(E)_i : i \in [n]\big)$$

maps the box $0 \le E_i \le \min\{D_i, \widetilde{\Delta}_i^Z\}$ into itself: its i-th coordinate is at most both $\alpha_i \Phi^Z(D)_i = D_i$ and $\Phi^Z(\widetilde{\Delta}^Z)_i = \widetilde{\Delta}_i^Z$. Brouwer's fixed-point theorem, as in Lemma 2.4, gives a fixed point E in this box. Its coordinates are positive because every coordinate of the map is positive. Since $E_i \le \min\{D_i, \widetilde{\Delta}_i^Z\}$,

$$G_E^Z \succeq G_D^Z, \qquad G_E^Z \succeq \widetilde{G}^Z.$$

Taking traces in the resolvent equations (2.7) and using $E_i / \Phi^Z(E)_i = \min\{\alpha_i, 1\}$ gives

$$\mathrm{Tr}\big((-Z)(G_E^Z - G_D^Z)\big) = \sum_i \big[\alpha_i - \min\{\alpha_i, 1\} - (D_i - E_i)\big] \le \sum_i \big(\alpha_i - \min\{\alpha_i, 1\}\big),$$
$$\mathrm{Tr}\Big((-Z)(G_E^Z - \widetilde{G}^Z)\Big) = \sum_i \big[1 - \min\{\alpha_i, 1\} - (\widetilde{\Delta}_i^Z - E_i)\big] \le \sum_i \big(1 - \min\{\alpha_i, 1\}\big).$$

Both differences are positive semidefinite, so their nuclear norms equal their traces. Using $-Z \succeq \eta_Z I_p$ and the triangle inequality gives

$$|G_D^Z - \widetilde{G}^Z|_* \le |G_E^Z - G_D^Z|_* + |G_E^Z - \widetilde{G}^Z|_* \le \frac{1}{\eta_Z} \sum_i |\alpha_i - 1| = \frac{1}{\eta_Z} \sum_i |D_i r_i(D)|.$$

□

3.2 Hilbert–Schmidt-norm route

We use $\check{G}^Z$ and $\check{\Delta}$ as defined in Section 3.1.3. We first bound $\mathbb{E}[G^Z] - \check{G}^Z$ and $r(\check{\Delta})$ in terms of $J(\check{\Delta})$, defined below. Under the additional assumption $|Z|_{\mathrm{op}} = O(1)$, we then show that $J(\check{\Delta}) = O(\sigma_\Sigma)$ and apply the Hilbert–Schmidt transfer.

For a fixed $Z \in \mathcal{K}$, define

$$J(D) := 1 + \frac{1}{n}\sum_{i=1}^{n} |D_i|\,|\Sigma_i G_D^Z|_{\mathrm{op}}, \qquad D \in \mathcal{D}_n(\mathcal{C}_Z). \tag{3.31}$$

Proposition 3.5 (Bias and inverse defect of the check pivot). *Assume that* $\sqrt{p}\,|\Sigma|_{\mathrm{HS}}^{\infty} = O(n)$. *Then*

$$|\mathbb{E}[G^Z] - \check{G}^Z|_{\mathrm{HS}} = O\left(J(\check{\Delta})\left[\frac{|\Sigma|_{\mathrm{HS}}^{\infty}}{n} + \min\left\{M_{\mathrm{HS}}^{(1)}, \frac{(M_{\mathrm{HS}}^{(2)})^2\sqrt{p}}{n}\right\}\right]\right), \tag{3.32}$$

$$|r(\check{\Delta})|_{\infty} = O\left(\frac{|\Sigma|_{\mathrm{HS}}^{\infty}}{n} J(\check{\Delta})\left[\frac{|\Sigma|_{\mathrm{HS}}^{\infty}}{n} + \min\left\{M_{\mathrm{HS}}^{(1)}, \frac{(M_{\mathrm{HS}}^{(2)})^2\sqrt{p}}{n}\right\}\right]\right). \tag{3.33}$$

Proof. Put

$$\varepsilon := \frac{|\Sigma|_{\mathrm{HS}}^{\infty}}{n} + \min\left\{M_{\mathrm{HS}}^{(1)}, \frac{(M_{\mathrm{HS}}^{(2)})^2\sqrt{p}}{n}\right\}.$$

We first record two consequences of the profile-size condition:

$$\frac{|\Sigma|_{*}^{\infty}}{n} = O(1), \qquad \frac{|\Sigma|_{\mathrm{HS}}^{\infty}}{n} = O\Big(\frac{1}{\sqrt{p}}\Big) = O(1).$$

The check-pivot definition in Section 3.1.3 and the uniform resolvent bound (3.8) give

$$|\check{\Delta}_i|^{-1} \le 1 + C\frac{\mathrm{Tr}(\Sigma_i)}{n} = O(1),$$

so the definition of J gives $\frac{1}{n}\sum_i |\Sigma_i \check{G}^Z|_{\mathrm{op}} = O\big(J(\check{\Delta})\big)$.

To estimate the bias by Hilbert–Schmidt duality, fix a deterministic matrix A with $|A|_{\mathrm{HS}} \le 1$, and set

$$t_i = x_i^\top G_{-i}^Z x_i, \qquad s_i = x_i^\top \check{G}^Z A G_{-i}^Z x_i.$$

Independence gives $\mathbb{E}s_i = \mathrm{Tr}(\Sigma_i \check{G}^Z A\,\mathbb{E}[G_{-i}^Z])$. The resolvent identity and $G^Z x_i = \Delta_i G_{-i}^Z x_i$ therefore give

$$\mathrm{Tr}\big(A(\mathbb{E}[G^Z] - \check{G}^Z)\big) = \frac{1}{n}\sum_i \big[\check{\Delta}_i\,\mathrm{Tr}\big(\Sigma_i \check{G}^Z A\,\mathbb{E}[G^Z]\big) - \mathbb{E}[\Delta_i s_i]\big] = \frac{1}{n}\sum_i (a_i + b_i),$$

where

$$a_i = \check{\Delta}_i\,\mathrm{Tr}\left(\Sigma_i \check{G}^Z A\,\mathbb{E}[G^Z - G_{-i}^Z]\right) \qquad \text{and} \qquad b_i = \mathbb{E}\left[s_i(\check{\Delta}_i - \Delta_i)\right].$$

For a_i, the mean leave-one-out estimate (3.15) gives

$$\max_i |\mathbb{E}[G^Z - G_{-i}^Z]|_{\mathrm{HS}} = O\left(\frac{|\Sigma|_{\mathrm{HS}}^{\infty}}{n}\right) = O(\varepsilon),$$

since $|\Sigma|_{\mathrm{HS}}^{\infty}/n \le \varepsilon$ by definition. The definition of J then yields

$$\frac{1}{n}\sum_i |a_i| = O\left(J(\check{\Delta})\frac{|\Sigma|_{\mathrm{HS}}^{\infty}}{n}\right).$$

For b_i, the check-pivot definition gives directly

$$b_i = \mathbb{E}\left[s_i\left(\frac{1}{1+\mathbb{E}t_i/n} - \frac{1}{1+t_i/n}\right)\right].$$

We use Lemma B.2 from Appendix B. Let $\mathbb{E}_i$ denote conditional expectation given X_{-i}, and write

$$s_i^\Sigma := \mathbb{E}_i s_i = \mathrm{Tr}(\Sigma_i \check{G}^Z A G_{-i}^Z), \qquad t_i^\Sigma := \mathbb{E}_i t_i = \mathrm{Tr}(\Sigma_i G_{-i}^Z).$$

Inserting the reciprocal at t_i^Σ and replacing s_i by its conditional mean in the second term gives

$$b_i = \mathbb{E}\left[s_i\left(\frac{1}{1+t_i^\Sigma/n} - \frac{1}{1+t_i/n}\right)\right] + \mathbb{E}\left[s_i^\Sigma\left(\frac{1}{1+\mathbb{E}t_i/n} - \frac{1}{1+t_i^\Sigma/n}\right)\right]. \tag{3.34}$$

We apply the lemma conditionally on X_{-i} to (s_i, t_i) in the first term, and without conditioning to (s_i^Σ, t_i^Σ) in the second. Its denominator hypotheses follow from Lemma 2.3, since t_i, t_i^Σ, and $\mathbb{E}t_i$ lie in $\overline{\mathcal{C}_Z}$.

Conditional quadratic-form control from Lemma B.3, Proposition 3.1, and Cauchy–Schwarz in the Hilbert–Schmidt norm give, for $q \in \{1,2\}$ with $M_{\mathrm{HS}}^{(q)} < \infty$,

$$\begin{aligned}
\|s_i - s_i^\Sigma\|_{L^q} &= O(M_{\mathrm{HS}}^{(q)}), & \|s_i^\Sigma - \mathbb{E}s_i\|_{L^q} &= O\left(c_{n,q}|\Sigma_i\check{G}^Z|_{\mathrm{op}} M_{\mathrm{HS}}^{(q)}\right),\\
\|t_i - t_i^\Sigma\|_{L^q} &= O(\sqrt{p}\, M_{\mathrm{HS}}^{(q)}), & \|t_i^\Sigma - \mathbb{E}t_i\|_{L^q} &= O\left(c_{n,q}|\Sigma|_{\mathrm{HS}}^\infty M_{\mathrm{HS}}^{(q)}\right),\\
|s_i^\Sigma| &= O\left(|\Sigma|_{\mathrm{HS}}^\infty\right), & |\mathbb{E}s_i| &= O\left(\sqrt{p}\,|\Sigma_i\check{G}^Z|_{\mathrm{op}}\right).
\end{aligned}$$

The bounds for $s_i - s_i^\Sigma$ and $t_i - t_i^\Sigma$ also hold for their conditional L^q norms given X_{-i}, uniformly almost surely. Using $c_{n,1} = 1$ and $\sqrt{p}\,|\Sigma|_{\mathrm{HS}}^\infty/n = O(1)$, the first-moment estimate of Lemma B.2 bounds the two terms in (3.34) by $O(M_{\mathrm{HS}}^{(1)})$ and $O\left(|\Sigma_i\check{G}^Z|_{\mathrm{op}} M_{\mathrm{HS}}^{(1)}\right)$, respectively. When $M_{\mathrm{HS}}^{(2)} < \infty$, its second-moment estimate and $c_{n,2} = n^{-1/2}$ give

$$\begin{aligned}
|b_i| &= O\left((M_{\mathrm{HS}}^{(2)})^2\left[\frac{\sqrt{p}}{n}\left(1 + \frac{\sqrt{p}\,|\Sigma|_{\mathrm{HS}}^\infty}{n}\right) + |\Sigma_i\check{G}^Z|_{\mathrm{op}}\left(\frac{|\Sigma|_{\mathrm{HS}}^\infty}{n^2} + \frac{\sqrt{p}\,(|\Sigma|_{\mathrm{HS}}^\infty)^2}{n^3}\right)\right]\right)\\
&= O\left(\left(1 + |\Sigma_i\check{G}^Z|_{\mathrm{op}}\right)\frac{(M_{\mathrm{HS}}^{(2)})^2\sqrt{p}}{n}\right),
\end{aligned}$$

since $|\Sigma|_{\mathrm{HS}}^\infty/n = O(p^{-1/2})$ and $p \geq 1$. Taking the better estimate and using $\frac{1}{n}\sum_i |\Sigma_i\check{G}^Z|_{\mathrm{op}} = O(J(\check{\Delta}))$ yields

$$\frac{1}{n}\sum_i |b_i| = O\left(J(\check{\Delta})\min\left\{M_{\mathrm{HS}}^{(1)}, \frac{(M_{\mathrm{HS}}^{(2)})^2\sqrt{p}}{n}\right\}\right).$$

Combining the bounds for a_i and b_i in the bias decomposition above gives $|\,\mathrm{Tr}(A(\mathbb{E}[G^Z] - \check{G}^Z))| = O(J(\check{\Delta})\varepsilon)$. Hilbert–Schmidt duality proves (3.32).

Finally, the exact check-pivot defect identity (3.18) gives

$$|r_i(\check{\Delta})| \leq \frac{|\Sigma_i|_{\mathrm{HS}}}{n}\left(|\mathbb{E}[G^Z] - \check{G}^Z|_{\mathrm{HS}} + |\mathbb{E}[G_{-i}^Z - G^Z]|_{\mathrm{HS}}\right) = O\left(\frac{|\Sigma_i|_{\mathrm{HS}}}{n} J(\check{\Delta})\varepsilon\right).$$

The last estimate uses (3.32) and (3.15) in Lemma 3.2, together with $J(\check{\Delta}) \geq 1$. Taking the maximum over i proves (3.33). □

Lemma 3.6 (Interaction bound). *Assume that $|Z|_{\mathrm{op}} = O(1)$. If $D \in \mathcal{D}_n(\mathcal{C}_Z)$ satisfies $|D|_\infty = O(1)$, then*

$$J(D) = O(\sigma_\Sigma).$$

To bound the contribution of a commuting approximation to $J(D)$, we first prove the following span estimate.

Lemma 3.7 (Positive matrices and span dimension). *Let $C_1, \dots, C_n$ be Hermitian positive semidefinite matrices such that $\sum_i C_i \preceq I_p$. Then*

$$\sum_{i=1}^{n} |C_i|_{\mathrm{op}} \le \dim_{\mathbb{R}} \operatorname{span}\{C_1, \dots, C_n\}.$$

Proof. For each i, choose a unit vector v_i such that $v_i^* C_i v_i = |C_i|_{\mathrm{op}}$, and define the entrywise nonnegative matrix

$$B_{ij} := v_j^* C_i v_j.$$

Every column sum is at most one, so its spectral radius satisfies

$$\rho(B) \le \max_{j \in [n]} \sum_{i \in [n]} |B_{ij}| \le 1.$$

Set $r := \dim_{\mathbb{R}} \operatorname{span}\{C_1, \dots, C_n\}$. The rows of B are the images of the C_i under the real linear map $C \mapsto (v_j^* C v_j)_{j=1}^n$, hence $\operatorname{rank}(B) \le r$. Counting eigenvalues with algebraic multiplicity, at most $\operatorname{rank}(B)$ of them are nonzero. Therefore

$$\sum_{i=1}^{n} |C_i|_{\mathrm{op}} = \operatorname{Tr}(B) \le \sum_{\lambda \in \operatorname{Sp}(B)} |\lambda| \le \operatorname{rank}(B) \le r.$$

□

Proof of Lemma 3.6. Fix any pairwise commuting positive-semidefinite family $T_1, \dots, T_n$ over which the infimum in (1.6) is taken. Write

$$r := \dim_{\mathbb{R}} \operatorname{span}\{T_1, \dots, T_n\}, \qquad \varepsilon := \frac{1}{n} \sum_i |\Sigma_i - T_i|_{\mathrm{op}}.$$

We first estimate the interaction within this commuting family. Set

$$R := \left(I_p + \frac{1}{n} \sum_i D_i T_i \right)^{-1}, \qquad C := I_p + \frac{1}{n} \sum_i |D_i| T_i.$$

The supporting-phase inequality gives

$$\mathfrak{C}_K(R^{-1}) = \mathfrak{C}_K(1) I_p + \frac{1}{n} \sum_i \mathfrak{C}_K(D_i) T_i \succeq c_K C.$$

In particular, the inverse defining R exists. Multiplying the preceding inequality on the left and right by $C^{-1/2}$, then applying the inverse norm bound, gives

$$|C^{1/2} R C^{1/2}|_{\mathrm{op}} \le c_K^{-1}.$$

Because the T_i commute, the matrices R, C, and all T_i are simultaneously diagonalizable; in particular $C^{1/2} R C^{1/2} = CR$. Hence $|CR|_{\mathrm{op}} \le c_K^{-1}$. Moreover, the matrices $|D_i| T_i C^{-1}/n$ are positive semidefinite, sum to $I_p - C^{-1} \preceq I_p$, and have real span dimension at most r. The span bound in Lemma 3.7 therefore gives

$$\frac{1}{n} \sum_i |D_i| \, |T_i R|_{\mathrm{op}} \le \frac{1}{c_K} \sum_i \left| \frac{|D_i|}{n} T_i C^{-1} \right|_{\mathrm{op}} \le \frac{r}{c_K}. \tag{3.35}$$

To compare this interaction with that of the original profile, set $E = \frac{1}{n}\sum_i D_i(\Sigma_i - T_i)$. Then $|E|_{\mathrm{op}} = O(\varepsilon)$, and the definitions of R and G_D^Z give

$$G_D^Z = R + R(Z + I_p - E)G_D^Z = R\big[I_p + (Z + I_p - E)G_D^Z\big].$$

Splitting $\Sigma_i = T_i + (\Sigma_i - T_i)$ therefore gives

$$\Sigma_i G_D^Z = T_i R\big[I_p + (Z + I_p - E)G_D^Z\big] + (\Sigma_i - T_i)G_D^Z.$$

Using (3.35), $|G_D^Z|_{\mathrm{op}} = O(1)$, and $|Z|_{\mathrm{op}} = O(1)$, we obtain

$$J(D) \le 1 + O\big(r(1+\varepsilon) + \varepsilon\big) = O\big((1+r)(1+\varepsilon)\big).$$

The implicit constant is independent of the chosen commuting family $T_1, \dots, T_n$. Taking the infimum in (1.6) proves the assertion. □

Stability and comparison. If $|Z|_{\mathrm{op}} = O(1)$, the bound $|\check{\Delta}|_\infty = O(1)$ and Lemma 3.6 give $J(\check{\Delta}) = O(\sigma_\Sigma)$. Combining Proposition 3.5 with part (i) of Proposition 3.3 gives the following bias comparison.

Proposition 3.8 (Hilbert–Schmidt comparison). *Assume that $\sqrt{p}\,|\Sigma|_{\mathrm{HS}}^\infty = O(n)$, and let $Z \in \mathcal{K}$ satisfy $|Z|_{\mathrm{op}} = O(1)$. For every deterministic $A \in \mathcal{M}_p$,*

$$\left|\mathrm{Tr}\left(A(\mathbb{E}[G^Z] - \widetilde{G}^Z)\right)\right| = O\left(\sigma_\Sigma |A|_{\mathrm{HS}}\left[\frac{|\Sigma|_{\mathrm{HS}}^\infty}{n} + \min\left\{M_{\mathrm{HS}}^{(1)}, \frac{(M_{\mathrm{HS}}^{(2)})^2\sqrt{p}}{n}\right\}\right]\right).$$

Proof. Put

$$\varepsilon := \frac{|\Sigma|_{\mathrm{HS}}^\infty}{n} + \min\left\{M_{\mathrm{HS}}^{(1)}, \frac{(M_{\mathrm{HS}}^{(2)})^2\sqrt{p}}{n}\right\}.$$

Lemma 3.6 and Proposition 3.5 give

$$|\mathbb{E}[G^Z] - \check{G}^Z|_{\mathrm{HS}} = O\left(\sigma_\Sigma \varepsilon\right), \qquad |r(\check{\Delta})|_\infty = O\left(\frac{|\Sigma|_{\mathrm{HS}}^\infty}{n}\sigma_\Sigma\varepsilon\right).$$

Since $|\Sigma|_*^\infty \le \sqrt{p}\,|\Sigma|_{\mathrm{HS}}^\infty = O(n)$, the Hilbert–Schmidt stability estimate (3.20) in Proposition 3.3 applies and yields

$$|\check{G}^Z - \widetilde{G}^Z|_{\mathrm{HS}} = O\left(\min\{|\Sigma|_{\mathrm{HS}}^\infty, \sqrt{p}\}|r(\check{\Delta})|_\infty\right) = O\left(\frac{|\Sigma|_{\mathrm{HS}}^\infty \min\{|\Sigma|_{\mathrm{HS}}^\infty, \sqrt{p}\}}{n}\sigma_\Sigma\varepsilon\right) = O\left(\sigma_\Sigma\varepsilon\right).$$

Here $\min\{|\Sigma|_{\mathrm{HS}}^\infty, \sqrt{p}\} \le \sqrt{p}$, and (1.7) gives $\sqrt{p}\,|\Sigma|_{\mathrm{HS}}^\infty/n = O(1)$. Combining the two resolvent comparisons gives

$$|\mathbb{E}[G^Z] - \widetilde{G}^Z|_{\mathrm{HS}} = O\left(\sigma_\Sigma\varepsilon\right).$$

The trace bound follows by the Hilbert–Schmidt Cauchy–Schwarz inequality. □

Proof of Theorem 1.6. Fix $Z \in \mathcal{K}$ with $|Z|_{\mathrm{op}} = O(1)$. Under (1.7), Propositions 3.8 and 3.1, combined by the triangle inequality in L^q, give for every deterministic $A \in \mathcal{M}_p$ and every $q \ge 1$,

$$\begin{aligned}&\left\|\mathrm{Tr}\Big(A(G^Z - \widetilde{G}^Z)\Big)\right\|_{L^q}\\ &\quad = O\left(|A|_{\mathrm{HS}}\left[c_{n,q}M_{\mathrm{HS}}^{(q)} + \sigma_\Sigma\left(\frac{|\Sigma|_{\mathrm{HS}}^\infty}{n} + \min\left\{M_{\mathrm{HS}}^{(1)}, \frac{(M_{\mathrm{HS}}^{(2)})^2\sqrt{p}}{n}\right\}\right)\right]\right).\end{aligned} \tag{3.36}$$

Now specialize to $Z = zI_p$, so that $\widetilde{G}^{zI_p} = \tilde{G}^z$. The additional norm bound holds uniformly for $z \in K$ by compactness. For an arbitrary compact subset of Ω, apply the

estimate on the finitely many compact pieces described at the beginning of Section 3 and take the maximum of the constants. For $q \geq 2$, $c_{n,q} = n^{-1/2}$; taking the second entry in the minimum gives (1.9). For $q \geq 1$, taking the first entry and using $c_{n,q} \leq 1$, $M_{\mathrm{HS}}^{(1)} \leq M_{\mathrm{HS}}^{(q)}$, and $\sigma_\Sigma \geq 1$ gives (1.8). □

Proof of Corollary 1.13. Fix a sequence in Θ with $p \to \infty$ and a compact set $K \subset \Omega$. The finite-cover argument in Section 3 makes the scalar comparison uniform on K. Here $\Sigma_i = I_p$ for every i, so $|\Sigma|_{\mathrm{HS}}^{\infty} = \sqrt{p}$ and $\sigma_\Sigma \leq 2$. For every $L > 0$, apply (B.6) from Proposition B.7 with $\sigma = 1$. Taking the supremum over $|A|_{\mathrm{HS}} \leq 1$ and dividing by $\sqrt{p}$ gives

$$\frac{M_{\mathrm{HS}}^{(1)}}{\sqrt{p}} \leq \frac{\sqrt{2} + L}{\sqrt{p}} + 2\mathbb{E}\big[\xi_{p,n}^2 \mathbf{1}_{\{|\xi_{p,n}|>L\}}\big].$$

Taking the limsup along the sequence and then letting $L \to \infty$ gives $M_{\mathrm{HS}}^{(1)}/\sqrt{p} \to 0$ by uniform integrability of the squared entries. Since $p = O(n)$, the profile satisfies (1.7). The first-moment estimate of Theorem 1.6, with $A = I_p/p$, therefore yields

$$\sup_{z \in K} \|g(z) - \tilde{g}(z)\|_{L^1} = O\left(\frac{M_{\mathrm{HS}}^{(1)}}{\sqrt{p}} + \frac{1}{n}\right) \longrightarrow 0.$$

For the identity profile, the matrix fixed-point equation gives $\tilde{G}^z = \tilde{g}(z) I_p$ and

$$\tilde{g}(z) = \left(-z + \frac{1}{1 + (p/n)\tilde{g}(z)}\right)^{-1}.$$

Equivalently,

$$\frac{p}{n} z \tilde{g}(z)^2 + \left(z + \frac{p}{n} - 1\right) \tilde{g}(z) + 1 = 0.$$

The Stieltjes solution of this equation is the transform of $\mu_{\mathrm{MP},p/n}$. If $p/n \to c \in [0, \infty)$, continuity of the Marchenko–Pastur Stieltjes transform with respect to the aspect ratio gives

$$\tilde{g}(z) = \tilde{g}_{\mathrm{MP},p/n}(z) \longrightarrow \tilde{g}_{\mathrm{MP},c}(z), \qquad \Im z > 0,$$

where

$$\tilde{g}_{\mathrm{MP},c}(z) := \int_{\mathbb{R}_+} \frac{\mu_{\mathrm{MP},c}(d\lambda)}{\lambda - z}, \qquad z \in \Omega.$$

For $c = 0$, the limiting transform is $(1 - z)^{-1}$, corresponding to $\delta_1 = \mu_{\mathrm{MP},0}$.

The preceding L^1 comparison now gives $g(z) \to \tilde{g}_{\mathrm{MP},c}(z)$ in probability for every z with $\Im z > 0$. From any subsequence, diagonal extraction gives a further subsequence with almost-sure convergence simultaneously on a fixed countable dense subset of the upper half-plane. The bound

$$|g(z) - g(w)| \leq \frac{|z - w|}{\Im z \, \Im w}, \qquad \Im z, \Im w > 0,$$

valid for every probability Stieltjes transform, extends convergence on this event to the entire upper half-plane. The Stieltjes-transform continuity theorem [BS10, Theorem B.9] gives almost-sure weak convergence along that further subsequence. The subsequence characterization of convergence in probability proves the claimed weak convergence in probability. □

3.3 Operator-norm route

To display the trace dependence in the estimates below, write

$$\rho_\Sigma := 1 + \frac{|\Sigma|_*^{\infty}}{n}. \tag{3.37}$$

We first prove one-class comparisons for parameters in $\mathcal{K}$ with $|Z|_{\mathrm{op}} = O(1)$. In the classwise application, we normalize each background before applying these comparisons. We replace the classes in a fixed order, using a conditional average pivot followed by its expectation. The last step compares the resulting deterministic pivot with $\widetilde{\Delta}^Z$.

3.3.1 One-class case

For auxiliary models with N columns, the notation $Z \in \mathcal{K}$ and the uniformity convention are understood over their index set of pairs (p, N). The statements below hold uniformly for every $p, N \geq 1$ and every individual parameter $Z_p \in \mathcal{K}_p$. No upper bound on $|Z|_{\mathrm{op}}$ is assumed unless stated. For arbitrary independent real columns $x_1, \ldots, x_N$ with finite second moments $\Sigma_i = \mathbb{E}[x_i x_i^\top]$, write

$$G = \left(\frac{1}{N} \sum_i x_i x_i^\top - Z \right)^{-1}, \qquad \Delta_i = \left(1 + \frac{x_i^\top G_{-i} x_i}{N} \right)^{-1},$$

where G_{-i} omits column i while retaining normalization $1/N$. For an auxiliary family of N columns, the modulus and trace factor are computed from that family and its normalization $1/N$. Given $q \geq 1$:

$$M_{\mathrm{op}}^{(q)} := \max_{i \in [N]} \sup_{|B|_{\mathrm{op}} \leq 1} \|x_i^\top B x_i - \mathrm{Tr}(\Sigma_i B)\|_{L^q}, \qquad \rho_\Sigma := 1 + \frac{|\Sigma|_*^\infty}{N}.$$

The preceding concentration and conditional replacement arguments apply with N in place of n, using the moduli of this auxiliary family.

The conditional versions below are understood with respect to a sigma-field $\mathscr{F}$ such that Z is $\mathscr{F}$-measurable and the selected columns are conditionally independent given $\mathscr{F}$. Their profiles are then $\Sigma_i = \mathbb{E}[x_i x_i^\top \mid \mathscr{F}]$, and their quadratic-form moduli are defined as above using conditional L^q norms. If all stated sectorial, profile, and moment hypotheses, including any bound on $|Z|_{\mathrm{op}}$, hold almost surely with the same deterministic constants, all expectations and L^q norms in the definitions and conclusions are taken conditionally on $\mathscr{F}$. The common sector and phase remain deterministic and fixed under this conditioning.

Lemma 3.9 (Expected pivots and partial pivot sums). *Every convex average d of the numbers $(\mathbb{E}[\Delta_i])_{i \in [N]}$ belongs to $\mathcal{C}_Z \setminus \{0\}$ and satisfies*

$$|d| = O(1), \qquad |d|^{-1} = O(\rho_\Sigma).$$

Moreover, for every deterministic $I \subset [N]$ and $q \geq 1$,

$$\left\| \sum_{i \in I} (\Delta_i - \mathbb{E}\Delta_i) \right\|_{L^q} = O\Big(c_{N,q} M_{\mathrm{op}}^{(q)} \Big). \tag{3.38}$$

Proof. By (3.3) and the resolvent bounds (3.8), Jensen's inequality gives

$$\mathfrak{C}_K(\mathbb{E}\Delta_i) \geq c_K \mathbb{E}\left[|\Delta_i|\right] \geq \frac{c_K}{1 + \mathrm{Tr}(\Sigma_i)/\big(N c_K \eta_K\big)}.$$

Together with the uniform upper pivot bound from Lemma 2.3, this proves the first assertion, including convex averages.

For the second assertion, we will apply Theorem B.1 to $\sum_{i \in I} \Delta_i$, viewed as a function of the N independent columns. It is enough to show that replacing any one column x_j by an independent copy changes this sum by $O(M_{\mathrm{op}}^{(q)}/N)$ in L^q, uniformly in $j \in [N]$. The Schur identity (3.10), with $n = N$, gives

$$\sum_{i \in I} (1 - \Delta_i) = \frac{1}{N} \sum_{i \in I} x_i^\top G x_i.$$

For a fixed $j \in [N]$, Sherman–Morrison gives the exact equality

$$\frac{1}{N} \sum_{i \in I} x_i^\top G x_i - \frac{1}{N} \sum_{i \in I \setminus \{j\}} x_i^\top G_{-j} x_i = \frac{x_j^\top B_j x_j / N}{1 + x_j^\top G_{-j} x_j / N},$$

where both sums retain the normalization $1/N$ and

$$B_j := \mathbf{1}_{\{j\in I\}} G_{-j} - \frac{1}{N}\sum_{i\in I\setminus\{j\}} G_{-j}x_i x_i^\top G_{-j}.$$

The matrix B_j depends only on the other columns. The positive semidefinite partial sum $N^{-1}\sum_{i\in I\setminus\{j\}} x_i x_i^\top$ is bounded above by the full leave-one-out sum in the Loewner order. The supporting-phase bound and the two inverse identities in Lemma 2.3, followed by Cauchy–Schwarz, therefore give

$$|B_j|_{\mathrm{op}} = O(1), \qquad |\operatorname{Tr}(\Sigma_j B_j)| = O(\mathfrak{C}_{\bar{K}}(\operatorname{Tr}(\Sigma_j G_{-j}))).$$

The reciprocal bounds of Lemma 2.3 therefore imply

$$\frac{|\operatorname{Tr}(\Sigma_j B_j)|}{N|1+\operatorname{Tr}(\Sigma_j G_{-j})/N|} = O(1).$$

Conditionally on the other columns, B_j and G_{-j} are fixed matrices with uniformly bounded operator norms, so the definition of $M_{\mathrm{op}}^{(q)}$ bounds the conditional L^q norms of $x_j^\top B_j x_j - \operatorname{Tr}(\Sigma_j B_j)$ and $x_j^\top G_{-j} x_j - \operatorname{Tr}(\Sigma_j G_{-j})$ by $O(M_{\mathrm{op}}^{(q)})$. Equation (3.13), applied conditionally on the other columns, now bounds an independent-copy replacement of the sum in L^q by $O(M_{\mathrm{op}}^{(q)}/N)$. Applying Theorem B.1 to $\sum_{i\in I}\Delta_i$ therefore yields

$$\left\|\sum_{i\in I}(\Delta_i - \mathbb{E}\Delta_i)\right\|_{L^q} = O\left(N^{1/(q\wedge 2)}\frac{M_{\mathrm{op}}^{(q)}}{N}\right) = O\Big(c_{N,q}M_{\mathrm{op}}^{(q)}\Big).$$

All constants are uniform in I and require no upper bound on $|Z|_{\mathrm{op}}$. □

Now assume $\Sigma_i = \Sigma$ for every i, and set

$$\widehat{\delta} := \frac{1}{N}\sum_{i=1}^N \mathbb{E}[\Delta_i], \qquad \widehat{G} := (\widehat{\delta}\Sigma - Z)^{-1}.$$

Lemma 3.9 gives $\widehat{\delta}\in\mathcal{C}_Z\setminus\{0\}$, so Lemma 2.3 makes $\widehat{G}$ well defined. The average is needed even within one class: equality of second moments does not imply equality of the column laws. For this common profile, $\rho_\Sigma = 1+\operatorname{Tr}(\Sigma)/N$.

The following comparison gives a common estimate for the first- and second-moment cases.

Proposition 3.10 (One-class comparison in a bounded sectorial background). *Assume that $\Sigma_i = \Sigma$ for every $i\in[N]$, and let $Z\in\mathcal{K}$ satisfy $|Z|_{\mathrm{op}} = O(1)$. For every $q\in\{1,2\}$ such that $M_{\mathrm{op}}^{(q)} < \infty$,*

$$|\mathbb{E}G - \widehat{G}|_* = O\Big(\frac{\rho_\Sigma}{N^{q-1}}\big(\min\{1,|\Sigma|_{\mathrm{op}}\} + M_{\mathrm{op}}^{(q)}\big)^q\Big). \tag{3.39}$$

To control the profile in the second-moment estimate, we first prove the following spectral-spread bound.

Lemma 3.11 (Spectral spread from quadratic-form fluctuations). *Let $x\in\mathbb{R}^p$ have finite second moments. Then*

$$|\Sigma|_{\mathrm{HS}} \le |\Sigma|_{\mathrm{op}} + \sup_{|B|_{\mathrm{op}}\le 1}\big\|x^\top Bx - \operatorname{Tr}(\Sigma B)\big\|_{L^2}, \qquad \Sigma := \mathbb{E}[xx^\top].$$

Proof. The result is immediate if the supremum is infinite, and it is trivial for $p=1$. Otherwise, let $\lambda_1(\Sigma)\ge\cdots\ge\lambda_p(\Sigma)\ge 0$ be the eigenvalues of Σ, choose an associated orthonormal eigenbasis $u_1,\dots,u_p$, and write $\xi_j := u_j^\top x$. Thus

$$\mathbb{E}[\xi_j\xi_k] = \lambda_j(\Sigma)\mathbf{1}_{\{j=k\}}.$$

For $1 \le \ell \le \lfloor p/2 \rfloor$, put $a = 2\ell - 1$ and $b = 2\ell$. Consider the two centered quadratic forms

$$Y_- := \xi_a^2 - \xi_b^2 - (\lambda_a(\Sigma) - \lambda_b(\Sigma)), \qquad Y_+ := 2\xi_a \xi_b,$$

associated respectively with the test matrices $B_- := u_a u_a^\top - u_b u_b^\top$ and $B_+ := u_a u_b^\top + u_b u_a^\top$. Their squared L^2 norms satisfy

$$\begin{aligned} \|Y_-\|_{L^2}^2 + \|Y_+\|_{L^2}^2 &= \mathbb{E}\left[(\xi_a^2 + \xi_b^2)^2\right] - (\lambda_a(\Sigma) - \lambda_b(\Sigma))^2 \\ &\ge (\lambda_a(\Sigma) + \lambda_b(\Sigma))^2 - (\lambda_a(\Sigma) - \lambda_b(\Sigma))^2 = 4\lambda_a(\Sigma)\lambda_b(\Sigma). \end{aligned}$$

Choose $B_\ell \in \{B_-, -B_-, B_+, -B_+\}$ whose centered quadratic form $Y_\ell := x^\top B_\ell x - \mathrm{Tr}(\Sigma B_\ell)$ has the largest L^2 norm, with its sign chosen so that

$$\mathbb{E}\left[Y_\ell \sum_{k<\ell} Y_k \right] \ge 0.$$

Set $B := \sum_{\ell=1}^{\lfloor p/2 \rfloor} B_\ell$. The matrices B_ℓ act on mutually orthogonal two-dimensional subspaces and have operator norm one, so $|B|_{\mathrm{op}} = 1$. Our choice of signs also gives

$$\begin{aligned} \left\| x^\top B x - \mathrm{Tr}(\Sigma B) \right\|_{L^2}^2 &= \sum_{\ell=1}^{\lfloor p/2 \rfloor} \|Y_\ell\|_{L^2}^2 + 2 \sum_{\ell=1}^{\lfloor p/2 \rfloor} \mathbb{E}\left[Y_\ell \sum_{k<\ell} Y_k \right] \ge \sum_{\ell=1}^{\lfloor p/2 \rfloor} \|Y_\ell\|_{L^2}^2 \\ &\ge 2 \sum_{\ell=1}^{\lfloor p/2 \rfloor} \lambda_{2\ell-1}(\Sigma)\lambda_{2\ell}(\Sigma) \ge 2 \sum_{\ell=1}^{\lfloor p/2 \rfloor} \lambda_{2\ell}(\Sigma)^2 \ge \sum_{j=2}^{p} \lambda_j(\Sigma)^2. \end{aligned} \tag{3.40}$$

The result follows from

$$|\Sigma|_{\mathrm{HS}} \le |\lambda_1(\Sigma) u_1 u_1^\top|_{\mathrm{HS}} + |\Sigma - \lambda_1(\Sigma) u_1 u_1^\top|_{\mathrm{HS}} \le |\Sigma|_{\mathrm{op}} + \sqrt{\sum_{j=2}^{p} \lambda_j(\Sigma)^2}.$$

□

Remark 3.12 (Second-moment size and quadratic-form fluctuations). The centered quadratic-form modulus does not control the size of the second-moment matrices. Indeed, if $x_i = \varepsilon_i v_i$, where $v_i \in \mathbb{R}^p$ is deterministic and ε_i is a Rademacher variable, then $\Sigma_i = v_i v_i^\top$ and

$$x_i^\top B x_i - \mathrm{Tr}(\Sigma_i B) = 0$$

almost surely for every B, whereas $|\Sigma_i|_{\mathrm{HS}} = |v_i|^2$ may be arbitrarily large. This shows that the largest eigenvalue cannot be included in the spectral sum in (3.40); profile sizes must therefore remain separate from the quadratic-form moduli.

Proof of Proposition 3.10. If $\Sigma = 0$, every column vanishes almost surely, and the conclusion is immediate. Assume henceforth that $\Sigma \neq 0$. All estimates involving $M_{\mathrm{op}}^{(2)}$ below are used only when this quantity is finite. Fix a deterministic matrix A with $|A|_{\mathrm{op}} \le 1$, and set

$$\delta := \frac{1}{N} \sum_{i=1}^{N} \Delta_i, \qquad s_i := x_i^\top \widehat{G} A G_{-i} x_i.$$

Let $\mathbb{E}_i$ denote expectation conditional on all columns except x_i. Since $\mathbb{E}_i s_i = \mathrm{Tr}(\Sigma \widehat{G} A G_{-i})$

and $\mathbb{E}[\delta - \widehat{\delta}] = 0$, the resolvent identity and $Gx_i = \Delta_i G_{-i} x_i$ give

$$\begin{aligned}\operatorname{Tr}\Big(A(\mathbb{E}G - \widehat{G})\Big) &= \widehat{\delta}\,\operatorname{Tr}\Big(\Sigma\widehat{G}A\,\mathbb{E}G\Big) - \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}[\Delta_i s_i] \\ &= \mathbb{E}\Big[\delta\,\operatorname{Tr}(\Sigma\widehat{G}AG)\Big] - \mathbb{E}\Big[(\delta-\widehat{\delta})\operatorname{Tr}\Big(\Sigma\widehat{G}A(G-\mathbb{E}G)\Big)\Big] \\ &\quad - \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\Big[\Delta_i\operatorname{Tr}(\Sigma\widehat{G}AG_{-i})\Big] - \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}[(\Delta_i - \mathbb{E}_i\Delta_i)(s_i - \mathbb{E}_i s_i)] \\ &= a + b + c, \end{aligned} \tag{3.41}$$

where

- $a := -\mathbb{E}\Big[(\delta-\widehat{\delta})\operatorname{Tr}\Big(\Sigma\widehat{G}A(G-\mathbb{E}G)\Big)\Big]$;
- $b := -\dfrac{1}{N}\displaystyle\sum_{i=1}^{N}\mathbb{E}[(\Delta_i - \mathbb{E}_i\Delta_i)(s_i - \mathbb{E}_i s_i)]$;
- $c := \dfrac{1}{N}\displaystyle\sum_{i=1}^{N}\mathbb{E}\Big[\Delta_i\operatorname{Tr}\Big(\Sigma\widehat{G}A(G-G_{-i})\Big)\Big]$.

We first bound $|\Sigma\widehat{G}|_{\mathrm{op}}$. The identity $\widehat{\delta}\Sigma\widehat{G} = I_p + Z\widehat{G}$ together with $|\widehat{\delta}|^{-1} = O(\rho_\Sigma)$, $|Z|_{\mathrm{op}} = O(1)$, and the uniform resolvent bounds, gives

$$|\Sigma\widehat{G}|_{\mathrm{op}} = O(\min\{|\Sigma|_{\mathrm{op}}, \rho_\Sigma\})\,. \tag{3.42}$$

We now bound a, b, and c in turn.

1. *Bound on a.* The uniform pivot bounds and Lemma 3.9 give

$$|\delta - \widehat{\delta}| = O(1), \qquad \|\delta - \widehat{\delta}\|_{L^2} = O\left(\frac{M_{\mathrm{op}}^{(2)}}{N\sqrt{N}}\right).$$

For $q \in \{1,2\}$ with $M_{\mathrm{op}}^{(q)} < \infty$, Proposition 3.1 gives

$$\Big\|\operatorname{Tr}\Big(\Sigma\widehat{G}A(G-\mathbb{E}G)\Big)\Big\|_{L^q} = O\Big(|\Sigma\widehat{G}|_{\mathrm{op}} N^{1/q-1} M_{\mathrm{op}}^{(q)}\Big)\,.$$

Using the boundedness of $\delta - \widehat{\delta}$ for the first estimate and Cauchy–Schwarz for the second, we obtain

$$|a| = O\Big(\min\{|\Sigma|_{\mathrm{op}}, \rho_\Sigma\} M_{\mathrm{op}}^{(1)}\Big) \qquad \text{and} \qquad |a| = O\left(\frac{\min\{|\Sigma|_{\mathrm{op}}, \rho_\Sigma\}(M_{\mathrm{op}}^{(2)})^2}{N^2}\right).$$

2. *Bound on b.* The uniform pivot bounds and the reciprocal Lipschitz estimate give

$$|\Delta_i - \mathbb{E}_i\Delta_i| = O(1), \qquad \|\Delta_i - \mathbb{E}_i\Delta_i\|_{L^2} = O\left(\frac{M_{\mathrm{op}}^{(2)}}{N}\right).$$

Since $\widehat{G}AG_{-i}$ is independent of x_i and has uniformly bounded operator norm, conditional quadratic-form control gives

$$\|s_i - \mathbb{E}_i s_i\|_{L^q} = O(M_{\mathrm{op}}^{(q)}), \qquad q \in \{1,2\}, \quad M_{\mathrm{op}}^{(q)} < \infty.$$

Therefore

$$|b| = O(M_{\mathrm{op}}^{(1)}) \qquad \text{and} \qquad |b| = O\left(\frac{(M_{\mathrm{op}}^{(2)})^2}{N}\right).$$

3. *Bound on c.* Set $T := G + GZG = N^{-1}GXX^\top G$. The Sherman–Morrison identities give

$$\sum_{i=1}^{N} \Delta_i (G - G_{-i}) = -\frac{1}{N} \sum_{i=1}^{N} \Delta_i^2 G_{-i} x_i x_i^\top G_{-i} = -T.$$

Therefore $c = -\frac{1}{N} \operatorname{Tr}\Big(\Sigma \widehat{G} A \, \mathbb{E}T\Big)$. The uniform resolvent bounds and $|Z|_{\mathrm{op}} = O(1)$ imply $|T|_{\mathrm{op}} = O(1)$. Since $\operatorname{rank}(T) \le N$, $|T|_* = O(N)$, and hence

$$|c| = O(|\Sigma \widehat{G}|_{\mathrm{op}}) = O(\min\{|\Sigma|_{\mathrm{op}}, \rho_\Sigma\}) \,.$$

For the second-moment bound, apply Lemma B.4 conditionally on the other columns. Together with the uniform pivot and resolvent bounds, this gives

$$|c| \le \frac{O(1)}{N^2} \sum_{i=1}^{N} \mathbb{E}\left[\left| x_i^\top G_{-i} \Sigma \widehat{G} A G_{-i} x_i \right|\right] = O\left(\frac{|\Sigma|_{\mathrm{HS}}^2}{N}\right) = O\left(\frac{|\Sigma|_{\mathrm{op}}^2 + (M_{\mathrm{op}}^{(2)})^2}{N}\right), \quad (3.43)$$

where the last bound follows from Lemma 3.11.

To refine this bound, choose a unit eigenvector v associated with the largest eigenvalue of Σ, set $P := vv^\top$, and write $\Sigma = |\Sigma|_{\mathrm{op}} P + \Sigma_0$. Set

$$G_0 := \left(\frac{1}{N}(I_p - P)XX^\top (I_p - P) - Z\right)^{-1}, \qquad T_0 := G_0 + G_0 Z G_0.$$

The common second-moment matrix of the projected columns $(I_p - P)x_i$ is Σ_0. Applying the same conditional quadratic-form estimate to these columns gives

$$\left|\operatorname{Tr}\Big(\Sigma_0 \widehat{G} A \, \mathbb{E}T_0\Big)\right| = O\big(|\Sigma_0|_{\mathrm{HS}}^2\big) = O\Big((M_{\mathrm{op}}^{(2)})^2\Big),$$

thanks to Lemma 3.11. Moreover,

$$\operatorname{rank}\big(XX^\top - (I_p - P)XX^\top (I_p - P)\big) \le 2.$$

The resolvent identity and the uniform operator-norm bounds therefore imply $|G - G_0|_* = O(1)$. Since

$$T - T_0 = (G - G_0) + (G - G_0)ZG + G_0 Z(G - G_0),$$

we also have $|T - T_0|_* = O(1)$. Finally, $P\Sigma\widehat{G}$ has rank at most one and $\Sigma_0 = (I_p - P)\Sigma$. Splitting $\Sigma = P\Sigma + \Sigma_0$, and then replacing T by $T_0 + (T - T_0)$ in the Σ_0 term, gives

$$\begin{aligned} N|c| &\le |P\Sigma\widehat{G}|_* \, |\mathbb{E}T|_{\mathrm{op}} + \left|\operatorname{Tr}\Big(\Sigma_0 \widehat{G} A \, \mathbb{E}T_0\Big)\right| + |\Sigma_0 \widehat{G}|_{\mathrm{op}} \, \mathbb{E}|T - T_0|_* \\ &\le O(|\Sigma\widehat{G}|_{\mathrm{op}}) + O\Big((M_{\mathrm{op}}^{(2)})^2\Big) + O(|\Sigma\widehat{G}|_{\mathrm{op}}) = O\Big(\min\{|\Sigma|_{\mathrm{op}}, \rho_\Sigma\} + (M_{\mathrm{op}}^{(2)})^2\Big) \,. \end{aligned}$$

Combining this with (3.43), we obtain

$$|c| = O\left(\frac{\min\{|\Sigma|_{\mathrm{op}}^2, |\Sigma|_{\mathrm{op}}, \rho_\Sigma\} + (M_{\mathrm{op}}^{(2)})^2}{N}\right). \quad (3.44)$$

For $q = 1$, the preceding estimates and $\rho_\Sigma \ge 1$ give

$$\begin{aligned} |a| + |b| + |c| &= O\Big(\min\{|\Sigma|_{\mathrm{op}}, \rho_\Sigma\} + \big(1 + \min\{|\Sigma|_{\mathrm{op}}, \rho_\Sigma\}\big) M_{\mathrm{op}}^{(1)}\Big) \\ &= O\Big(\rho_\Sigma \big(\min\{1, |\Sigma|_{\mathrm{op}}\} + M_{\mathrm{op}}^{(1)}\big)\Big) \,. \end{aligned}$$

For $q = 2$, use (3.44) and

$$\min\{|\Sigma|_{\mathrm{op}}^2, |\Sigma|_{\mathrm{op}}, \rho_\Sigma\} \le \rho_\Sigma \min\{1, |\Sigma|_{\mathrm{op}}^2\}, \qquad 1 + \frac{\min\{|\Sigma|_{\mathrm{op}}, \rho_\Sigma\}}{N} \le 2\rho_\Sigma.$$

It follows that

$$|a| + |b| + |c| = O\Big(\frac{\rho_\Sigma}{N}\big(\min\{1, |\Sigma|_{\mathrm{op}}^2\} + (M_{\mathrm{op}}^{(2)})^2\big)\Big),$$

which implies (3.39) for $q = 2$. All bounds are uniform over deterministic A with $|A|_{\mathrm{op}} \le 1$, so nuclear duality completes the proof. □

3.3.2 Multiple classes

Return to the original n-column family and the classes $I_1, \dots, I_k$ introduced in the main-results section. Thus

$$\mathbb{E}[x_i x_i^\top] = \Sigma^{(\ell)} \quad \text{for } i \in I_\ell, \qquad n_\ell := |I_\ell|, \qquad \pi_\ell := \frac{n_\ell}{n}, \qquad S_\ell := \frac{1}{n}\sum_{i \in I_\ell} x_i x_i^\top.$$

Fix $Z \in \mathcal{K}$ and a deterministic ordering $I_1, \dots, I_k$ of the classes. Put

$$\mathcal{F}^\ell := \sigma(x_i : i \in I_{\ell+1} \cup \cdots \cup I_k), \qquad Z^{(0)} := Z, \qquad G^{(0)} := G^Z.$$

At each stage, we first average the Schur pivots conditionally on the unreplaced classes. We then take the expectation of this conditional average and retain the resulting deterministic coefficient at subsequent stages. Given $Z^{(\ell-1)}$, define, for $i \in I_\ell$,

$$G_{-i}^{(\ell-1)} := \left(\sum_{b \ge \ell} S_b - \frac{x_i x_i^\top}{n} - Z^{(\ell-1)}\right)^{-1},$$

and

$$\widehat{\delta}_{\mathcal{F}^\ell}^{(\ell)} := \frac{1}{n_\ell}\sum_{i \in I_\ell} \mathbb{E}\left[\left(1 + \frac{x_i^\top G_{-i}^{(\ell-1)} x_i}{n}\right)^{-1} \middle| \mathcal{F}^\ell\right], \qquad \widehat{\delta}^{(\ell)} := \mathbb{E}\big[\widehat{\delta}_{\mathcal{F}^\ell}^{(\ell)}\big]. \tag{3.45}$$

Define the conditional and deterministic parameters by

$$Z_{\mathcal{F}^\ell}^{(\ell)} := Z - \sum_{b<\ell} \pi_b \widehat{\delta}^{(b)} \Sigma^{(b)} - \pi_\ell \widehat{\delta}_{\mathcal{F}^\ell}^{(\ell)} \Sigma^{(\ell)}, \qquad Z^{(\ell)} = Z - \sum_{b \le \ell} \pi_b \widehat{\delta}^{(b)} \Sigma^{(b)}. \tag{3.46}$$

The corresponding resolvents are

$$G_{\mathcal{F}^\ell}^{(\ell)} := \left(\sum_{b>\ell} S_b - Z_{\mathcal{F}^\ell}^{(\ell)}\right)^{-1}, \qquad G^{(\ell)} := \left(\sum_{b>\ell} S_b - Z^{(\ell)}\right)^{-1}. \tag{3.47}$$

The parameters $Z^{(\ell)}$ are deterministic. Both $G_{\mathcal{F}^\ell}^{(\ell)}$ and $G^{(\ell)}$ are $\mathcal{F}^\ell$-measurable and may depend on the unreplaced classes. In particular:

$$G^{(k)} = (-Z^{(k)})^{-1} = G_{\widehat{\Delta}}^{Z} \qquad \text{where} \quad \widehat{\Delta}_i := \widehat{\delta}^{(a)} \qquad \text{for } i \in I_a.$$

Lemma 3.13 (Iterated pivots). *For every $\ell \in [k]$, $Z^{(\ell)} \in \mathcal{K}_p$ and $Z_{\mathcal{F}^\ell}^{(\ell)} \in \mathcal{K}_p$. All stage resolvents are well defined. Moreover, $\widehat{\delta}_{\mathcal{F}^\ell}^{(\ell)}, \widehat{\delta}^{(\ell)} \in \mathcal{C}_Z \setminus \{0\}$, and*

$$\Big|\widehat{\delta}_{\mathcal{F}^\ell}^{(\ell)}\Big| + \Big|\widehat{\delta}^{(\ell)}\Big| = O(1), \qquad \Big|\widehat{\delta}_{\mathcal{F}^\ell}^{(\ell)}\Big|^{-1} + \Big|\widehat{\delta}^{(\ell)}\Big|^{-1} = O(\rho_\Sigma).$$

In addition,

$$n_\ell \left\| \widehat{\delta}^{(\ell)}_{\mathcal{F}^\ell} - \widehat{\delta}^{(\ell)} \right\|_{L^1} = O\Big(\pi_\ell \tau_\Sigma M^{(1)}_{\mathrm{op}} \Big), \tag{3.48}$$

$$n_\ell \left\| \widehat{\delta}^{(\ell)}_{\mathcal{F}^\ell} - \widehat{\delta}^{(\ell)} \right\|_{L^2} = O\left(\pi_\ell \left[\frac{\tau_\Sigma M^{(2)}_{\mathrm{op}}}{\sqrt{n}} + \frac{(M^{(2)}_{\mathrm{op}})^2}{n} \right] \right). \tag{3.49}$$

If $Z = Z^* \prec 0$, all stage resolvents are positive definite and all stage pivots are positive.

Proof. Starting from $Z^{(0)} = Z$, proceed by induction with the fixed common phase. If $Y = Z - \sum_j d_j S_j$, where $d_j \in \mathcal{C}_Z$ and $S_j \succeq 0$, then $\mathcal{C}_Y \subseteq \mathcal{C}_Z \subseteq \mathcal{C}_K$ and

$$\mathfrak{C}_K(-Y) = \mathfrak{C}_K(-Z) + \sum_j \mathfrak{C}_K(d_j) S_j \succeq c_K \left(\eta_K I_p + \sum_j |d_j| S_j \right).$$

Consequently, $Y \in \mathcal{K}_p$. This applies to the retained pivot terms and to the unreplaced sample blocks, for which $d_j = 1$. At stage ℓ, condition on $\mathcal{F}^\ell$ and apply Lemmas 2.3 and 3.9 with $N = n$, zero columns outside I_ℓ, and parameter $Z^{(\ell-1)} - \sum_{b>\ell} S_b$. This gives the pivot inclusions and bounds and ensures that the stage resolvents are well defined. The definitions (3.46) and convexity of $\mathcal{C}_Z$ give the sector inclusions and preserve membership in $\mathcal{K}_p$ for $Z^{(\ell)}$ and $Z^{(\ell)}_{\mathcal{F}^\ell}$; positivity is preserved when $Z = Z^* \prec 0$.

It remains to prove (3.48) and (3.49). Set

$$\Delta_{i,\ell} := \left(1 + \frac{x_i^\top G^{(\ell-1)}_{-i} x_i}{n} \right)^{-1}, \qquad i \in I_\ell.$$

Since

$$n_\ell \widehat{\delta}^{(\ell)}_{\mathcal{F}^\ell} = \mathbb{E}\left[\sum_{i \in I_\ell} \Delta_{i,\ell} \middle| \mathcal{F}^\ell \right], \qquad n_\ell \widehat{\delta}^{(\ell)} = \mathbb{E}\left[\sum_{i \in I_\ell} \Delta_{i,\ell} \right],$$

apply (3.38) with $N = n$, $I = I_\ell$, parameter $Z^{(\ell-1)}$, and zero columns in the already replaced classes. Conditional Jensen then gives

$$n_\ell \left\| \widehat{\delta}^{(\ell)}_{\mathcal{F}^\ell} - \widehat{\delta}^{(\ell)} \right\|_{L^q} = O\Big(c_{n,q} M^{(q)}_{\mathrm{op}} \Big), \qquad q = 1, 2. \tag{3.50}$$

To obtain a bound involving $|\Sigma^{(\ell)}|_{\mathrm{op}}$, define

$$\widehat{\delta}^{(\ell)}_\Sigma := \frac{1}{n_\ell} \sum_{i \in I_\ell} \mathbb{E}\left[\left(1 + \frac{1}{n} \operatorname{Tr}(\Sigma^{(\ell)} G^{(\ell-1)}_{-i}) \right)^{-1} \middle| \mathcal{F}^\ell \right].$$

This quantity remains $\mathcal{F}^\ell$-measurable: only the quadratic form inside the reciprocal has been replaced by its conditional mean. Apply both estimates of Lemma B.2 with $s = 1$ and $t = x_i^\top G^{(\ell-1)}_{-i} x_i$, conditionally on all columns other than x_i. The conditional mean of t is $\operatorname{Tr}(\Sigma^{(\ell)} G^{(\ell-1)}_{-i})$, and Lemma 2.3 bounds the reciprocal denominators. Conditional quadratic-form control from Lemma B.3, after taking expectations given $\mathcal{F}^\ell$ and averaging over $i \in I_\ell$, gives, uniformly in the conditional background,

$$\left| \widehat{\delta}^{(\ell)}_{\mathcal{F}^\ell} - \widehat{\delta}^{(\ell)}_\Sigma \right| = O\left(\min\left\{ \frac{M^{(1)}_{\mathrm{op}}}{n}, \frac{(M^{(2)}_{\mathrm{op}})^2}{n^2} \right\} \right).$$

Moreover, combine the reciprocal Lipschitz bound (B.3) in the proof of Lemma B.2 with Proposition 3.1 applied to $\operatorname{Tr}(\Sigma^{(\ell)} G^{(\ell-1)}_{-i})$ in the same auxiliary n-column model. Conditional Jensen then gives

$$\left\| \widehat{\delta}^{(\ell)}_\Sigma - \mathbb{E}[\widehat{\delta}^{(\ell)}_\Sigma] \right\|_{L^q} = O\left(\frac{c_{n,q} |\Sigma^{(\ell)}|_{\mathrm{op}} M^{(q)}_{\mathrm{op}}}{n} \right), \qquad q = 1, 2.$$

Centering $\widehat{\delta}^{(\ell)}_{\mathcal{F}^\ell} - \widehat{\delta}^{(\ell)}_{\Sigma}$ and combining the two preceding bounds by the triangle inequality, with $n_\ell = n\pi_\ell$, gives

$$n_\ell \left\| \widehat{\delta}^{(\ell)}_{\mathcal{F}^\ell} - \widehat{\delta}^{(\ell)} \right\|_{L^1} = O\Big(\pi_\ell(1 + |\Sigma^{(\ell)}|_{\mathrm{op}}) M^{(1)}_{\mathrm{op}}\Big),$$
$$n_\ell \left\| \widehat{\delta}^{(\ell)}_{\mathcal{F}^\ell} - \widehat{\delta}^{(\ell)} \right\|_{L^2} = O\left(\pi_\ell |\Sigma^{(\ell)}|_{\mathrm{op}} \frac{M^{(2)}_{\mathrm{op}}}{\sqrt{n}} + \pi_\ell \frac{(M^{(2)}_{\mathrm{op}})^2}{n}\right).$$

Taking the better of these estimates and (3.50), and using

$$\min\{1, \pi_\ell(1 + |\Sigma^{(\ell)}|_{\mathrm{op}})\} \le 2\pi_\ell\tau_\Sigma, \qquad \min\{1, \pi_\ell|\Sigma^{(\ell)}|_{\mathrm{op}}\} \le 2\pi_\ell\tau_\Sigma,$$

proves (3.48) and (3.49). □

Throughout the three results below, set

$$\varepsilon := \min\left\{\tau_\Sigma\big(1 + M^{(1)}_{\mathrm{op}}\big), \frac{\tau_\Sigma M^{(2)}_{\mathrm{op}}}{\sqrt{n}} + \frac{\tau_\Sigma^2 + (M^{(2)}_{\mathrm{op}})^2}{n}\right\}. \tag{3.51}$$

The second entry is interpreted as $+\infty$ when $M^{(2)}_{\mathrm{op}} = \infty$. In these three results, write

$$\widehat{G} := G^Z_{\widehat{\Delta}} = G^{(k)}.$$

Proposition 3.14. *Let $Z \in \mathcal{K}$ satisfy $|Z|_{\mathrm{op}} = O(1)$. Then*

$$|\mathbb{E}[G^Z] - \widehat{G}|_* = O(\rho_\Sigma\varepsilon). \tag{3.52}$$

Proof. The proof relies on the telescoping decomposition:

$$\widehat{G} - \mathbb{E}[G^Z] = \sum_{\ell=1}^{k} \mathbb{E}[R_\ell] \qquad \text{with} \quad R_\ell := G^{(\ell)} - \mathbb{E}[G^{(\ell-1)} \mid \mathcal{F}^\ell]. \tag{3.53}$$

For each $\ell \in [k]$, $G^{(\ell)}$ and $G^{(\ell-1)}$ both share the $\mathcal{F}^\ell$-measurable parameter matrix

$$B_\ell := \sum_{b>\ell} S_b - Z^{(\ell-1)}.$$

Indeed, the definitions of the stage parameters (3.46) and resolvents (3.47) give

$$G^{(\ell-1)} = (S_\ell + B_\ell)^{-1}, \qquad G^{(\ell)} = \big(\pi_\ell\widehat{\delta}^{(\ell)}\Sigma^{(\ell)} + B_\ell\big)^{-1}. \tag{3.54}$$

The background B_ℓ need not be uniformly bounded. Set

$$U_\ell := \left(I_p + \sum_{b>\ell} S_b + \sum_{b<\ell} \pi_b|\widehat{\delta}^{(b)}|\Sigma^{(b)}\right)^{-1/2}.$$

We will prove, uniformly in ℓ, that $\mathbb{E}\left|U_\ell^{-1} R_\ell U_\ell^{-1}\right|_* = O(\rho_\Sigma\pi_\ell\varepsilon)$, and then sum over the classes.

Fix $\ell \in [k]$ and condition on $\mathcal{F}^\ell$. The matrix U_ℓ is real symmetric, $\mathcal{F}^\ell$-measurable, and satisfies $|U_\ell|_{\mathrm{op}} \le 1$. First note that:

$$\sum_{b>\ell} U_\ell S_b U_\ell + \sum_{b<\ell} \pi_b|\widehat{\delta}^{(b)}|U_\ell\Sigma^{(b)}U_\ell = I_p - U_\ell^2 \preceq I_p.$$

Hence, by Cauchy–Schwarz,

$$\sup_{|\theta_b|\le 1}\left|\sum_{b>\ell}\theta_b U_\ell S_b U_\ell+\sum_{b<\ell}\theta_b\pi_b\widehat{\delta}^{(b)}U_\ell\Sigma^{(b)}U_\ell\right|_{\mathrm{op}}\le 1. \tag{3.55}$$

In particular,

$$|U_\ell B_\ell U_\ell|_{\mathrm{op}}\le 1+|Z|_{\mathrm{op}}=O(1).$$

Moreover, $-U_\ell B_\ell U_\ell\in\mathcal{K}_p$ and therefore satisfies the conditional one-class comparison with uniform deterministic constants. Indeed $\mathcal{C}_{-U_\ell B_\ell U_\ell}\subseteq\mathcal{C}_Z\subseteq\mathcal{C}_K$ and

$$\begin{aligned}\mathfrak{C}_K(U_\ell B_\ell U_\ell)=U_\ell\mathfrak{C}_K(B_\ell)U_\ell&\succeq c_K U_\ell\left(\sum_{b>\ell}S_b+\sum_{b<\ell}\pi_b|\widehat{\delta}^{(b)}|\Sigma^{(b)}+\eta_K I_p\right)U_\ell\\&=c_K\big[I_p-(1-\eta_K)U_\ell^2\big]\succeq c_K\eta_K I_p.\end{aligned} \tag{3.56}$$

The last inequality uses $0<\eta_K\le 1$, as defined in (3.1).

Apply Proposition 3.10 conditionally to

$$y_i:=\sqrt{\pi_\ell}\,U_\ell x_i,\qquad i\in I_\ell,$$

with $N=n_\ell$ and parameter $-U_\ell B_\ell U_\ell$. With normalization $1/n_\ell$, these columns give the sample matrix $U_\ell S_\ell U_\ell$ and the same conditional average pivot $\widehat{\delta}^{(\ell)}_{\mathcal{F}^\ell}$. The random and comparison resolvents are, respectively,

$$U_\ell^{-1}G^{(\ell-1)}U_\ell^{-1},\qquad U_\ell^{-1}G^{(\ell)}_{\mathcal{F}^\ell}U_\ell^{-1}.$$

For these columns, the common conditional second moment, quadratic-form moduli, and trace factor satisfy $|\pi_\ell U_\ell\Sigma^{(\ell)}U_\ell|_{\mathrm{op}}\le\pi_\ell|\Sigma^{(\ell)}|_{\mathrm{op}}$, and

$$M^{(q)}_{\mathrm{op}}(y_i:i\in I_\ell\mid\mathcal{F}^\ell)\le\pi_\ell M^{(q)}_{\mathrm{op}},\qquad 1+\frac{\mathrm{Tr}(\pi_\ell U_\ell\Sigma^{(\ell)}U_\ell)}{n_\ell}\le\rho_\Sigma.$$

Using $\min\{1,\pi_\ell|\Sigma^{(\ell)}|_{\mathrm{op}}\}\le 2\pi_\ell\tau_\Sigma$ and $n_\ell=\pi_\ell n$, the unified estimate (3.39) gives, for $q\in\{1,2\}$ such that $M^{(q)}_{\mathrm{op}}<\infty$,

$$\left|U_\ell^{-1}\left(\mathbb{E}[G^{(\ell-1)}\mid\mathcal{F}^\ell]-G^{(\ell)}_{\mathcal{F}^\ell}\right)U_\ell^{-1}\right|_*=O\Big(\frac{\rho_\Sigma\pi_\ell}{n^{q-1}}\big(\tau_\Sigma+M^{(q)}_{\mathrm{op}}\big)^q\Big).$$

Since $\tau_\Sigma\ge 1$, the $q=1$ bound is $O(\rho_\Sigma\pi_\ell\tau_\Sigma(1+M^{(1)}_{\mathrm{op}}))$. When $M^{(2)}_{\mathrm{op}}<\infty$, the $q=2$ bound is $O(\rho_\Sigma\pi_\ell(\tau_\Sigma^2+(M^{(2)}_{\mathrm{op}})^2)/n)$. Taking the better of the available estimates therefore gives

$$\left|U_\ell^{-1}\left(\mathbb{E}[G^{(\ell-1)}\mid\mathcal{F}^\ell]-G^{(\ell)}_{\mathcal{F}^\ell}\right)U_\ell^{-1}\right|_*=O(\rho_\Sigma\pi_\ell\varepsilon) \tag{3.57}$$

almost surely, with uniform deterministic constants.

The resolvent identity gives

$$G^{(\ell)}_{\mathcal{F}^\ell}-G^{(\ell)}=\pi_\ell\big(\widehat{\delta}^{(\ell)}-\widehat{\delta}^{(\ell)}_{\mathcal{F}^\ell}\big)G^{(\ell)}_{\mathcal{F}^\ell}\Sigma^{(\ell)}G^{(\ell)}.$$

The uniform bounds from Lemma 2.3, applied with the common phase to the normalized resolvents, imply, pointwise,

$$\left|U_\ell^{-1}\big(G^{(\ell)}_{\mathcal{F}^\ell}-G^{(\ell)}\big)U_\ell^{-1}\right|_*\le C(\rho_\Sigma-1)n_\ell\left|\widehat{\delta}^{(\ell)}_{\mathcal{F}^\ell}-\widehat{\delta}^{(\ell)}\right|.$$

Taking expectations and applying (3.48) or (3.49) from Lemma 3.13, with $\|\cdot\|_{L^1}\le\|\cdot\|_{L^2}$ for the second-moment bound, yields finally

$$\mathbb{E}\left|U_\ell^{-1}R_\ell U_\ell^{-1}\right|_*=O(\rho_\Sigma\pi_\ell\varepsilon)\,. \tag{3.58}$$

One can then conclude with (3.53) since $|U_\ell|_{\mathrm{op}}\le 1$ and $\sum_\ell\pi_\ell=1$. □

For stability, we need to bound the weighted inverse defect $\sum_i |\widehat{\Delta}_i r_i(\widehat{\Delta})|$ of Proposition 3.3, rather than only the resolvent difference or an unweighted bound on $r(\widehat{\Delta})$.

Lemma 3.15 (Weighted inverse defect). *Let $Z \in \mathcal{K}$ satisfy $|Z|_{\mathrm{op}} = O(1)$. Then*

$$\sum_{i=1}^{n} |\widehat{\Delta}_i r_i(\widehat{\Delta})| = O(\rho_\Sigma \varepsilon)\,. \tag{3.59}$$

Proof. For $i \in I_a$, the inverse defect is constant on the class; write $r^{(a)}(\widehat{\Delta}) := r_i(\widehat{\Delta})$. Starting from the definition given in (2.10), we have

$$\sum_{i=1}^{n} |\widehat{\Delta}_i r_i(\widehat{\Delta})| = \sum_{a=1}^{k} n_a |\widehat{\delta}^{(a)} r^{(a)}(\widehat{\Delta})| = \sum_{a=1}^{k} \left| n_a(1-\widehat{\delta}^{(a)}) - \pi_a \widehat{\delta}^{(a)} \operatorname{Tr}(\Sigma^{(a)} \widehat{G}) \right|. \tag{3.60}$$

The Schur identity (3.10), applied at stage a with the pivots (3.45), gives

$$n_a(1-\widehat{\delta}^{(a)}) = \mathbb{E}[\operatorname{Tr}(S_a G^{(a-1)})].$$

Moreover, with the definition of R_ℓ in (3.53):

$$\widehat{G} = \mathbb{E}[G^{(a)}] - \sum_{\ell > a} \mathbb{E}[G^{(\ell-1)} - G^{(\ell)}] = \mathbb{E}[G^{(a)}] + \sum_{\ell > a} \mathbb{E}[R_\ell].$$

Finally, taking traces in the identities

$$G^{(\ell-1)}(S_\ell + B_\ell) = I_p = G^{(\ell)}\big(\pi_\ell \widehat{\delta}^{(\ell)} \Sigma^{(\ell)} + B_\ell\big),$$

and using the $\mathcal{F}^\ell$-measurability of B_ℓ and $G^{(\ell)}$, we obtain:

$$\mathbb{E}[\operatorname{Tr}(S_\ell G^{(\ell-1)})] - \pi_\ell \widehat{\delta}^{(\ell)} \operatorname{Tr}\Big(\Sigma^{(\ell)} \mathbb{E}[G^{(\ell)}]\Big) = \mathbb{E}[\operatorname{Tr}(B_\ell R_\ell)].$$

Therefore, (3.60) gives

$$\begin{aligned}
\sum_{i=1}^{n} |\widehat{\Delta}_i r_i(\widehat{\Delta})| &= \sum_{a=1}^{k} \left| \mathbb{E}[\operatorname{Tr}(B_a R_a)] - \pi_a \widehat{\delta}^{(a)} \sum_{\ell > a} \operatorname{Tr}(\Sigma^{(a)} \mathbb{E}[R_\ell]) \right| \\
&\le \sum_{\ell=1}^{k} |\mathbb{E}[\operatorname{Tr}(B_\ell R_\ell)]| + \sum_{\ell=1}^{k} \sum_{a<\ell} \pi_a |\widehat{\delta}^{(a)}|\, |\operatorname{Tr}(\Sigma^{(a)} \mathbb{E}[R_\ell])| \\
&\le \sum_{\ell=1}^{k} \mathbb{E}\left[\left| U_\ell^{-1} R_\ell U_\ell^{-1} \right|_* \left(|U_\ell B_\ell U_\ell|_{\mathrm{op}} + \left| \sum_{a<\ell} \theta_{a,\ell} \pi_a \widehat{\delta}^{(a)} U_\ell \Sigma^{(a)} U_\ell \right|_{\mathrm{op}} \right) \right],
\end{aligned}$$

where, for each ℓ, we choose deterministic unit phases $\theta_{a,\ell}$, $a < \ell$, such that

$$\theta_{a,\ell} \widehat{\delta}^{(a)} \operatorname{Tr}(\Sigma^{(a)} \mathbb{E}[R_\ell]) \in \mathbb{R}_+.$$

By (3.55),

$$\left| \sum_{a<\ell} \theta_{a,\ell} \pi_a \widehat{\delta}^{(a)} U_\ell \Sigma^{(a)} U_\ell \right|_{\mathrm{op}} \le 1.$$

The result follows from (3.58), $|U_\ell B_\ell U_\ell|_{\mathrm{op}} = O(1)$, and $\sum_\ell \pi_\ell = 1$. □

Proposition 3.16 (Classwise comparison with the fixed-point resolvent). *Let $Z \in \mathcal{K}$ satisfy $|Z|_{\mathrm{op}} = O(1)$. Then*

$$|\mathbb{E}[G^Z] - \widetilde{G}^Z|_* = O\Big(\rho_\Sigma^{3/2} \varepsilon\Big)\,. \tag{3.61}$$

If $Z = Z^* \prec 0$, one can replace $\rho_\Sigma^{3/2}$ by ρ_Σ in this bound.

Proof. Lemma 3.13 gives $\widehat{\Delta} \in \mathcal{D}_n(\mathcal{C}_Z \setminus \{0\})$, $|\widehat{\Delta}|_\infty = O(1)$ and $\max_i |\widehat{\Delta}_i|^{-1} = O(\rho_\Sigma)$. The fixed-point equation (2.7) and Proposition 2.7 also give $|\widetilde{\Delta}|_\infty = O(1)$ and

$$|\widetilde{\Delta}_i|^{-1} = \left|1 + \frac{1}{n}\operatorname{Tr}(\Sigma_i \widetilde{G}^Z)\right| \le 1 + \frac{\operatorname{Tr}(\Sigma_i)}{n}|\widetilde{G}^Z|_{\mathrm{op}} = O(\rho_\Sigma).$$

Therefore (3.21) in Proposition 3.3, followed by Lemma 3.15, yields

$$|\widehat{G} - \widetilde{G}^Z|_* = O\left(\rho_\Sigma^{1/2} \sum_{i=1}^n |\widehat{\Delta}_i r_i(\widehat{\Delta})|\right) = O\left(\rho_\Sigma^{3/2}\varepsilon\right).$$

Combining this with Proposition 3.14 and $\rho_\Sigma \ge 1$ proves (3.61).

If $Z = Z^* \prec 0$, the pivots are positive by Lemma 3.13 and one can use Corollary 3.4. □

Proof of Theorem 1.4. For every deterministic $A \in \mathcal{M}_p$, every $q \ge 1$, and every $Z \in \mathcal{K}$,

$$\left\|\operatorname{Tr}\left(A(G^Z - \widetilde{G}^Z)\right)\right\|_{L^q} \le \left\|\operatorname{Tr}\big(A(G^Z - \mathbb{E}[G^Z])\big)\right\|_{L^q} + |A|_{\mathrm{op}}|\mathbb{E}[G^Z] - \widetilde{G}^Z|_*.$$

Assume now that $|Z|_{\mathrm{op}} = O(1)$. Proposition 3.1 and (3.61) then give

$$\begin{aligned}&\left\|\operatorname{Tr}\left(A(G^Z - \widetilde{G}^Z)\right)\right\|_{L^q} \\ &\quad = O\left(|A|_{\mathrm{op}}\left[c_{n,q}M_{\mathrm{op}}^{(q)} + \rho_\Sigma^{3/2}\min\left\{\tau_\Sigma\big(1 + M_{\mathrm{op}}^{(1)}\big),\ \frac{\tau_\Sigma M_{\mathrm{op}}^{(2)}}{\sqrt{n}} + \frac{\tau_\Sigma^2 + (M_{\mathrm{op}}^{(2)})^2}{n}\right\}\right]\right). \quad (3.62)\end{aligned}$$

The minimum uses the same $+\infty$ convention as (3.51) when $M_{\mathrm{op}}^{(2)} = \infty$. Specialize (3.62) to $Z = zI_p$. The additional norm bound holds uniformly for $z \in K$ by compactness. When $z < 0$, Proposition 3.16 allows us to replace the factor $\rho_\Sigma^{3/2}$ by ρ_Σ. □

3.4 Resolvent sandwiches

Wagner, Couillet, Debbah, and Slock [WCDS12] derive two-resolvent equivalents by differentiating a deterministic equivalent with respect to the parameter of an additive matrix perturbation. Hoydis, ten Brink, and Debbah [HtBD13] state a normalized-trace equivalent for uniformly bounded positive semidefinite insertions. In a random-feature model, Louart, Liao, and Couillet [LLC18] obtain operator-norm error bounds for the expected sandwich by a direct resolvent expansion. Xiao and Slock [XS26] formulate the differentiation argument in a matrix-Dyson-equation setting using a one-resolvent approximation uniform under perturbation.

Here Cauchy's integral formula and Minkowski's inequality transfer the preceding one-resolvent L^q estimates to resolvent sandwiches, with explicit dependence on the dimension and second-moment profile.

We retain the compact set K and the constant η_K from (3.1). For $z \in K$, we have $\eta_z \ge \eta_K > 0$, and compactness gives $\sup_{z\in K}|z| = O(1)$. The finite-cover reduction following (3.1) extends the estimates to arbitrary compact subsets of Ω.

For a deterministic matrix $A \in \mathcal{M}_p$ and $t \in \mathbb{C}$, define Z_t and, whenever the inverse exists, G^{Z_t} by

$$Z_t := zI_p - tA, \qquad G^{Z_t} = \left(\frac{1}{n}XX^\top + tA - zI_p\right)^{-1}.$$

Note that

$$d\big(\mathcal{W}(Z_t), \mathbb{R}_+\big) \ge \eta_z - |t|\,|A|_{\mathrm{op}}. \quad (3.63)$$

Now choose $r := \frac{\eta_K}{4}$. Uniformly for $z \in K$, $|t| < 2r$, and $|A|_{\mathrm{op}} \le 1$, (3.63) gives

$$d\big(\mathcal{W}(Z_t), \mathbb{R}_+\big) \ge \eta_K/2, \qquad |Z_t|_{\mathrm{op}} \le \sup_{z\in K}|z| + 2r = O(1).$$

Set $K^+ := K + \{w \in \mathbb{C} : |w| \le 2r\}$. Then $d(\operatorname{conv} K^+, \mathbb{R}_+) \ge \eta_K/2$ and $\mathcal{W}(Z_t) \subseteq K^+$, so the preceding estimates apply with K^+ in place of K. Proposition 2.7 gives

$$\widetilde{G}^{Z_t} = \left(tA - zI_p + \frac{1}{n}\sum_{i=1}^{n} \frac{\Sigma_i}{1 + \frac{1}{n}\operatorname{Tr}(\Sigma_i \widetilde{G}^{Z_t})} \right)^{-1}. \tag{3.64}$$

Proposition 2.9 shows that $t \mapsto \widetilde{G}^{Z_t}$ is holomorphic on $|t| < 2r$.

Proposition 3.17 (Deterministic equivalent of the resolvent sandwich). *For every $z \in \Omega$ and every deterministic matrix $A \in \mathcal{M}_p$, the linear fixed-point equation* (1.10) *has a unique solution, and*

$$\widetilde{\mathcal{S}}^{\,z}[A] = -\left.\frac{\mathrm{d}}{\mathrm{d}t}\widetilde{G}^{Z_t}\right|_{t=0}. \tag{3.65}$$

The solution depends complex-linearly on A. Moreover, $\widetilde{\mathcal{S}}^{\,\bar{z}}[\overline{A}] = \overline{\widetilde{\mathcal{S}}^{\,z}[A]}$.

Proof. If $A = 0$, the path $Z_t = zI_p$ is constant. Otherwise apply the preceding disk construction to $A/|A|_{\mathrm{op}}$ and the scalar parameter $s = t|A|_{\mathrm{op}}$. This gives holomorphic dependence in a neighborhood of $t = 0$ for every $A \in \mathcal{M}_p$. Put

$$T := -\left.\partial_t \widetilde{G}^{Z_t}\right|_{t=0}.$$

Differentiating (3.64) gives

$$T = \tilde{G}^z A \tilde{G}^z + \frac{1}{n^2}\sum_i (\widetilde{\Delta}_i^{zI_p})^2 \operatorname{Tr}(\Sigma_i T)\tilde{G}^z \Sigma_i \tilde{G}^z.$$

Since $\widetilde{\Delta}_i^{zI_p} = (1 + \frac{1}{n}\operatorname{Tr}(\Sigma_i \tilde{G}^z))^{-1}$, this is (1.10).

For uniqueness, let U solve the homogeneous equation and set

$$s_i := (\widetilde{\Delta}_i^{zI_p})^2 \operatorname{Tr}(\Sigma_i U).$$

Taking the weighted traces in that equation gives

$$s = \Psi^{zI_p}(\widetilde{\Delta}^{zI_p}, \widetilde{\Delta}^{zI_p}) s.$$

Proposition 2.6 implies $s = 0$. The homogeneous equation then gives $U = 0$, proving uniqueness and $T = \widetilde{\mathcal{S}}^{\,z}[A]$. Since (1.10) is complex-linear in the unknown matrix and its forcing term is complex-linear in A, uniqueness implies that $A \mapsto \widetilde{\mathcal{S}}^{\,z}[A]$ is complex-linear. Finally, conjugate the linear equation, using the reality of the profiles Σ_i and (2.16). The conjugated equation has spectral parameter $\bar{z}$ and insertion $\overline{A}$, so uniqueness gives the asserted conjugation identity for the sandwich. □

Proof of Corollary 1.7. The case $A = 0$ is immediate. We first prove the estimates for $|A|_{\mathrm{op}} \le 1$. For a deterministic test matrix B, set

$$F(t) := \operatorname{Tr}\Big(B[G^{Z_t} - \widetilde{G}^{Z_t}] \Big).$$

The inverse defining G^{Z_t} and Proposition 2.9 show that F is holomorphic pathwise on $|t| < 2r$. On the integration circle, the resolvent bounds in Lemma 2.3 dominate $|F(t)|$ by a deterministic multiple of $|B|_*$, so Cauchy's integral formula and Minkowski's inequality apply. They give

$$\|F'(0)\|_{L^q} \le \frac{1}{2\pi r}\int_0^{2\pi} \|F(re^{\mathrm{i}\theta})\|_{L^q}\, d\theta \le \frac{1}{r}\sup_{|t|=r} \|F(t)\|_{L^q}.$$

Differentiating the random resolvent and applying Proposition 3.17 gives

$$-\partial_t G^{Z_t}\big|_{t=0} = G^z A G^z, \qquad -\partial_t \widetilde{G}^{Z_t}\Big|_{t=0} = \widetilde{\mathcal{S}}^z[A],$$

so we obtain the common transfer bound

$$\left\|\mathrm{Tr}\Big(B[\mathcal{S}^z[A] - \widetilde{\mathcal{S}}^z[A]]\Big)\right\|_{L^q} \le \frac{1}{r}\sup_{|t|=r}\left\|\mathrm{Tr}\Big(B[G^{Z_t} - \widetilde{G}^{Z_t}]\Big)\right\|_{L^q}. \tag{3.66}$$

Apply (3.62) and (3.36) with test matrix B, uniformly on the contour. The same simplifications used in the proofs of the one-resolvent theorems give the two stated bounds, since $1/r = 4/\eta_K = O(1)$. The general operator comparison is used even when $z < 0$, because the contour is not a family of negative Hermitian parameters.

For general $A \neq 0$, apply the preceding estimates to $A/|A|_{\mathrm{op}}$. Both $\mathcal{S}^z[A]$ and $\widetilde{\mathcal{S}}^z[A]$ depend complex-linearly on A, so multiplication by $|A|_{\mathrm{op}}$ gives the stated bounds.

□

A Notation index

This appendix indexes the principal notation used in the paper.

Symbol	Meaning	Definition
$M_{\mathrm{HS}}^{(q)}, M_{\mathrm{op}}^{(q)}$	Quadratic-form moduli for the Hilbert–Schmidt and operator test balls	(1.2)
I_a, n_a, π_a	Exact second-moment classes, their sizes, and their proportions	Section 1.2.2
τ_Σ	Operator-size-adjusted exact-class coefficient	(1.3)
$\lvert\Sigma\rvert_\nu^\infty$	Maximum of $\lvert\Sigma_i\rvert_\nu$, for $\nu \in \{\mathrm{op}, \mathrm{HS}, *\}$	(0.1)
ρ_Σ	Trace factor $1 + \frac{\lvert\Sigma\rvert_*^\infty}{n}$	(3.37)
σ_Σ	Commuting-approximation coefficient of the finite profile	(1.6)
$c_{n,q}$	$n^{1/(q\wedge 2)-1}$	(3.11)
Ω	Scalar parameter domain $\mathbb{C} \setminus [0,\infty)$	Section 2
$\mathcal{W}(Z)$	Numerical range $\{u^*Zu : \lvert u\rvert = 1\}$	Section 2
η_Z, η_z	Distances $d(\mathcal{W}(Z), \mathbb{R}_+)$ and $d(z, \mathbb{R}_+)$, respectively	Section 2
$\mathcal{C}_Z$	$\mathrm{cone}(\mathbb{R}_+ \cup -\mathcal{W}(Z))$, closed for admissible Z	(2.1)
θ_Z^-, θ_Z^+	Boundary arguments of $\mathcal{C}_Z$	Lemma 2.1
θ_Z, ω_Z, c_Z	Bisecting angle and phase; cosine of half the aperture	(2.3)
$\mathfrak{C}_Z, \mathfrak{C}_{\bar{Z}}$	Rotated real-part maps for the sector and its conjugate	(2.9)
K	Compact scalar set determining the common sector; in the probabilistic proof setup, $d(\mathrm{conv}\, K, \mathbb{R}_+) > 0$	(3.1)
η_K	Convex-hull separation capped at one: $1 \wedge d(\mathrm{conv}\, K, \mathbb{R}_+)$	(3.1)
$\mathcal{C}_K, \overline{\mathcal{C}_K}$	Common closed sector $\mathrm{cone}(\mathbb{R}_+ \cup (-K))$ and its conjugate	(3.2)
ω_K, c_K	Bisecting unit phase and supporting constant of $\mathcal{C}_K$	(3.3)
$\mathfrak{C}_K, \mathfrak{C}_{\bar{K}}$	Rotated real-part maps for the common sector and its conjugate	(3.4)
$\mathcal{K}_p, \mathcal{K}$	Matrix class in dimension p and corresponding parameter families over Θ; no upper-norm restriction	(3.5)
G^Z, G_{-i}^Z	Random resolvent and its i-th leave-one-out version	Section 3.1.1
$G_D^Z, \Phi^Z(D)$	Deterministic resolvent at a diagonal profile and its fixed-point map	(2.7)
$\Psi^Z(D, D')$	Deterministic secant kernel	Lemma 2.5
$\widetilde{\Delta}^Z, \widetilde{G}^Z$	Fixed point of Φ^Z and its deterministic resolvent	Proposition 2.7
Δ_i^Z	Random Schur pivot	(3.7)
$\check{\Delta}$	Diagonal profile with entries $(\mathbb{E}[\Delta_i^{-1}])^{-1}$	Section 3.1.3
$\widehat{\delta}_{\mathcal{F}^\ell}^{(\ell)}, \widehat{\delta}^{(\ell)}$	Conditional average Schur pivot for class ℓ and its expectation	(3.45)
$Z_{\mathcal{F}^\ell}^{(\ell)}, Z^{(\ell)}$	Class-replacement parameters using the conditional and expected pivots	(3.46)
$G_{\mathcal{F}^\ell}^{(\ell)}, G^{(\ell)}$	Class-replacement resolvents using the conditional and expected pivots	(3.47)

Symbol	Meaning	Definition
$r(D)$	Inverse defect of a diagonal pivot	(2.10)
$J(D)$	One plus the average weighted profile interaction	(3.31)
$\tilde{\Gamma}^z$	Companion diagonal profile for the scalar parameter z, $\tilde{\Gamma}^z = -z^{-1}\widetilde{\Delta}^{zI_p}$	Corollary 2.10
$\mathcal{S}^z[A], \widetilde{\mathcal{S}}^z[A]$	Random resolvent sandwich and its deterministic equivalent	Section 1.2.4

B Auxiliary results

This appendix collects the moment inequalities, quadratic-form estimates, and Stieltjes-transform facts used in the proofs and applications.

For the replacement-moment inequality, fix $q \geq 1$. Let $U_1, \ldots, U_n$ be independent random variables, and let

$$F = f(U_1, \ldots, U_n) \in L^q$$

be real- or complex-valued. For each $i \in [n]$, let U_i' be an independent copy of U_i, independent of everything else, and define

$$F^{(i)} := f(U_1, \ldots, U_{i-1}, U_i', U_{i+1}, \ldots, U_n).$$

Thus $F^{(i)}$ has the same law as F and belongs to L^q.

Theorem B.1 (Martingale replacement-moment inequality). *If $1 \leq q \leq 2$, then*

$$\|F - \mathbb{E}[F]\|_{L^q} = O\left(\left(\sum_{i=1}^n \|F - F^{(i)}\|_{L^q}^q\right)^{1/q}\right) = O\left(n^{1/q} \sup_{i\in[n]} \|F - F^{(i)}\|_{L^q}\right).$$

If $q \geq 2$, then

$$\|F - \mathbb{E}[F]\|_{L^q} = O\left(\left(\sum_{i=1}^n \|F - F^{(i)}\|_{L^q}^2\right)^{1/2}\right) = O\left(\sqrt{n} \sup_{i\in[n]} \|F - F^{(i)}\|_{L^q}\right).$$

Proof. We first assume that F is real-valued. Set

$$\mathcal{F}_i := \sigma(U_1, \ldots, U_i), \qquad M_i := \mathbb{E}[F \mid \mathcal{F}_i], \qquad D_i := M_i - M_{i-1}, \qquad V_i := F - F^{(i)},$$

with $\mathcal{F}_0$ trivial. Then $(D_i)_{i=1}^n$ is a martingale difference sequence and

$$F - \mathbb{E}[F] = \sum_{i=1}^n D_i.$$

Since U_i' has the same law as U_i, and since $U_i', U_{i+1}, \ldots, U_n$ are independent of $\mathcal{F}_i$,

$$\mathbb{E}[F^{(i)} \mid \mathcal{F}_i] = \mathbb{E}[F \mid \mathcal{F}_{i-1}] = M_{i-1}.$$

Therefore

$$D_i = \mathbb{E}[F - F^{(i)} \mid \mathcal{F}_i] = \mathbb{E}[V_i \mid \mathcal{F}_i],$$

and conditional Jensen's inequality gives, for every $q \geq 1$,

$$\|D_i\|_{L^q} \leq \|V_i\|_{L^q}.$$

For $1 < q \leq 2$, set $S_0 := 0$ and $S_i := \sum_{j=1}^i D_j$. The scalar inequality underlying the von Bahr–Esseen estimate [vBE65] is

$$|x + y|^q \leq |x|^q + q|x|^{q-1}\operatorname{sgn}(x)y + C_q|y|^q, \qquad x, y \in \mathbb{R}.$$

Apply it with $x = S_{i-1}$ and $y = D_i$. The linear term is integrable by Hölder's inequality and has conditional mean zero given $\mathcal{F}_{i-1}$. Thus

$$\mathbb{E}[|S_i|^q \mid \mathcal{F}_{i-1}] \le |S_{i-1}|^q + C_q \mathbb{E}[|D_i|^q \mid \mathcal{F}_{i-1}].$$

Taking expectations and iterating from $S_0 = 0$ gives

$$\mathbb{E}\left[\left|\sum_{i=1}^n D_i\right|^q\right] \le C_q \sum_{i=1}^n \mathbb{E}[|D_i|^q].$$

For $q = 1$, the corresponding bound follows directly from the triangle inequality. Hence, for $1 \le q \le 2$,

$$\|F - \mathbb{E}[F]\|_{L^q} = O\left(\left(\sum_{i=1}^n \|V_i\|_{L^q}^q\right)^{1/q}\right).$$

For $q \ge 2$, Burkholder's square-function inequality [Bur88] gives

$$\left\|\sum_{i=1}^n D_i\right\|_{L^q} \le (q-1) \left\|\left(\sum_{i=1}^n D_i^2\right)^{1/2}\right\|_{L^q}.$$

Moreover, Minkowski's inequality in $L^{q/2}$ gives

$$\left\|\left(\sum_{i=1}^n D_i^2\right)^{1/2}\right\|_{L^q}^2 = \left\|\sum_{i=1}^n D_i^2\right\|_{L^{q/2}} \le \sum_{i=1}^n \|D_i\|_{L^q}^2.$$

It follows that

$$\|F - \mathbb{E}[F]\|_{L^q} \le (q-1)\left(\sum_{i=1}^n \|V_i\|_{L^q}^2\right)^{1/2} \le (q-1)\sqrt{n} \sup_{i\in[n]} \|V_i\|_{L^q}.$$

If F is complex-valued, apply the real result separately to $\Re(F)$ and $\Im(F)$. Since the corresponding replacement differences have L^q norm at most $\|V_i\|_{L^q}$, the triangle inequality introduces at most a factor of two. □

Lemma B.2 (Reciprocal difference estimate). *Let s and t be integrable complex random variables, with $s, t \in L^2$ for the second estimate. If*

$$\left|1 + \frac{\mathbb{E}[t]}{n}\right|^{-1} = O(1), \qquad \left|1 + \frac{t}{n}\right|^{-1} = O(1)$$

almost surely, then

$$\left|\mathbb{E}\left[s\left(\frac{1}{1+\mathbb{E}[t]/n} - \frac{1}{1+t/n}\right)\right]\right| = O\left(\|s - \mathbb{E}[s]\|_{L^1} + \frac{|\mathbb{E}[s]|}{n}\|t - \mathbb{E}[t]\|_{L^1}\right),$$
$$\left|\mathbb{E}\left[s\left(\frac{1}{1+\mathbb{E}[t]/n} - \frac{1}{1+t/n}\right)\right]\right| = O\left(\frac{\|s - \mathbb{E}[s]\|_{L^2}\|t - \mathbb{E}[t]\|_{L^2}}{n} + \frac{|\mathbb{E}[s]|\|t - \mathbb{E}[t]\|_{L^2}^2}{n^2}\right).$$

Proof. Put

$$d := \frac{1}{1+\mathbb{E}[t]/n} - \frac{1}{1+t/n}.$$

The exact expansions

$$d = \frac{t - \mathbb{E}[t]}{n(1+\mathbb{E}[t]/n)(1+t/n)} \tag{B.1}$$
$$= \frac{t - \mathbb{E}[t]}{n(1+\mathbb{E}[t]/n)^2} - \frac{(t-\mathbb{E}[t])^2}{n^2(1+\mathbb{E}[t]/n)^2(1+t/n)} \tag{B.2}$$

and the denominator bounds give

$$|d| = O(1), \qquad |d| = O\left(\frac{|t - \mathbb{E}[t]|}{n}\right). \tag{B.3}$$

Taking expectations in (B.1) gives

$$|\mathbb{E}[d]| = O\left(\frac{\|t - \mathbb{E}[t]\|_{L^1}}{n}\right).$$

If $t \in L^2$, the linear term in (B.2) has mean zero, so

$$|\mathbb{E}[d]| = O\left(\frac{\|t - \mathbb{E}[t]\|_{L^2}^2}{n^2}\right).$$

Finally,

$$\mathbb{E}[sd] = \mathbb{E}\left[(s - \mathbb{E}[s])d\right] + \mathbb{E}[s]\,\mathbb{E}[d].$$

The first conclusion follows from $|d| = O(1)$ and the first bound on $|\mathbb{E}[d]|$. For the second, apply Cauchy–Schwarz to the centered term and use the second bounds on $|d|$ and $|\mathbb{E}[d]|$. □

For $B \in \mathcal{M}_p$, write $B_{\mathrm{s}} := (B + B^\top)/2$.

Lemma B.3 (Complex and conditional test reduction). *Let $x \in \mathbb{R}^p$ have finite second moments, and set $\Sigma := \mathbb{E}[xx^\top]$. Then*

$$x^\top B x - \operatorname{Tr}(\Sigma B) = x^\top B_{\mathrm{s}} x - \operatorname{Tr}(\Sigma B_{\mathrm{s}}), \qquad |B_{\mathrm{s}}|_{\mathrm{op}} \le |B|_{\mathrm{op}}, \qquad |B_{\mathrm{s}}|_{\mathrm{HS}} \le |B|_{\mathrm{HS}}.$$

Consequently, each modulus in (1.2) may be applied to a complex, nonsymmetric test after symmetrization. If a bound has only been established for real symmetric tests, applying it to the real and imaginary parts gives the complex version with at most twice the numerical constant. Finally, if B is independent of x and lies almost surely in the relevant deterministic test ball, the same L^q bound follows by conditioning on B and then taking expectations.

Proof. The skew-transpose part of B has zero quadratic form against a real vector, and its trace against the real symmetric matrix Σ is zero. The two norm inequalities follow from the triangle inequality and the invariance of the operator and Hilbert–Schmidt norms under transpose. Write $B_{\mathrm{s}} = C + \mathbf{i}D$ with C, D real symmetric. The real and imaginary parts of the centered quadratic form are the centered quadratic forms associated with C and D, respectively; moreover $|C|_{\mathrm{op}}, |D|_{\mathrm{op}} \le |B_{\mathrm{s}}|_{\mathrm{op}}$, and the same inequalities hold with the Hilbert–Schmidt norm in place of the operator norm. □

Lemma B.4 (Uncentered quadratic-form bound). *For every $i \in [n]$ and every deterministic $B \in \mathcal{M}_p$,*

$$\mathbb{E}\left[|x_i^\top B x_i|\right] \le |\Sigma_i|_{\mathrm{HS}} |B|_{\mathrm{HS}}, \qquad \mathbb{E}\left[|x_i^\top B x_i|\right] \le |B|_{\mathrm{op}} \operatorname{Tr}(\Sigma_i). \tag{B.4}$$

Consequently,

$$M_{\mathrm{HS}}^{(1)} \le 2|\Sigma|_{\mathrm{HS}}^{\infty}, \qquad M_{\mathrm{op}}^{(1)} \le 2|\Sigma|_{*}^{\infty}.$$

Proof. Polar decomposition and Cauchy–Schwarz give the pointwise inequality

$$|x_i^* B x_i| \le \left(x_i^*(BB^*)^{1/2} x_i\right)^{1/2} \left(x_i^*(B^*B)^{1/2} x_i\right)^{1/2}.$$

Since x_i is real, $x_i^* B x_i = x_i^\top B x_i$. Taking expectations and applying Cauchy–Schwarz, followed by Hilbert–Schmidt duality, gives

$$\mathbb{E}\left[|x_i^\top B x_i|\right] \le \sqrt{\operatorname{Tr}\left(\Sigma_i (BB^*)^{1/2}\right) \operatorname{Tr}\left(\Sigma_i (B^*B)^{1/2}\right)} \le |\Sigma_i|_{\mathrm{HS}} |B|_{\mathrm{HS}}.$$

Here both positive square-root matrices have Hilbert–Schmidt norm $|B|_{\mathrm{HS}}$. For the operator norm, the pointwise bound $|x_i^\top B x_i| \le |B|_{\mathrm{op}}|x_i|^2$ gives

$$\mathbb{E}\left[|x_i^\top B x_i|\right] \le |B|_{\mathrm{op}}\mathbb{E}[|x_i|^2] = |B|_{\mathrm{op}}\operatorname{Tr}(\Sigma_i).$$

Since $\operatorname{Tr}(\Sigma_i B) = \mathbb{E}[x_i^\top B x_i]$, the triangle inequality gives

$$\left\|x_i^\top B x_i - \operatorname{Tr}(\Sigma_i B)\right\|_{L^1} \le 2\mathbb{E}\left[|x_i^\top B x_i|\right].$$

Taking the supremum over $i \in [n]$ and over the Hilbert–Schmidt or operator unit ball proves the two modulus bounds. □

Theorem B.5 (Von Bahr–Esseen inequality [vBE65]). *Let $1 \le q \le 2$, and let $Y_1, \dots, Y_m$ be independent centered real random variables with finite qth absolute moments. Then*

$$\mathbb{E}\left|\sum_{j=1}^m Y_j\right|^q \le 2\sum_{j=1}^m \mathbb{E}|Y_j|^q.$$

Proposition B.6 (Quadratic forms with independent coordinates). *Let $x \in \mathbb{R}^p$ have centered independent coordinates, and put $\Sigma := \mathbb{E}[xx^\top]$. Let $r \ge 1$. If*

$$K_{2r} := \sup_{i\in[p]} \|x_i\|_{L^{2r}} < \infty,$$

then, for every deterministic $A \in \mathcal{M}_p$,

$$\alpha := \max\left\{\frac{1}{r} - \frac{1}{2}, 0\right\}, \qquad \left\|x^\top A x - \operatorname{Tr}\big(A\Sigma\big)\right\|_{L^1} = O\big(K_{2r}^2 p^\alpha |A|_{\mathrm{HS}}\big).$$

For $r = 1$, one has the explicit bound

$$\left\|x^\top A x - \operatorname{Tr}\big(A\Sigma\big)\right\|_{L^1} \le 2K_2^2\sqrt{p}\,|A|_{\mathrm{HS}}.$$

Proof of Proposition B.6. For $r = 1$, the matrix Σ is diagonal and has Hilbert–Schmidt norm at most $K_2^2\sqrt{p}$. Thus Lemma B.4 and the triangle inequality give the explicit bound.

Now let $r > 1$. By Lemma B.3, we may replace A by its complex symmetric part. Write $A = D + R$, where $D = \operatorname{Diag}(d_1, \dots, d_p)$ and R has zero diagonal. Independence and centering give the off-diagonal identity

$$\mathbb{E}\left[|x^\top R x|^2\right] = 4\sum_{1\le i<j\le p} |R_{ij}|^2 \mathbb{E}[x_i^2]\mathbb{E}[x_j^2]. \tag{B.5}$$

This identity uses only second moments; the absolute squares are retained because R may be complex.

Put $K_2 := \sup_{i\in[p]} \|x_i\|_{L^2} \le K_{2r}$. For the diagonal part, apply Theorem B.5 with $q := r \wedge 2$. Then $1 < q \le 2$ and $2q \le 2r$, so

$$\left\|x_i^2 - \mathbb{E}[x_i^2]\right\|_{L^q} \le 2\|x_i\|_{L^{2q}}^2 \le 2K_{2r}^2.$$

The centered squares are independent real random variables. Since the coefficients d_i may be complex, apply the theorem separately to $\Re(d_i)(x_i^2 - \mathbb{E}[x_i^2])$ and $\Im(d_i)(x_i^2 - \mathbb{E}[x_i^2])$, then combine the bounds by the triangle inequality in L^q. Using $\|\cdot\|_{L^1} \le \|\cdot\|_{L^q}$ gives

$$\left\|x^\top D x - \operatorname{Tr}(D\Sigma)\right\|_{L^1} = O\left(K_{2r}^2\left(\sum_{i\in[p]} |d_i|^q\right)^{1/q}\right) = O\big(K_{2r}^2 p^\alpha |D|_{\mathrm{HS}}\big).$$

The last step follows from Hölder's inequality:

$$\left(\sum_{i\in[p]}|d_i|^q\right)^{1/q} \le p^{1/q-1/2}\left(\sum_{i\in[p]}|d_i|^2\right)^{1/2} = p^{\alpha}|D|_{\mathrm{HS}},$$

since $1/q - 1/2 = \alpha$. Equation (B.5) gives

$$\mathbb{E}\left[|x^\top Rx|^2\right] \le 2K_2^4|R|_{\mathrm{HS}}^2.$$

Consequently,

$$\|x^\top Rx\|_{L^1} \le \sqrt{2}K_2^2|R|_{\mathrm{HS}}.$$

Since $p^\alpha \ge 1$, and $|D|_{\mathrm{HS}} + |R|_{\mathrm{HS}} \le \sqrt{2}|A|_{\mathrm{HS}}$, the two estimates prove the proposition. □

Proposition B.7 (Quadratic forms under finite variance). *Let the coordinates of $x \in \mathbb{R}^p$ be independent copies of a centered real random variable ξ with $\sigma^2 := \mathbb{E}[\xi^2] < \infty$, where $\sigma \ge 0$. For every deterministic $A \in \mathcal{M}_p$ and every $L > 0$,*

$$\left\|x^\top Ax - \sigma^2\operatorname{Tr}(A)\right\|_{L^1} \le |A|_{\mathrm{HS}}\left(\sqrt{2}\,\sigma^2 + L\sigma + 2\sqrt{p}\,\mathbb{E}\left[\xi^2\mathbf{1}_{\{|\xi|>L\}}\right]\right). \tag{B.6}$$

Consequently, if all entries of X are independent copies of ξ, and ξ has a fixed law with finite variance, $M_{\mathrm{HS}}^{(1)} = o(\sqrt{p})$ as $p \to \infty$.

Proof. Use the symmetrization and decomposition $A = D + R$ from the proof of Proposition B.6, with $D = \mathrm{Diag}(d_1, \dots, d_p)$ and R having zero diagonal. For the diagonal part, decompose

$$x_j^2 - \sigma^2 = \left(x_j^2\mathbf{1}_{\{|x_j|\le L\}} - \mathbb{E}\left[\xi^2\mathbf{1}_{\{|\xi|\le L\}}\right]\right) + \left(x_j^2\mathbf{1}_{\{|x_j|>L\}} - \mathbb{E}\left[\xi^2\mathbf{1}_{\{|\xi|>L\}}\right]\right).$$

The sum of the bounded centered terms has L^2 norm at most

$$\left(\sum_{j=1}^p |d_j|^2\mathbb{E}\left[\xi^4\mathbf{1}_{\{|\xi|\le L\}}\right]\right)^{1/2} \le L\sigma\,|D|_{\mathrm{HS}},$$

whereas the tail sum has L^1 norm at most

$$2\sum_{j=1}^p |d_j|\mathbb{E}\left[\xi^2\mathbf{1}_{\{|\xi|>L\}}\right] \le 2\sqrt{p}\,|D|_{\mathrm{HS}}\mathbb{E}\left[\xi^2\mathbf{1}_{\{|\xi|>L\}}\right].$$

Equation (B.5) also gives

$$\|x^\top Rx\|_{L^1} \le \sqrt{2}\,\sigma^2|R|_{\mathrm{HS}}.$$

Since $|D|_{\mathrm{HS}}, |R|_{\mathrm{HS}} \le |A|_{\mathrm{HS}}$, combining these three bounds proves (B.6). Taking the supremum over $|A|_{\mathrm{HS}} \le 1$ and dividing by $\sqrt{p}$ gives

$$\frac{M_{\mathrm{HS}}^{(1)}}{\sqrt{p}} \le \frac{\sqrt{2}\,\sigma^2 + L\sigma}{\sqrt{p}} + 2\mathbb{E}\left[\xi^2\mathbf{1}_{\{|\xi|>L\}}\right].$$

For each fixed L, taking the upper limit as $p \to \infty$ yields

$$\limsup_{p\to\infty}\frac{M_{\mathrm{HS}}^{(1)}}{\sqrt{p}} \le 2\mathbb{E}\left[\xi^2\mathbf{1}_{\{|\xi|>L\}}\right].$$

Letting $L \to \infty$ proves the conclusion, since ξ^2 is integrable. □

Recall that $B_{\mathrm{s}} = (B + B^\top)/2$ denotes the symmetric part of $B \in \mathcal{M}_p$.

Lemma B.8 (Geometry of the two test balls). *For every real symmetric positive semidefinite matrix $\Sigma \in \mathcal{M}_p$,*

$$\sup_{\substack{B\in\mathcal{M}_p \\ |B|_{\mathrm{HS}}\le 1}} |\Sigma^{1/2} B_{\mathrm{s}} \Sigma^{1/2}|_{\mathrm{HS}} = |\Sigma|_{\mathrm{op}}, \qquad \sup_{\substack{B\in\mathcal{M}_p \\ |B|_{\mathrm{op}}\le 1}} |\Sigma^{1/2} B_{\mathrm{s}} \Sigma^{1/2}|_{\mathrm{HS}} = |\Sigma|_{\mathrm{HS}}.$$

Proof. Lemma B.3 gives $|B_{\mathrm{s}}|_{\mathrm{HS}} \le |B|_{\mathrm{HS}}$. Also, $|AXB|_{\mathrm{HS}} \le |A|_{\mathrm{op}} |X|_{\mathrm{HS}} |B|_{\mathrm{op}}$. Hence

$$|\Sigma^{1/2} B_{\mathrm{s}} \Sigma^{1/2}|_{\mathrm{HS}} \le |\Sigma|_{\mathrm{op}} |B_{\mathrm{s}}|_{\mathrm{HS}} \le |\Sigma|_{\mathrm{op}} |B|_{\mathrm{HS}}.$$

For the other test ball, work in a real orthonormal eigenbasis of Σ, with eigenvalues $(\lambda_j)_{j\in[p]}$. Then

$$|\Sigma^{1/2} B_{\mathrm{s}} \Sigma^{1/2}|_{\mathrm{HS}}^2 = \sum_{j,k=1}^{p} \lambda_j \lambda_k |(B_{\mathrm{s}})_{jk}|^2 \le \frac{1}{2} \sum_{j,k=1}^{p} (\lambda_j^2 + \lambda_k^2) |(B_{\mathrm{s}})_{jk}|^2 \le |\Sigma|_{\mathrm{HS}}^2 |B|_{\mathrm{op}}^2.$$

Every row and column of B_{s} has Euclidean norm at most $|B_{\mathrm{s}}|_{\mathrm{op}} \le |B|_{\mathrm{op}}$. This proves the last inequality. For the reverse inequalities, take respectively $B = vv^\top$, where v is a unit eigenvector corresponding to $|\Sigma|_{\mathrm{op}}$, and $B = I_p$. □

Lemma B.9 (Independent-coordinate quadratic-form moduli). *Let $x \in \mathbb{R}^p$ have second-moment matrix $\Sigma := \mathbb{E}[xx^\top]$, and assume that $x = \Sigma^{1/2} y$, where y has independent, centered, standardized coordinates. If*

$$K_4 := \sup_{j\in[p]} \|y_j\|_{L^4} < \infty,$$

then

$$\sup_{\substack{B\in\mathcal{M}_p \\ |B|_{\mathrm{HS}}\le 1}} \left\| x^\top B x - \mathrm{Tr}(\Sigma B) \right\|_{L^2} = O\big(K_4^2 |\Sigma|_{\mathrm{op}}\big),$$
$$\sup_{\substack{B\in\mathcal{M}_p \\ |B|_{\mathrm{op}}\le 1}} \left\| x^\top B x - \mathrm{Tr}(\Sigma B) \right\|_{L^2} = O\big(K_4^2 |\Sigma|_{\mathrm{HS}}\big).$$

Proof. Quadratic forms depend only on B_{s} by Lemma B.3. For a fixed test matrix B, put

$$C := \Sigma^{1/2} B_{\mathrm{s}} \Sigma^{1/2}.$$

Then

$$x^\top B x - \mathrm{Tr}(\Sigma B) = y^\top C y - \mathrm{Tr}(C).$$

Writing $C = (c_{jk})_{j,k\in[p]}$, independence, centering, and standardization give

$$\left\| y^\top C y - \mathrm{Tr}(C) \right\|_{L^2}^2 = \sum_{j=1}^{p} |c_{jj}|^2 \, \mathbb{V}(y_j^2) + 4 \sum_{1\le j<k\le p} |c_{jk}|^2 \le 2K_4^4 |C|_{\mathrm{HS}}^2.$$

Taking square roots and applying Lemma B.8 to the two test balls proves the result. □

For columns $x_i = \Sigma_i^{1/2} y_i$ satisfying these hypotheses with $\sup_{i\in[n],\, j\in[p]} \|(y_i)_j\|_{L^4} \le K_4 < \infty$, taking the maximum over $i \in [n]$ gives

$$M_{\mathrm{HS}}^{(2)} = O\big(K_4^2 |\Sigma|_{\mathrm{op}}^{\infty}\big), \qquad M_{\mathrm{op}}^{(2)} = O\big(K_4^2 |\Sigma|_{\mathrm{HS}}^{\infty}\big).$$

Lemma B.10 (Gaussian quadratic-form moduli). *Let $x \sim \mathcal{N}(0, \Sigma)$, where $\Sigma \in \mathcal{M}_p$ is real symmetric and positive semidefinite. For every fixed $q \geq 2$,*

$$\sqrt{2}|\Sigma|_{\mathrm{op}} \leq \sup_{\substack{B \in \mathcal{M}_p \\ |B|_{\mathrm{HS}} \leq 1}} \left\| x^\top B x - \mathrm{Tr}(\Sigma B) \right\|_{L^q} = O(|\Sigma|_{\mathrm{op}}), \tag{B.7}$$

$$\sqrt{2}|\Sigma|_{\mathrm{HS}} \leq \sup_{\substack{B \in \mathcal{M}_p \\ |B|_{\mathrm{op}} \leq 1}} \left\| x^\top B x - \mathrm{Tr}(\Sigma B) \right\|_{L^q} = O(|\Sigma|_{\mathrm{HS}}). \tag{B.8}$$

For $q = 2$, both lower bounds are equalities.

Proof. Write $x = \Sigma^{1/2} g$, where $g \sim \mathcal{N}(0, I_p)$, and let $B_{\mathrm{s}} = (B + B^\top)/2$. Wick's identity gives, also for complex test matrices,

$$\left\| x^\top B x - \mathrm{Tr}(\Sigma B) \right\|_{L^2} = \sqrt{2}\, |\Sigma^{1/2} B_{\mathrm{s}} \Sigma^{1/2}|_{\mathrm{HS}}.$$

For $q \geq 2$, monotonicity of the L^q norms and Gaussian hypercontractivity for second-order chaos yield

$$\sqrt{2}\, |\Sigma^{1/2} B_{\mathrm{s}} \Sigma^{1/2}|_{\mathrm{HS}} \leq \left\| x^\top B x - \mathrm{Tr}(\Sigma B) \right\|_{L^q} = O\Big(|\Sigma^{1/2} B_{\mathrm{s}} \Sigma^{1/2}|_{\mathrm{HS}} \Big).$$

Taking the two test-ball suprema and applying Lemma B.8 proves (B.7) and (B.8). At $q = 2$, Wick's identity and the two exact geometric suprema give the asserted identities. □

For Gaussian columns $x_i \sim \mathcal{N}(0, \Sigma_i)$, taking the maximum over $i \in [n]$ gives $\sqrt{2}|\Sigma|_{\mathrm{op}}^{\infty} \leq M_{\mathrm{HS}}^{(q)} = O(|\Sigma|_{\mathrm{op}}^{\infty})$ and $\sqrt{2}|\Sigma|_{\mathrm{HS}}^{\infty} \leq M_{\mathrm{op}}^{(q)} = O(|\Sigma|_{\mathrm{HS}}^{\infty})$ for every fixed $q \geq 2$.

The deterministic construction uses the following characterization; see [Bol97, KA16].

Theorem B.11 (Classical Stieltjes-transform characterization). *Let f be holomorphic on the upper half-plane, with $\Im f(z) > 0$ there. If*

$$\lim_{y \to +\infty} iy\, f(iy) = -1,$$

then f is the Stieltjes transform of a probability measure μ on $\mathbb{R}$:

$$f(z) = \int_{\mathbb{R}} \frac{\mu(d\lambda)}{\lambda - z}.$$

Moreover, for $a < b$ such that $\mu(\{a\}) = \mu(\{b\}) = 0$,

$$\mu([a, b]) = \lim_{y \to 0^+} \frac{1}{\pi} \int_a^b \Im(f(x + iy))\, dx.$$

If, in addition, $\Im(z f(z)) \geq 0$ for every z with $\Im z > 0$, then $\mu(\mathbb{R} \setminus \mathbb{R}_+) = 0$, and f admits a holomorphic continuation to $\mathbb{C} \setminus \mathbb{R}_+$.

References

[AEK17] Johannes Alt, László Erdős, and Torben Krüger. Local law for random gram matrices. *Electronic Journal of Probability*, 22(25):1–41, 2017.

[AEK19] Oskari H. Ajanki, László Erdős, and Torben Krüger. *Quadratic Vector Equations on Complex Upper Half-Plane*, volume 261 of *Memoirs of the American Mathematical Society*. American Mathematical Society, 2019.

[BGC16] Florent Benaych-Georges and Romain Couillet. Spectral analysis of the gram matrix of mixture models. *ESAIM: Probability and Statistics*, 20:217–237, 2016.

[Bol97] Vladimir Bolotnikov. On a general moment problem on the half axis. *Linear Algebra and its Applications*, 255:57–112, 1997.

[BS10] Zhidong D. Bai and Jack W. Silverstein. *Spectral Analysis of Large Dimensional Random Matrices*. Springer, New York, 2 edition, 2010.

[Bur88] Donald L. Burkholder. Sharp inequalities for martingales and stochastic integrals. In *Colloque Paul Lévy sur les processus stochastiques*, number 157–158 in Astérisque, pages 75–94. Société mathématique de France, 1988.

[BZ08] Zhidong D. Bai and Wang Zhou. Large sample covariance matrices without independence structures in columns. *Statistica Sinica*, 18(2):425–442, 2008.

[Cho22] Clément Chouard. Quantitative deterministic equivalent of sample covariance matrices with a general dependence structure, 2022.

[CM24] Chen Cheng and Andrea Montanari. Dimension free ridge regression. *The Annals of Statistics*, 52(6):2879–2912, 2024.

[DKL22] Alicja Dembczak-Kołodziejczyk and Anna Lytova. On the empirical spectral distribution for certain models related to sample covariance matrices with different correlations. *Random Matrices: Theory and Applications*, 11(3):2250030, 2022.

[DS07] R. Brent Dozier and Jack W. Silverstein. On the empirical distribution of eigenvalues of large dimensional information-plus-noise-type matrices. *Journal of Multivariate Analysis*, 98(4):678–694, 2007.

[DW25] Xiucai Ding and Zhenggang Wang. Global and local CLTs for linear spectral statistics of general sample covariance matrices when the dimension is much larger than the sample size with applications. *IEEE Transactions on Information Theory*, 71(8):6248–6296, 2025 (could only access Arxiv version arXiv:2308.08646).

[EK09] Noureddine El Karoui. Concentration of measure and spectra of random matrices: applications to correlation matrices, elliptical distributions and beyond. *The Annals of Applied Probability*, 19(6):2362–2405, 2009.

[FMPW26] Zhou Fan, Renyuan Ma, Elliot Paquette, and Zhichao Wang. Anisotropic local law for non-separable sample covariance matrices, 2026.

[HLN07] Walid Hachem, Philippe Loubaton, and Jamal Najim. Deterministic equivalents for certain functionals of large random matrices. *The Annals of Applied Probability*, 17(3):875–930, 2007.

[HLNV13] Walid Hachem, Philippe Loubaton, Jamal Najim, and Pascal Vallet. On bilinear forms based on the resolvent of large random matrices. *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques*, 49(1):36–63, 2013.

[HtBD13] Jakob Hoydis, Stephan ten Brink, and Mérouane Debbah. Massive MIMO in the UL/DL of cellular networks: How many antennas do we need? *IEEE Journal on Selected Areas in Communications*, 31(2):160–171, 2013.

[KA16] Abla Kammoun and Mohamed-Slim Alouini. No eigenvalues outside the limiting support of generally correlated gaussian matrices. *IEEE Transactions on Information Theory*, 62(7):4312–4326, 2016.

[KY17] Antti Knowles and Jun Yin. Anisotropic local laws for random matrices. *Probability Theory and Related Fields*, 169(1–2):257–352, 2017.

[LLC18] Cosme Louart, Zhenyu Liao, and Romain Couillet. A random matrix approach to neural networks. *Annals of Applied Probability*, 28(2):1190–1248, 2018.

[MM26] Renyuan Ma and Theodor Misiakiewicz. The anisotropic local law for sample covariance matrices under quadratic-form concentration, 2026. arXiv:2609.09440.

[MP67] V. A. Marčenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967.

[MS24] Theodor Misiakiewicz and Basil Saeed. A non-asymptotic theory of Kernel Ridge Regression: deterministic equivalents, test error, and GCV estimator, 2024.

[MWY23] Tianxing Mei, Chen Wang, and Jianfeng Yao. On singular values of data matrices with general independent columns. *The Annals of Statistics*, 51(2):624–645, 2023.

[PS11] Leonid Pastur and Mariya Shcherbina. *Eigenvalue Distribution of Large Random Matrices*, volume 171 of *Mathematical Surveys and Monographs*. American Mathematical Society, 2011.

[vBE65] Bengt von Bahr and Carl-Gustav Esseen. Inequalities for the rth absolute moment of a sum of random variables, $1 \leq r \leq 2$. *Annals of Mathematical Statistics*, 36(1):299–303, 1965.

[WCDS12] Sebastian Wagner, Romain Couillet, Mérouane Debbah, and Dirk T. M. Slock. Large system analysis of linear precoding in correlated MISO broadcast channels under limited feedback. *IEEE Transactions on Information Theory*, 58(7):4509–4537, 2012.

[XS26] Fangqing Xiao and Dirk T. M. Slock. Gaussian-process dynamics of diagonal expectation propagation under variance-profile gaussian measurements, 2026.

[Yas14] Pavel Yaskov. The universality principle for spectral distributions of sample covariance matrices. *arXiv preprint arXiv:1410.5190*, 2014.

[Yas16] Pavel Yaskov. Necessary and sufficient conditions for the marchenko–pastur theorem. *Electronic Communications in Probability*, 21:Paper No. 73, 8, 2016.

[Yin20] Yanqing Yin. On the singular value distribution of large-dimensional data matrices whose columns have different correlations. *Statistics*, 54(2):353–374, 2020.

[ZZXS26] Zeyan Zhuang, Xin Zhang, Dongfang Xu, and Shenghui Song. No eigenvalues outside the limiting support of generally correlated and noncentral sample covariance matrices. *IEEE Transactions on Information Theory*, 2026.